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A federated many-to-one Hopfield model for associative neural networks

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ABSTRACT

Federated learning (FL) enables collaborative training without centralizing raw data, but remains notoriously challenged by severe client heterogeneity and catastrophic forgetting under streaming distribution shifts. We propose a federated many-to-one (M2O) associative-memory framework designed for the unsupervised recovery of shared archetypes in continual, non-i.i.d. regimes. Departing from standard gradient-exchange methods, each client compresses its local experience into a low-rank Hebbian correlation operator. A central server then aggregates and factorizes these operators into global archetypes via an L-layer Associative Memory (LAM) architecture, strictly preserving raw-data locality. We rigorously cast the server-side aggregation as a low-rank-plus-noise spectral inference problem, utilizing random matrix theory to derive BBP phase transitions for archetype detectability and retrieval robustness: exact at initialization and, in the feedback-driven continual phase, extended through a conditional decomposition of the aggregated operator that leaves the detection threshold unchanged up to an explicit, server-observable inflation factor. To manage the distributed stability-plasticity dilemma, we introduce a local, label-free entropy controller that adaptively modulates client updates. Finally, we show that our adaptive framework, extended to federated Boolean-Gaussian Restricted Boltzmann Machines (RBMs), significantly outperforms standard FedAvg on structured non-IID datasets, validating the spectral mechanism of federated consolidation.

1. Introduction

Federated Learning (FL) enables collaborative learning across multiple data owners without centralizing raw data, by iterating local updates and server-side aggregation. This paradigm is attractive in privacy- and governance-constrained domains, but it is notoriously challenged by *statistical heterogeneity*: client data can be reasonably modeled as independent across clients but, within a client, they are often scarce and unbalanced. Such heterogeneity may induce biased or high-variance updates, slow convergence, and yield unstable global models (Hsu et al., 2019; Kairouz et al., 2021; Li et al., 2020; McMahan et al., 2017). These issues are exacerbated in streaming or open-world settings, where new patterns emerge over time and catastrophic forgetting becomes a primary failure mode.

A complementary line of work studies *memory systems* and continual adaptation. In neuroscience-inspired accounts such as Complementary Learning Systems (CLS), fast episodic storage and slow integrative learning jointly support generalization under distribution shifts (Kumaran et al., 2016; O'Reilly et al., 2014). In machine learning, continual learning methods seek to mitigate catastrophic forgetting by regulating the stability–plasticity trade-off (e.g. via regularization or replay), yet most approaches rely on centralized access to data or curated rehearsal buffers. In FL, the stability–plasticity dilemma becomes *distributed*: even under independence, heterogeneous client marginals and local streaming dynamics interact with drift and novelty, while privacy constraints strictly preclude global rehearsal.

In this context, *associative memories* provide a principled abstraction for storing and retrieving prototypical patterns from partial or noisy cues. Classical Hopfield-type models define content-addressable retrieval as energy minimization, with well-studied capacity and robustness properties (Amit et al., 1985; Hopfield, 1982; Personnaz et al., 1985). Beyond their original formulations, modern Hopfield networks have been linked to attention-like updates and large-capacity retrieval rules, renewing interest in associative mechanisms as building blocks for representation learning (Ramsauer et al., 2021). However, deploying associative memories in federated regimes raises new questions: which

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information should be stored locally, what should be aggregated, and how can a central server infer *global archetypes* from fragmented, heterogeneous local memories without ever accessing raw samples?

This work addresses these questions by proposing a *federated many-to-one associative memory* that learns and maintains a shared set of archetypes under heterogeneous, streaming (i.n.i.d.) regimes. Departing from standard gradient-exchange methods, each client compresses its local experience into a low-rank Hebbian operator, while the server aggregates these operators and factorizes the resulting global memory to recover the underlying archetypes. Our approach is motivated by two key observations. First, storing *operators* instead of raw examples enables the communication of batch-averaged correlation statistics that remain meaningful even when local datasets are small or drifting. While inherently preserving raw-data locality—since individual examples are never transmitted—these operators remain data-dependent and encapsulate aggregate archetypal information. Second, archetype discovery can be formally posed as a spectral inference problem: the aggregated operator concentrates around a low-rank signal plus noise under independence assumptions, allowing random-matrix theory to rigorously predict when global archetypes become spectrally detectable and stably retrievable (Baik et al., 2005; Benaych-Georges & Nadakuditi, 2011).

Although our core theory is developed in an idealized setting where latent archetypes are explicitly modeled, the same federated-memory principle applies more broadly to structured datasets where these archetypes are implicitly embedded and must be learned from data. To illustrate this extension, Section 5.5 presents a federated Boolean–Gaussian Restricted Boltzmann Machine (RBM) experiment on MNIST (with full implementation details deferred to Appendix B). While MNIST is traditionally a supervised classification benchmark rather than an archetype-reconstruction task, it serves as an effective testbed to demonstrate how unsupervised, federated energy-based learning extracts latent prototypical representations from real-world structured data. Ultimately, this RBM experiment provides a complementary outlook, confirming that our protocol readily generalizes beyond synthetic or teacher-generated patterns to scenarios where the archetypal structure must be discovered autonomously.

Related work

Our contribution lies at the intersection of four research lines. We review each of them in turn and then delineate precisely how the proposed method departs from and advances beyond them.

Federated learning and client heterogeneity. Federated learning was introduced through the FedAvg algorithm, which alternates several local stochastic-gradient steps with a server-side weighted average of the resulting model parameters (McMahan et al., 2017). When client data are not identically distributed, prolonged local optimization pulls the parameters toward client-specific optima, producing the client-drift phenomenon that slows convergence and destabilizes the global model (Hsu et al., 2019; Kairouz et al., 2021). Two influential correctives are FedProx, which augments the local objective with a proximal term anchoring each update to the current global model (Li et al., 2020), and SCAFFOLD, which cancels the drift through control variates in a variance-reduction scheme (Karimireddy et al., 2020). Crucially, all these methods communicate gradients or weights of a single differentiable predictor and optimize a supervised loss. Our setting differs fundamentally in the communicated object itself: clients transmit second-order Hebbian correlation operators rather than parameter vectors, and the global goal is the unsupervised recovery of shared archetypes, with no gradient exchange and no common parametric classifier.

Continual learning and catastrophic forgetting. Training a network on a sequence of distributions induces catastrophic forgetting—the rapid erosion of previously acquired competence once the parameters are overwritten to fit new data (Parisi et al., 2019). Regularization-based methods, such as elastic weight consolidation, slow down learning along the directions deemed important for past tasks, with importance typically estimated through the Fisher information (Kirkpatrick et al., 2017); replay-based methods, such as gradient episodic memory, instead store a buffer of past exemplars and constrain the update so that it does not increase the loss on them (Lopez-Paz & Ranzato, 2017). The empirical difficulty of the problem depends sharply on the evaluation protocol, and pure parameter regularization degrades markedly in the class-incremental regime, where task identity must itself be inferred (van de Ven & Tolias, 2019). These mechanisms echo the complementary learning systems account of fast and slow memory stores in the brain (Kumaran et al., 2016; O'Reilly et al., 2014), yet they generally presuppose centralized access to data or to a global replay buffer—precisely what a privacy-constrained federation forbids. Our stability and plasticity control is fully decentralized and replay-free: it operates at the operator level through the local entropy diagnostic introduced in Section 4.2, computed independently by each client from its own Hebbian operator and the broadcast server reconstruction.

Modern Hopfield networks and associative memory. Classical Hopfield networks store patterns as attractors of a quadratic energy, with a storage capacity limited to a linear fraction of the number of neurons (Amit et al., 1985; Hopfield, 1982). Dense associative memories lift this limit by replacing the quadratic interaction with higher-order energies (Krotov & Hopfield, 2016), and continuous modern Hopfield networks render the retrieval rule differentiable and formally equivalent to the attention mechanism of transformers (Ramsauer et al., 2021). The L-directional Associative Memory (LAM) architecture couples several Hopfield layers through a shared synaptic matrix with imitative intra-layer and anti-imitative inter-layer interactions, enabling the disentanglement of mixtures of hidden patterns directly at the synaptic level (Agliari et al., 2025b,c). Furthermore, the duality between Hopfield networks and Boolean–Gaussian restricted Boltzmann machines connects Hebbian couplings to energy-based generative models (Barra et al., 2012, 2018). We emphasize that the LAM factorization itself is not a contribution of the present work: rather, our contribution is to embed it within a federated, continual setting and to characterize, by spectral means, when its inputs are reliable.

Spectral methods and random matrix theory in machine learning. When the number of samples and the dimension grow proportionally, the spectrum of an empirical covariance matrix is governed by the Marchenko–Pastur law, and a planted low-rank signal becomes spectrally visible only once its strength exceeds the Baik–Ben Arous–Péché (BBP) phase transition (Baik et al., 2005; Baik & Silverstein, 2006; Johnstone, 2001). The companion theory for eigenvectors quantifies the asymptotic alignment between the outlier eigenvector and the true signal direction (Benaych-Georges & Nadakuditi, 2011; Paul, 2007), and random matrix theory has thus matured into a constructive design tool for high-dimensional learning (Couillet & Liao, 2022). While these tools are routinely applied to gradient or kernel covariance matrices, we are not aware of prior work that uses a BBP analysis to predict when archetypes become detectable inside a *federated Hebbian* estimator.

Positioning. The proposed framework departs from each of these lines along a distinct axis. Unlike gradient-based federated optimization (FedAvg, FedProx, SCAFFOLD), it replaces the exchange of model parameters with the exchange of Hebbian correlation operators, recasting federated consolidation as an unsupervised archetype-recovery problem rather than a synchronized parametric optimization. Unlike continual-learning methods built on parameter regularization or replay (elastic weight consolidation, gradient episodic memory), it attains stability and plasticity through a local, label-free entropy controller that needs neither a Fisher matrix nor stored exemplars, remaining fully compatible with raw-data locality. Unlike the monolithic LAM factorization, it distributes the disentanglement along a federated L-directional axis, re-

Table 1
Main notation used throughout the paper.

Symbol	Meaning
N	Number of neurons/features in each binary vector.
K	Number of latent archetypes in the teacher model.
L	Number of active clients and number of LAM layers.
λ	strength of anti-imitative interactions in LAM layers.
H	strength of external fields acting on LAM layers.
$\boldsymbol{\xi}^{\mu}$	Latent archetype μ, with $\mu = 1, \dots, K$.
$\boldsymbol{\eta}_c^{\prime,m}$	Example m observed by client c at round t.
r	Quality/correlation parameter of the noisy-example channel.
M_c^t	Number of examples observed by client c at round t.
$\boldsymbol{J}_c^{(t)}$	Client-side Hebbian operator at round t.
$\boldsymbol{J}_s^{(t)}$	Server-side aggregated Hebbian operator at round t.
$\widehat{\boldsymbol{J}}_s^{(t)}$	Server-side operator reconstructed from the recovered archetypes.
$\widehat{K}_{\mathrm{rec}}(t)$	Number of archetypes detected/reconstructed at round t.
δ_{dup}	Mutual-overlap threshold used for duplicate removal.
τ_{acc}	Quadratic-score threshold used for candidate acceptance.
$\tau_{\mathrm{MP}}(t)$	MP/BBP threshold applied to the unsharpened aggregated operator.
τ_{spec}	Post-sharpening eigenvalue threshold used for seed selection.
$\pi_{t,c}(\mu)$	Probability that client c observes archetype μ at round t.
$\pi_t(\mu)$	Global round-level mixture probability of archetype μ.
$e_\mu(t)$	Empirical exposure of archetype μ at round t.
$w_c(t)$	Plasticity weight used by client c at round t.
$\bar{w}_t$	Round-level plasticity, $\bar{w}_t = L^{-1}\sum_c w_c(t)$.
$\pi_t^{\mathrm{eff}}(\mu)$	Plasticity-weighted exposure of archetype μ at round t, cf. (4.15).
ρ_t	Plasticity dispersion $w_{\max}(t)/w_{\min}(t)$ of round t, cf. (4.22).
$\boldsymbol{P}_t,\ \boldsymbol{F}_{t-1}$	Plastic and feedback parts of the round-t operator, cf. (4.16).

constructing global archetypes from statistically partial client views in the absence of any global observer. Finally, it elevates random matrix theory from an *a posteriori* analytical lens to an operational component of the pipeline: the BBP detectability thresholds determine the effective number of archetypes, and the eigenvector-alignment law supplies the seeds for the reconstruction dynamics.

The remainder of the paper is organized as follows: we formalize the model, the federated setting, and the reconstruction pipeline in Section 2; we develop the theoretical analysis in Section 4; and we present experiments and ablations, including the RBM extension to structured datasets, in Section 5.

2. Problem setting and notation

We now introduce the problem setting and notation (see Table 1) and formalize the federated framework studied in this work. Our goal is to exploit federated interactions among multiple associative memories to reconstruct latent *archetypes* from noisy observations, particularly in the realistic regime where data are partitioned across clients according to heterogeneous (non-i.i.d.) distributions. Before presenting the federated pipeline, we recall the basic building blocks—Hopfield-type associative memories and their heteroassociative multilayer extension—and specify the generative model used throughout.

2.1. Associative and heteroassociative neural networks

Central to our method are Hopfield-like associative memories. We briefly recall the classical formulation, which serves as the local learner at each client, and then introduce the multilayer associative memory (LAM) architecture utilized by the server.

Hopfield associative memory and RBM duality. The key components of our pipeline are Hopfield-like associative memories: binary neural networks whose neurons interact through Hebbian synaptic couplings (Agliari et al., 2023). Their primary purpose is to store and retrieve a collection of entities, usually referred to as patterns or archetypes. In the present setting, we consider a collection of K Boolean patterns of length N, denoted by $\{\boldsymbol{\xi}^{\mu}\}_{\mu=1}^{K}$, whose entries are i.i.d. Rademacher random variables, namely

$$\mathbb{P}(\xi_i^{\mu}) = \frac{1}{2}\delta(\xi_i^{\mu} - 1) + \frac{1}{2}\delta(\xi_i^{\mu} + 1), \qquad \forall\, i = 1, \dots, N, \quad \mu = 1, \dots, K. \tag{2.1}$$

These random variables are used to construct the Hebbian interaction matrix

$$J_{ij}^{\mathrm{Hebb}}(\boldsymbol{\xi}) = \frac{1}{N}\sum_{\mu=1}^{K}\xi_i^{\mu}\xi_j^{\mu}, \qquad i \neq j, \tag{2.2}$$

with $J_{ii}^{\mathrm{Hebb}} = 0$. This matrix encodes the pairwise correlations among the stored patterns and models the effective interactions among neurons in the network.

The neural configuration is described by

$$\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_N) \in \Omega, \qquad \Omega = \{-1, +1\}^N, \tag{2.3}$$

and the system is governed by the Hopfield Hamiltonian

$$\mathcal{H}_{\mathrm{Hop}}(\boldsymbol{\sigma} \mid \boldsymbol{\xi}) = -\frac{1}{2}\sum_{\substack{i,j=1 \\ i\neq j}}^{N} J_{ij}^{\mathrm{Hebb}}\,\sigma_i\sigma_j. \tag{2.4}$$

The configuration space Ω is endowed with the random Boltzmann–Gibbs measure

$$\mathbb{P}_{\mathrm{Hop}}(\boldsymbol{\sigma} \mid \boldsymbol{\xi}) = \frac{\exp\left[-\beta\,\mathcal{H}_{\mathrm{Hop}}(\boldsymbol{\sigma} \mid \boldsymbol{\xi})\right]}{Z_{\mathrm{Hop}}(\beta \mid \boldsymbol{\xi})}, \tag{2.5}$$

where

$$Z_{\mathrm{Hop}}(\beta \mid \boldsymbol{\xi}) = \sum_{\boldsymbol{\sigma}\in\Omega}\exp\left[-\beta\,\mathcal{H}_{\mathrm{Hop}}(\boldsymbol{\sigma} \mid \boldsymbol{\xi})\right] \tag{2.6}$$

is the partition function and $\beta := 1/T \in \mathbb{R}^+$ is the inverse temperature.

The dynamical network update rules can be written as

$$\sigma_i(t+1) = \mathrm{sign}\left[\tanh\left(\beta h_i(\boldsymbol{\sigma}(t) \mid \boldsymbol{\xi})\right) + u_i(t)\right], \tag{2.7}$$

where $u_i(t) \overset{\text{i.i.d.}}{\sim} \mathcal{U}([-1,1])$ for $i = 1, \dots, N$, and $h_i(\boldsymbol{\sigma} \mid \boldsymbol{\xi}) = \sum_{\substack{j=1 \\ j\neq i}}^{N} J_{ij}^{\mathrm{Hebb}}\,\sigma_j$ is the local effective field acting on the ith neuron.

Remark 1. The level of stochasticity in the system is governed by the temperature $T = \beta^{-1}$. In the zero-temperature limit $T \to 0$ (equivalently $\beta \to \infty$), the update becomes deterministic and neurons align perfectly with their local fields,

$$\sigma_i(t+1) = \mathrm{sign}\left[h_i(\boldsymbol{\sigma}(t) \mid \boldsymbol{\xi})\right]. \tag{2.8}$$

Conversely, in the infinite-temperature limit $T \to \infty$ (equivalently $\beta \to 0$), the local field becomes irrelevant and the dynamics reduces to pure noise.

The recurrent formulation above also admits an equivalent dual representation in terms of a binary–Gaussian (or hybrid) Restricted Boltzmann Machine (RBM) (Barra et al., 2012, 2018). In this dual framework, the visible layer represents the binary neural configuration $\boldsymbol{\sigma} \in \{-1,+1\}^N$, while the hidden layer consists of K continuous Gaussian variables $\boldsymbol{z} = (z_1, \dots, z_K) \in \mathbb{R}^K$, each associated with one stored pattern $\boldsymbol{\xi}^{\mu}$. By absorbing the noise level β into the visible–hidden couplings at unit inverse temperature, the RBM energy reads

$$\mathcal{H}_{\mathrm{RBM}}(\boldsymbol{\sigma}, \boldsymbol{z} \mid \boldsymbol{\xi}) = \frac{1}{2}\sum_{\mu=1}^{K} z_\mu^2 - \sqrt{\frac{\beta}{N}}\sum_{i=1}^{N}\sum_{\mu=1}^{K}\xi_i^{\mu}\sigma_i z_\mu. \tag{2.9}$$

The corresponding joint Boltzmann–Gibbs measure is given by

$$\mathbb{P}_{\mathrm{RBM}}(\boldsymbol{\sigma}, \boldsymbol{z} \mid \boldsymbol{\xi}) = \frac{\exp\left[-\mathcal{H}_{\mathrm{RBM}}(\boldsymbol{\sigma}, \boldsymbol{z} \mid \boldsymbol{\xi})\right]}{Z_{\mathrm{RBM}}(\beta \mid \boldsymbol{\xi})}, \tag{2.10}$$

with the partition function defined as

$$Z_{\mathrm{RBM}}(\beta \mid \boldsymbol{\xi}) = \sum_{\boldsymbol{\sigma}\in\Omega}\int_{\mathbb{R}^K}\prod_{\mu=1}^{K}\frac{dz_\mu}{\sqrt{2\pi}}\,\exp\left[-\mathcal{H}_{\mathrm{RBM}}(\boldsymbol{\sigma}, \boldsymbol{z} \mid \boldsymbol{\xi})\right]. \tag{2.11}$$

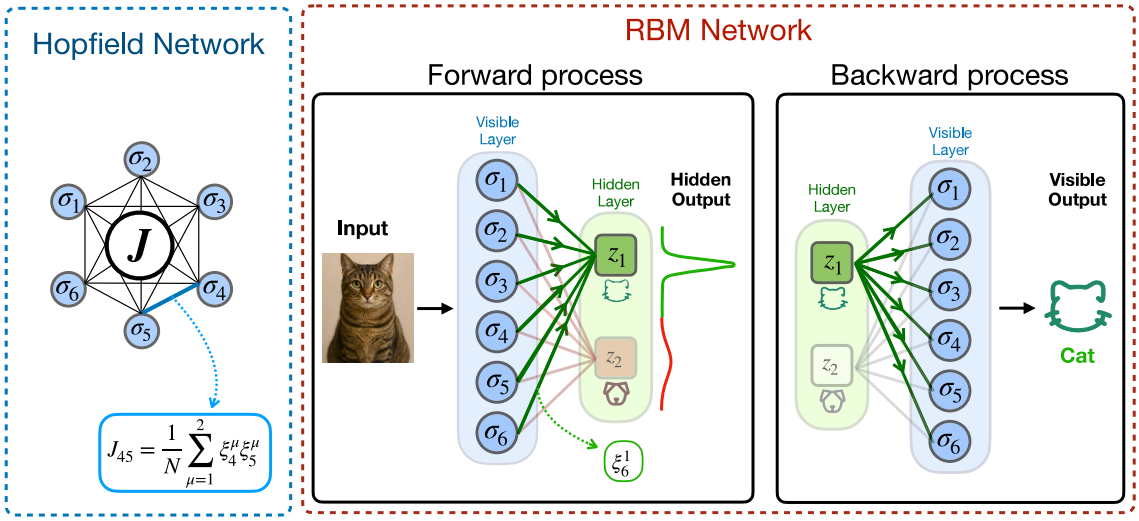


Fig. 1. Hopfield–RBM dual representation of associative retrieval. On the Hopfield side, the visible neurons are fully connected through the Hebbian matrix $\boldsymbol{J}^{\text{Hebb}}$, and an input cue is iteratively mapped into an output attractor via recurrent relaxation. On the RBM side, the same interaction is represented by a bipartite architecture with binary visible units $\boldsymbol{\sigma}$ and continuous Gaussian hidden variables $\boldsymbol{z}$. Visible units activate hidden pattern variables through the weights ξ_i^μ, and the hidden layer feeds back a reconstructed visible configuration. Marginalizing over the hidden variables yields the recurrent Hopfield coupling $J_{ij}^{\text{Hebb}} = N^{-1}\sum_{\mu=1}^{K}\xi_i^\mu\xi_j^\mu$, as illustrated by the isolated edge in the figure.

The equivalence with the Hopfield model naturally emerges by marginalizing over the Gaussian hidden variables. By applying the elementary Gaussian identity $\int_{\mathbb{R}}\frac{dz}{\sqrt{2\pi}}\exp(-z^2/2+az)=\exp(a^2/2)$, we obtain

$$\begin{aligned}\mathbb{P}_{\text{RBM}}(\boldsymbol{\sigma}\mid\boldsymbol{\xi}) &\propto \int_{\mathbb{R}^K}\prod_{\mu=1}^{K}\frac{dz_\mu}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}\sum_{\mu=1}^{K}z_\mu^2+\sqrt{\frac{\beta}{N}}\sum_{i=1}^{N}\sum_{\mu=1}^{K}\xi_i^\mu\sigma_i z_\mu\right]\\ &=\exp\left[\frac{\beta}{2N}\sum_{\mu=1}^{K}\left(\sum_{i=1}^{N}\xi_i^\mu\sigma_i\right)^2\right]\\ &=\exp\left[\frac{\beta}{2}\sum_{i,j=1}^{N}\left(\frac{1}{N}\sum_{\mu=1}^{K}\xi_i^\mu\xi_j^\mu\right)\sigma_i\sigma_j\right].\end{aligned} \tag{2.12}$$

Up to the diagonal contribution $i=j$ (which is constant since $\sigma_i^2=1$), the marginal distribution over the visible layer perfectly coincides with the Hopfield Boltzmann–Gibbs measure, namely $\mathbb{P}_{\text{RBM}}(\boldsymbol{\sigma}\mid\boldsymbol{\xi})\propto\exp\left[-\beta\,\mathcal{H}_{\text{Hop}}(\boldsymbol{\sigma}\mid\boldsymbol{\xi})\right]$. This demonstrates that the Hebbian recurrent network and the binary–Gaussian RBM are simply two equivalent representations of the same equilibrium model.

In the recurrent formulation, pattern information is encoded directly within the effective pairwise couplings J_{ij}^{Hebb}; conversely, in the RBM formulation, these same correlations are mediated by the Gaussian hidden layer associated with the stored archetypes.

This dual viewpoint is highly instructive for the present work. The Hopfield perspective emphasizes the recurrent relaxation of the system toward attractors in the energy landscape, where low-energy states correspond to internally consistent neural configurations. In the retrieval regime, the dynamics drives an initial cue toward local minima of Eq. (2.4), naturally aligning the configuration with the stored patterns. As a result, in suitable regimes of K versus N, each pattern $\boldsymbol{\xi}^\mu$ acts as a stable (or metastable) attractor, allowing the system to reconstruct a prototype even from a corrupted initialization. On the other hand, the RBM perspective makes the corresponding input–feature–output information flow explicit: a visible configuration activates hidden pattern variables, which then feed back to reconstruct the visible configuration. This underlying duality is schematically illustrated in Fig. 1.

Archetypes, examples, and client-side learning. In the federated setting, we refer to the latent patterns $\{\boldsymbol{\xi}^\mu\}_{\mu=1}^K$ as *archetypes*, acting as prototypical data-generating anchors. Each client observes only *examples*—noisy and partial realizations of these archetypes—and aims to learn exclusively from its local, unlabeled dataset.

The examples are generated via a signal-plus-noise scenario: first, an archetype index μ is selected, and then a corrupted realization $\boldsymbol{\eta}\in\{-1,+1\}^N$ is produced via independent bit flips:

$$\mathbb{P}\big(\eta_i=\xi_i^\mu\mid\mu\big)=\tfrac{1+r}{2},\qquad \mathbb{P}\big(\eta_i=-\xi_i^\mu\mid\mu\big)=\tfrac{1-r}{2}, \tag{2.13}$$

where $r\in[0,1]$ quantifies the *dataset quality*.As $r\to1$, the examples perfectly coincide with their generating archetype. Conversely, the endpoint $r=0$ is not treated as an informative data-generating regime; rather, it corresponds to examples that are statistically independent of their generating archetype. We treat this endpoint strictly as a degenerate limiting case, or as an effective description of corrupted/adversarial clients whose local batches carry no archetypal information.

In our federated architecture, the number of clients, L, is deliberately matched to the number of layers in the server-side block. Therefore, throughout this paper, L denotes both the number of federated clients and the number of layers in the server's multilayer associative memory. To prevent ambiguity, we index clients by $c=1,\dots,L$ and the server-side layers by $a,b=1,\dots,L$. These two sets are kept in one-to-one correspondence within the proposed many-to-one framework.

Each client c stores M_c examples and locally constructs a Hopfield-like synaptic matrix based on its local dataset. This is achieved via an unsupervised adaptation of the Hebbian rule, given by

$$\left(J_c\right)_{ij}=\frac{1}{N\,M_c}\sum_{m=1}^{M_c}\left(\eta_c\right)_i^m\left(\eta_c\right)_j^m,\qquad c=1,\dots,L, \tag{2.14}$$

where the total dataset size across the federation is $M=\sum_{c=1}^{L}M_c$. Even though the data are assumed to be mutually independent, their random allocation implies that a given client may not have access to all archetypes, but only to a strict subset of them.

Clients do not communicate directly. Coordination is mediated solely by a central server responsible for aggregating synaptic-level information and reconstructing the underlying archetypes. To this end, the server is equipped with a heteroassociative generalization of Hopfield networks, namely the *L-layer Associative Memory* (LAM) model introduced in Agliari et al. (2025b). A key feature of LAM is its ability to

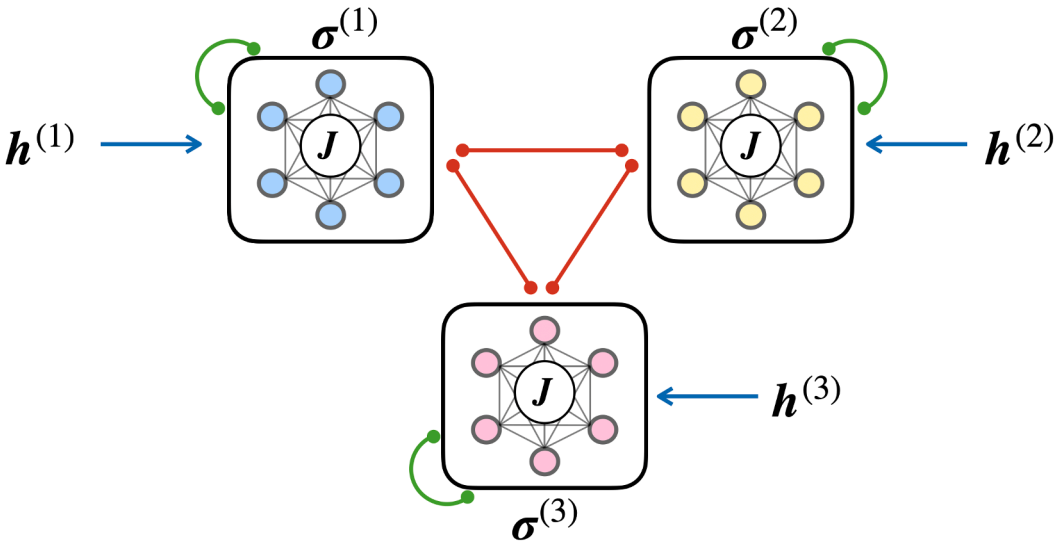


Fig. 2. Schematic illustration of the model for $L = 3$ layers. Each layer $a \in \{1, 2, 3\}$ acts as a Hopfield network (with state $\boldsymbol{\sigma}^{(a)}$), and all layers share the same synaptic coupling matrix. The three contributions to the Hamiltonian in (2.15) are highlighted: imitative intra-layer interactions (green self-loop), anti-imitative inter-layer interactions (red links), and the coupling to an external field $\boldsymbol{h}^{(a)}$ (blue arrow).

operate directly at the synaptic level, enabling full reconstruction from the aggregated interaction matrices alone (Agliari et al., 2025c).

We use the acronym *LAM* to denote the general L-directional (or L-layer) associative-memory architecture. The term *TAM* is strictly reserved for the three-layer tripartite case ($L = 3$), making it a specific instance of LAM rather than a distinct model. In this work, the server-side reconstruction module operates as a LAM block whose layer count, L, matches the number of federated clients. Naturally, when $L = 3$, this setup recovers the TAM framework explored in previous studies.

LAM: an L-layer associative memory. We now briefly recall the LAM model to fix notation and emphasize the mechanism that will be exploited by the server.

Consider L layers, each composed of N binary neurons. The configuration of layer a is denoted by $\boldsymbol{\sigma}^{(a)} = (\sigma_1^{(a)}, \dots, \sigma_N^{(a)}) \in \{-1, +1\}^N$ for $a = 1, \dots, L$. Layers interact through a shared synaptic matrix $\boldsymbol{J}^{\text{Hebb}}$ and a coupling matrix $\boldsymbol{g} = \{g_{ab}\}_{a,b=1}^{L} \in \mathbb{R}^{L\times L}$ that controls the intensity and polarity (imitative vs. anti-imitative) of intra- and inter-layer interactions. The Hamiltonian takes the form

$$\mathcal{H}(\boldsymbol{\sigma} \mid \boldsymbol{g}, \boldsymbol{J}, H, \boldsymbol{h}) = -\sum_{a,b=1}^{L} g_{ab} \sum_{i,j=1}^{N} \sigma_i^{(a)} J_{ij}^{\text{Hebb}} \sigma_j^{(b)} - H \sum_{a=1}^{L} \sum_{i=1}^{N} h_i^{(a)} \sigma_i^{(a)}. \tag{2.15}$$

The last term accounts for a layer-dependent external field $\boldsymbol{h}^{(a)}$, modulated by a global strength H, which can be used to drive the dynamics.

The sign of g_{ab} determines whether layers tend to align or anti-align: if $g_{ab} > 0$ (resp. $g_{ab} < 0$), configurations that maximize (resp. minimize) the overlap $\boldsymbol{\sigma}^{(a)} \cdot \boldsymbol{\sigma}^{(b)}$ are energetically favored. In the following, we focus on the simple yet expressive choice

$$g_{ab} = \begin{cases} 1, & a = b, \\ -\lambda, & a \neq b, \end{cases} \tag{2.16}$$

where $\lambda \in \left[0, (L-1)^{-1}\right)$ guarantees that $\boldsymbol{g}$ is positive definite, effectively making $\boldsymbol{g}$ an equicorrelation matrix. This structure yields imitative Hopfield-like interactions within each layer, promoting coherent retrieval states, while simultaneously inducing anti-imitative couplings across distinct layers, discouraging configurations where all layers retrieve the exact same pattern (see Fig. 2).

This built-in bias perfectly matches our target task: starting from synaptic-level information (an interaction matrix assumed to be Hebbian), the model promotes a *factorization* of the stored content across the multiple layers, ultimately enabling the stable reconstruction of the underlying archetypes.

2.1.1. Factorization procedure

We now introduce the factorization procedure embedded in the LAM dynamics, which constitutes the core mechanism enabling the reconstruction of latent patterns from an observed synaptic matrix.

The LAM model (2.15), for a given realization of hidden patterns $\{\boldsymbol{\xi}^{\mu}\}_{\mu=1}^{K}$, can disentangle mixed configurations such as classical spurious states. For instance, when $L = 3$, one may consider the mixture $\boldsymbol{x} = \text{sgn}\left(\sum_{\ell=1}^{3} \boldsymbol{\xi}^{\ell}\right)$, where $\text{sgn}(\cdot)$ acts componentwise. By feeding $\boldsymbol{x}$ as input to each layer, the dynamics can relax to configurations of the form

$$(\boldsymbol{\sigma}^{(1)}, \boldsymbol{\sigma}^{(2)}, \boldsymbol{\sigma}^{(3)}) = (\boldsymbol{\xi}^{1}, \boldsymbol{\xi}^{2}, \boldsymbol{\xi}^{3}) \quad \text{(up to permutations)},$$

thus successfully recovering the constituents of the mixture (Agliari et al., 2025b).

In Agliari et al. (2025c), it is further shown that a closely related architecture—obtained by modifying the heteroassociative component—can tackle even more challenging reconstruction tasks. Specifically, consider the Hamiltonian

$$\begin{aligned} \widetilde{\mathcal{H}}(\boldsymbol{\sigma}; \lambda, H, \boldsymbol{J}^{\text{Hebb}}, \boldsymbol{h}) = &-\sum_{a=1}^{L} \sum_{i,j=1}^{N} J_{ij}^{\text{Hebb}} \sigma_i^{(a)} \sigma_j^{(a)} \\ &+ \frac{\lambda}{N} \sum_{\substack{a,b=1 \\ a\neq b}}^{L} \sum_{i,j,k,l=1}^{N} J_{ij}^{\text{Hebb}} J_{kl}^{\text{Hebb}} \sigma_i^{(a)} \sigma_j^{(b)} \sigma_k^{(a)} \sigma_l^{(b)} \\ &- H \sum_{a=1}^{L} \sum_{i=1}^{N} h_i^{(a)} \sigma_i^{(a)}. \end{aligned} \tag{2.17}$$

Exploiting the mean-field structure of the model, the Hamiltonian (2.17) can be conveniently recast as $\mathcal{E}_{N,\Xi}(\boldsymbol{\sigma}) = -\sum_{a=1}^{L} \sum_{i=1}^{N} \hat{h}_i^{(a)} \sigma_i^{(a)}$, where the effective field $\hat{\boldsymbol{h}}^{(a)}$ acting on neurons in layer a is given by:

$$\hat{\boldsymbol{h}}^{(a)}(\boldsymbol{\sigma}) = \boldsymbol{h}^{a\to a}(\boldsymbol{\sigma}) + \sum_{\substack{b=1 \\ b\neq a}}^{L} \boldsymbol{h}^{b\to a}(\boldsymbol{\sigma}) + H\, \boldsymbol{h}^{(a)}. \tag{2.18}$$

This effective field comprises three distinct contributions: the intra-module (auto-associative) field, the inter-module (anti-imitative) field,

and the external field. The definitions of the first two contributions are derived directly from Eq. (2.17), yielding:

$$\boldsymbol{h}^{a\to a}(\boldsymbol{\sigma}) = \boldsymbol{J}^{\text{Hebb}} \cdot \boldsymbol{\sigma}^{(a)}, \tag{2.19}$$

$$\boldsymbol{h}^{b\to a}(\boldsymbol{\sigma}) = -\frac{\lambda}{N}\left(\boldsymbol{J}^{\text{Hebb}} \cdot \boldsymbol{\sigma}^{(b)}\right)\left((\boldsymbol{\sigma}^{(b)})^T \boldsymbol{J}^{\text{Hebb}} \boldsymbol{\sigma}^{(a)}\right). \tag{2.20}$$

Here the fields (2.19)-(2.20) are written in the cavity normalization inherited from the LAM construction of Agliari et al. (2025b,c): both the intra- and the inter-layer field equal one half of $-\partial\widetilde{\mathcal{H}}/\partial\sigma_i^{(a)}$, i.e. the symmetric mean-field drift acting on a spin. This factor is uniform across the two contributions and is immaterial for the dynamics, since it is reabsorbed into the inverse temperature β entering the update rule (2.21).

We now set up a dynamics that evolves configurations toward lower-energy states. Allowing for stochastic noise controlled by the inverse temperature $\beta \in \mathbb{R}_+$, the neuronal configuration $\boldsymbol{\sigma}(t)$ at (discrete) time t is updated synchronously according to

$$\boldsymbol{\sigma}^{(a)}(t+1) = \text{sign}\left[\tanh\left(\beta\hat{\boldsymbol{h}}^{(a)}(\boldsymbol{\sigma}(t))\right) + \boldsymbol{u}^{(a)}(t)\right], \tag{2.21}$$

where $\hat{\boldsymbol{h}}^{(a)}$ is the effective field of the ath layer, computed at each time-step t via Eqs. (2.18–2.20), and $\boldsymbol{u}^{(a)}(t) \overset{\text{i.i.d.}}{\sim} \mathcal{U}([-1,1]^N)$ for all $a = 1, \dots, L$.

From now on, we assume that the Hebbian kernel $\boldsymbol{J}^{\text{Hebb}}$ is known (or can be reliably recovered from data), while its factorization in terms of the underlying patterns $\{\boldsymbol{\xi}^\mu\}$ remains *unknown*. Our goal is to reconstruct the hidden patterns using only $\boldsymbol{J}^{\text{Hebb}}$ and, when available, a set of mixed inputs.

Candidate generation from mixed inputs. In Agliari et al. (2025c), the model (2.17) is studied in a setting where one has access to the synaptic matrix alongside m mixed inputs of the form

$$\boldsymbol{x}^\alpha = \text{sgn}\left[f\left(\sum_{\mu=1}^{K} c_\mu^\alpha \boldsymbol{\xi}^\mu\right)\right], \qquad \alpha = 1, \dots, m, \tag{2.22}$$

where $f(\cdot)$ is a componentwise nonlinearity (possibly the identity) and the coefficients $\{c_\mu^\alpha\}$ depend on the specific scenario.[1]

For each α, we clamp the external fields to the same input across all layers (i.e., $\boldsymbol{h}^{(a)} = \boldsymbol{x}^\alpha$ for all $a = 1, \dots, L$) and let the dynamics (2.21) evolve to stationarity. Collecting the final configurations across all layers and repeating this for the m mixtures yields Lm candidates $\{\bar{\boldsymbol{\sigma}}^\ell\}_{\ell=1}^{Lm}$. Unlike approaches that require $L \geq K$ in a single run, here one can trade layers for repetitions: it suffices to choose m such that $Lm \gtrsim K$, promoting adequate coverage of the distinct sources. However, two main issues may arise: (i) the emergence of *duplicate* candidates (configurations with large mutual overlap), and (ii) convergence to *spurious equilibria* (stable states unrelated to the true patterns).

Because not every run guarantees a disentangled solution, we introduce an acceptance criterion that can be evaluated entirely without access to the ground truth.

Acceptance criterion. The acceptance test combines two distinct filters to address both duplicates and spurious equilibria.

(i) **De-duplication via mutual overlap.** We compute the pairwise overlap

$$q_{\ell k} = \frac{1}{N}\sum_{i=1}^{N} \bar{\sigma}_i^\ell \bar{\sigma}_i^k, \tag{2.23}$$

and discard duplicates whenever $q_{\ell k} > \delta_{\text{dup}}$, a conservative threshold for random patterns. A threshold of $\delta_{\text{dup}} = 0.5$ safely identifies candidates that are strongly correlated and thus likely represent the same recovered pattern (up to noise). Since the overlap between independent random Boolean patterns concentrates around zero for large N, this value provides a robust margin to merge near-identical reconstructions without erroneously fusing distinct patterns.[2]

(ii) **Spectral filter via a pseudo-inverse kernel.** Let $\boldsymbol{C} \in \mathbb{R}^{K\times K}$ be the pattern correlation matrix, where $C_{\mu\nu} = \frac{1}{N}\sum_{i=1}^{N} \xi_i^\mu \xi_i^\nu$. We define the pseudo-inverse kernel (Kanter & Sompolinsky, 1987; Personnaz et al., 1985) as

$$J_{ij}^{KS} = \frac{1}{N}\sum_{\mu,\nu=1}^{K} \xi_i^\mu \, C_{\mu\nu}^{-1} \, \xi_j^\nu. \tag{2.24}$$

It can be shown (Kanter & Sompolinsky, 1987; Personnaz et al., 1985) that the true patterns are eigenvectors of $\boldsymbol{J}^{KS}$ with an eigenvalue of 1, namely

$$\sum_{j=1}^{N} J_{ij}^{KS}\xi_j^\mu = \xi_i^\mu, \qquad \frac{1}{N}\sum_{i,j=1}^{N} \xi_i^\mu J_{ij}^{KS} \xi_j^\mu = 1.$$

Consequently, a theoretically ideal, yet infeasible, acceptance criterion would necessitate

$$\frac{1}{N}(\bar{\boldsymbol{\sigma}}^\ell)^\top \boldsymbol{J}^{KS} \bar{\boldsymbol{\sigma}}^\ell = 1. \tag{2.25}$$

Although $\boldsymbol{J}^{KS}$ depends on the unknown patterns, it can be approximated directly from the observed Hebbian kernel via an iterative unlearning process (Fachechi et al., 2019):

$$\boldsymbol{J}_{k+1} = \boldsymbol{J}_k + \frac{\epsilon}{1+\epsilon k}\left(\boldsymbol{J}_k - \boldsymbol{J}_k^2\right), \quad \epsilon < \left[\left(1 + \sqrt{\text{Tr}\,\boldsymbol{J}_0/N}\right)^2 - 1\right]^{-1}. \tag{2.26}$$

Crucially, to correctly capture the network load, this iterative procedure is initialized with the unhollowed empirical correlation matrix $\boldsymbol{J}_0$, which preserves the diagonal self-overlaps $(\boldsymbol{J}_0)_{ii} \approx K/N$ so that $\text{Tr}\,\boldsymbol{J}_0/N \approx K/N$. Once the sequence converges to a fixed point, its diagonal is explicitly set to zero to prevent self-reinforcement during the LAM retrieval dynamics (as detailed in Appendix D). We denote this hollowed numerical approximation by $\hat{\boldsymbol{J}}^{KS}$.

We denote the corresponding unique fixed point (or its numerical approximation) by $\hat{\boldsymbol{J}}^{KS}$.

Since exact equality in (2.25) is rarely achieved in simulations—due to finite fractions of flipped bits in $\bar{\boldsymbol{\sigma}}^\ell$ and the approximate nature of $\hat{\boldsymbol{J}}^{KS}$—we accept a candidate whenever

$$\frac{1}{N}(\bar{\boldsymbol{\sigma}}^\ell)^\top \hat{\boldsymbol{J}}^{KS} \bar{\boldsymbol{\sigma}}^\ell > \tau_{\text{acc}}, \tag{2.27}$$

where τ_{acc} is the acceptance threshold acting on the quadratic score of a reconstructed candidate. Notice that τ_{acc} is distinct from the eigenvalue threshold τ_{spec} introduced below: the former acts on reconstructed configurations, whereas the latter acts on the spectrum of the sharpened operator.

Overall, candidates are examined sequentially and $\bar{\boldsymbol{\sigma}}^\ell$ is retained if and only if

$$\text{(i)} \ \max_{k\in\mathcal{R}_\ell} q_{\ell k} < \delta_{\text{dup}} \qquad \text{and} \qquad \text{(ii)} \ \frac{1}{N}(\bar{\boldsymbol{\sigma}}^\ell)^\top \hat{\boldsymbol{J}}^{KS} \bar{\boldsymbol{\sigma}}^\ell > \tau_{\text{acc}}, \tag{2.28}$$

where $\mathcal{R}_\ell$ is the set of candidates already retained when $\bar{\boldsymbol{\sigma}}^\ell$ is examined, with the convention $\max_{k\in\emptyset}(\cdot) = -\infty$. Comparing each candidate with the already retained ones, rather than with all the others, guarantees that every group of duplicates contributes exactly one representative instead of being discarded as a whole; this is the greedy rule implemented in Algorithm 2. The accepted, distinct candidates form the final set of reconstructed patterns $\{\boldsymbol{\xi}_R^\ell\}_{\ell=1}^{\hat{K}_{\text{rec}}}$, where $\hat{K}_{\text{rec}}$ denotes the total number of successfully reconstructed archetypes.

[1] Equivalently, $x_i^\alpha = \text{sgn}\left(f(\sum_{\mu=1}^{K} c_\mu^\alpha \xi_i^\mu)\right)$ defines a nonlinear random mapping from the K-dimensional feature vector $(\xi_i^1, \dots, \xi_i^K)$ to m outputs; this can be interpreted as a perceptron-like transformation, with the site index i labeling datapoints.

[2] For two configurations differing by a fraction p of spins, the overlap is $q_{\ell k} \approx 1 - 2p$; hence, $q_{\ell k} > 0.5$ implies $p < 0.25$. By contrast, the overlap between independent random patterns is typically $\mathcal{O}(N^{-1/2})$, making values above 0.5 overwhelmingly unlikely unless the solutions are indeed duplicates.

Generating initial configurations from $\boldsymbol{J}^{\text{Hebb}}$ alone. In our target setting, the only accessible object is the synaptic coupling matrix $\boldsymbol{J}^{\text{Hebb}}$, whereas the LAM dynamics inherently requires input configurations that resemble spurious mixtures (i.e., sign combinations of the original patterns). We therefore need a mechanism to generate suitable initial states directly from $\boldsymbol{J}^{\text{Hebb}}$.

To this end, we exploit the spectral structure of the pseudo-inverse coupling $\boldsymbol{J}^{KS}$ defined in (2.24). The eigenspace associated with eigenvalue 1 is K-dimensional and is spanned by linear combinations of the true patterns. Furthermore, we can approximate $\boldsymbol{J}^{KS}$ starting from $\boldsymbol{J}^{\text{Hebb}}$ via the unlearning iteration (2.26). These observations lead to the following fully unsupervised factorization pipeline:

1. **Approximate the pseudo-inverse kernel.** Starting from $\boldsymbol{J}^{\text{Hebb}}$, iterate
$$\boldsymbol{J}_{k+1} = \boldsymbol{J}_k + \frac{\epsilon}{1+\epsilon k}(\boldsymbol{J}_k - \boldsymbol{J}_k^2), \qquad \boldsymbol{J}_0 = \boldsymbol{J}^{\text{Hebb}},$$
until convergence to $\widehat{\boldsymbol{J}}^{KS}$.
2. **Spectral filtering.** Compute the eigendecomposition of the sharpened operator $\widehat{\boldsymbol{J}}^{KS}$ and retain the eigenvectors associated with eigenvalues larger than the post-sharpening threshold τ_{spec}. This yields:
   - an estimate of the effective number of stored patterns,
   $$\widehat{K}_{\text{spec}} = \#\{i : \lambda_i(\widehat{\boldsymbol{J}}^{KS}) > \tau_{\text{spec}}\},$$
   which acts as a proxy for the true load K. Here τ_{spec} is an operational threshold on the sharpened spectrum and should not be identified with the Marchenko–Pastur upper edge, which is defined for the unsharpened aggregated operator (see Section 4.1 and Appendix D);
   - a set of $\widehat{K}_{\text{spec}}$ orthogonalized linear combinations of the true patterns,
   $$\tilde{x}_i^{\delta} = \sum_{\mu=1}^{K} \tilde{c}_{\mu}^{\delta}\, \xi_i^{\mu}, \qquad \delta = 1, \dots, \widehat{K}_{\text{spec}}, \tag{2.29}$$
   where $\tilde{c}_{\mu}^{\delta}$ are random coefficients (e.g., uniform, gaussian or binary), spanning the same subspace as $\{\boldsymbol{\xi}^{\mu}\}$.

   The number m of synthetic mixtures generated at the next step is itself not a free parameter. It is fixed by the coverage heuristic of Eq. (D.11) (Appendix D), which schedules a number of candidate blocks as a function of the estimated spectral rank $\widehat{K}_{\text{spec}}$ and of the number of layers, following a standard balls-into-bins coverage argument. This ensures, with high probability, that every spectrally detectable direction is represented among the initial mixtures before the pruning and acceptance steps are applied.
3. **Generate synthetic spurious mixtures.** Construct nonlinear mixtures to be used as inputs for the LAM:
$$x_i^{\alpha} = \operatorname{sign}\left(\sum_{\delta=1}^{\widehat{K}_{\text{spec}}} c_{\delta}^{\alpha}\, \tilde{x}_i^{\delta}\right), \qquad \alpha = 1, \dots, m, \tag{2.30}$$
where c_{δ}^{α} are random binary coefficients.
4. **Run LAM and accept reconstructions.** Feed the synthetic mixtures $\{\boldsymbol{x}^{\alpha}\}_{\alpha=1}^{m}$ into the LAM, run the dynamics, and apply the acceptance criterion (2.28) to extract the final set of archetypes $\{\boldsymbol{\xi}_R^{\ell}\}_{\ell=1}^{\widehat{K}_{\text{rec}}}$.

This complete procedure is summarized in Algorithms 1-2 of Appendix A.

Remark 2 (Spectral rank versus reconstructed rank)**.** The quantities $\widehat{K}_{\text{spec}}$ and $\widehat{K}_{\text{rec}}$ serve distinct but complementary purposes. The former, $\widehat{K}_{\text{spec}} = \#\{i : \lambda_i(\widehat{\boldsymbol{J}}^{KS}) > \tau_{\text{spec}}\}$, is the operational post-sharpening estimate of the dimension of the signal subspace used for seed generation. It should be distinguished from the BBP/MP count $\widehat{K}_{\text{MP}}$ defined on the unsharpened aggregated operator. The latter, $\widehat{K}_{\text{rec}}$, represents the actual number of archetypes retained after running the LAM dynamics and applying the subsequent pruning steps. In regular regimes—where the supercritical eigenspace is well separated from the bulk and the LAM dynamics successfully disentangles all detectable directions—these two quantities typically coincide. However, in more challenging finite-size or low-signal regimes, certain spectrally detectable directions might fail to pass the dynamical or acceptance criteria, leading to $\widehat{K}_{\text{rec}} \le \widehat{K}_{\text{spec}}$. Thus, $\widehat{K}_{\text{spec}}$ should be interpreted as an upper bound on the number of potentially recoverable archetypes, whereas $\widehat{K}_{\text{rec}}$ denotes the effective number of archetypes reliably reconstructed by the pipeline.

In summary, this procedure enables an entirely unsupervised factorization of a Hebbian kernel into its underlying patterns, without requiring external samples or side information. The core concept relies on leveraging the spectral geometry of the pseudo-inverse estimate to generate self-consistent synthetic mixtures, which the LAM dynamics then elegantly disentangles into the fundamental components.

For brevity, from now on we drop the superscript "Hebb" and denote the Hebbian synaptic matrix simply by $\boldsymbol{J}$.

2.2. Dataset partition and round definition

To fully specify our federated learning setting, we consider L clients participating in a federation over T communication rounds. To model robustness under strong data heterogeneity, we define $K_c \subseteq \{1, \dots, K\}$ as the *effective support* of archetypes observed by client c over the considered time horizon. We assume that the federation collectively covers the entire latent population, namely $\bigcup_{c=1}^{L} K_c = \{1, \dots, K\}$.[3]

Local batches and class mixtures. At round $t \in \{0, 1, \dots, T\}$, each client $c \in \{1, \dots, L\}$ receives a local batch of M_c^t examples, $\{(\boldsymbol{\eta}_c^t)^m\}_{m=1}^{M_c^t}$, drawn according to the generative model (2.13). Here $t = 0$ labels the initialization round of step (i) of Section 3, in which no server feedback is available yet, while $t = 1, \dots, T$ label the communication rounds proper; each client thus observes $T + 1$ batches in total. We denote the corresponding class-mixing distribution by $\boldsymbol{\pi}_{t,c}$; specifically, $\pi_{t,c}(\mu)$ is the probability that an example observed by client c at round t is generated from archetype μ. Consistent with the definition of K_c, this mixture satisfies:
$$\pi_{t,c}(\mu) = 0 \qquad \forall\, t = 0, 1, \dots, T,\ \forall\, \mu \notin K_c. \tag{2.31}$$

Consequently, different clients observe distinct subsets K_c of archetypes and may operate under different noise levels r_c. As a result, local datasets reflect only partial facets of the global archetypal population, establishing a rigorously non-i.i.d. federated setting (Hsu et al., 2019; Kairouz et al., 2021).

Crucially, the condition $\pi_{t,c}(\mu) = 0$ for $\mu \notin K_c$ reflects the *statistical realization* of the local datasets rather than a structural limitation. We do not assume that client c is fundamentally incapable of observing archetype μ (e.g., due to sensor blindness). Instead, the local generative process is highly heterogeneous and sparse, effectively leaving specific classes out of the local sampling budget during the training rounds. This perspective aligns with realistic non-i.i.d. federated scenarios—such as constrained local datasets or regional variations in prevalence—and naturally accommodates the emergence of novel archetypes as the mixture distributions evolve over time.

Round-level aggregate mixture and sample budget. We define the global round-level mixture as the client-wise average
$$\pi_t(\mu) := \frac{1}{L}\sum_{c=1}^{L} \pi_{t,c}(\mu), \tag{2.32}$$
which summarizes the class-mixing distribution across the entire federation at round t. In particular, it dictates whether archetype μ is globally present or absent during that specific round.

[3] This union represents the complete latent space. The actual presence of specific archetypes at round t is governed by the time-varying mixtures $\pi_{t,c}(\mu)$, allowing for scenarios where new classes emerge mid-training.

We also define the total federation sample budget, which quantifies the overall volume of data processed across all clients and rounds:

$$M_{\text{tot}} = \sum_{t=0}^{T} \sum_{c=1}^{L} M_c^t, \tag{2.33}$$

the sum running over the initialization round $t = 0$ as well, since the batch used in step (i) of Section 3 is part of the data consumed by the federation. The coverage (2.35) is understood with the same convention, i.e. $C(t) = \{\mu : \sum_{s=0}^{t} e_\mu(s) > 0\}$.

Exposure and coverage. We conclude by introducing two quantities that, while not explicitly used by the algorithm, fundamentally govern the reconstruction dynamics (as detailed in Section 4). The first is the *federated exposure* of archetype μ at round t:

$$e_\mu(t) = \frac{1}{L} \sum_{c=1}^{L} \frac{1}{M_c^t} \sum_{m=1}^{M_c^t} \mathbf{1}\{(\boldsymbol{\eta}_c^t)^m \text{ was generated from archetype } \mu\}. \tag{2.34}$$

Conceptually, $e_\mu(t)$ is the empirical counterpart of the global mixture probability $\pi_t(\mu)$. It measures the average fraction of local examples generated from archetype $\boldsymbol{\xi}^\mu$ at round t across all clients. By the law of large numbers, $e_\mu(t) \to \pi_t(\mu)$ as the local batch sizes $M_c^t \to \infty$.

The second quantity is the *coverage* up to round t:

$$C(t) = \left\{ \mu \in \{1, \dots, K\} : \sum_{s=1}^{t} e_\mu(s) > 0 \right\}, \tag{2.35}$$

which represents the set of all archetypes that have been observed at least once across the entire federation up to round t.

In the following subsection, we detail how the federated dynamics unfolds across communication rounds, outline the components involved, and define the metrics used to assess the quality of the reconstructed archetypes.

3. Main model architecture

We now describe how the associative memory models discussed earlier can be embedded into a federated setting. Before detailing the pipeline, it is crucial to clarify the intended scope of these models. Hopfield-like architectures, such as the one presented here, are inherently designed for *reconstruction* tasks—that is, extracting and stabilizing latent archetypes (or centroids) from a noisy dataset. They are fundamentally distinct from standard discriminative or predictive models, and are therefore not intended for typical classification or forecasting tasks.

The overall workflow, sketched in Fig. 3, is organized into four steps and repeats over T communication rounds. Throughout this process, raw examples remain strictly local to the clients; only aggregated statistics (synaptic operators) are exchanged.

Pipeline overview. At each round t, clients build (or update) a local synaptic matrix from their current batch, upload it to the server, and receive back a server-side reconstruction encoded as a synaptic operator. Clients then fuse this global information with their new local evidence via a convex combination, and the process iterates.

(i) **Initialization (local operator at $t = 0$).** At the initial round, each client c computes an unsupervised synaptic matrix from its first batch of M_c^0 examples:

$$\left(J_c^{(0)}\right)_{ij} = \frac{1}{N M_c^0} \sum_{m=1}^{M_c^0} \left(\eta_c^{(0)}\right)_i^m \left(\eta_c^{(0)}\right)_j^m, \qquad c = 1, \dots, L. \tag{3.1}$$

Each client then transmits $\boldsymbol{J}_c^{(0)}$ to the server.

(ii) **Server aggregation and factorization.** The server aggregates the received operators (here, via an equally weighted average) to obtain a global synaptic matrix:

$$\left(J_s^{(t)}\right)_{ij} = \frac{1}{L} \sum_{c=1}^{L} \left(J_c^{(t)}\right)_{ij}. \tag{3.2}$$

The server then runs the unsupervised factorization procedure (Section 2.1.1), using $\boldsymbol{J}_s^{(t)}$ as input to the LAM-based reconstruction mechanism, yielding a set of reconstructed patterns:

$$\boldsymbol{J}_s^{(t)} \longrightarrow \text{LAM} \longrightarrow \left\{ \hat{\boldsymbol{\xi}}_{s,\mu}^{(t)} \right\}_{\mu=1}^{\widehat{K}_{\text{rec}}(t)},$$

where $\widehat{K}_{\text{rec}}(t)$ is the effective number of retrieved archetypes identified by the spectral filtering and acceptance steps. In general, $\widehat{K}_{\text{rec}}(t)$ may differ from the true number K (typically $\widehat{K}_{\text{rec}}(t) \le K$ in difficult regimes).

(iii) **Broadcast and client fusion.** The reconstructed patterns are broadcast back to the clients in operator form, via the server synaptic matrix:

$$\left(\hat{J}_s^{(t)}\right)_{ij} = \frac{1}{N} \sum_{\mu=1}^{\widehat{K}_{\text{rec}}(t)} \left(\hat{\xi}_{s,\mu}^{(t)}\right)_i \left(\hat{\xi}_{s,\mu}^{(t)}\right)_j. \tag{3.3}$$

Upon receiving $\hat{\boldsymbol{J}}_s^{(t)}$, each client c processes its *new* batch at round $t + 1$ by first computing the associated fresh Hebbian correlator:

$$\widetilde{\boldsymbol{J}}_c^{(t+1)} = \frac{1}{N M_c^{t+1}} \sum_{m=1}^{M_c^{t+1}} (\boldsymbol{\eta}_c^{(t+1)})^m \left[(\boldsymbol{\eta}_c^{(t+1)})^m\right]^\top. \tag{3.4}$$

These fresh operators are uploaded to the server *before* any local fusion occurs. The server forms the unweighted fresh aggregate:

$$\boldsymbol{J}_{\text{fresh}}^{(t+1)} = \frac{1}{L} \sum_{c=1}^{L} \widetilde{\boldsymbol{J}}_c^{(t+1)}. \tag{3.5}$$

The operator $\boldsymbol{J}_{\text{fresh}}^{(t+1)}$ is used only for the preliminary spectral-rank diagnostic described in Section 4.2. In particular, the server applies the spectral preprocessing of Section 2.1.1 to estimate the fresh post-sharpening rank $\widehat{K}_{\text{spec}}^{\text{fresh}}(t+1)$ and compares it with the number $\widehat{K}_{\text{rec}}(t)$ of archetypes reconstructed at the preceding round. This comparison determines whether the current data provide evidence of an expansion of the represented archetypal subspace. The full LAM factorization is not applied to $\boldsymbol{J}_{\text{fresh}}^{(t+1)}$ at this preliminary stage. The resulting rank-expansion indicator is then returned to the clients and used to condition the quality estimate entering the adaptive controller.

Once this feedback is received, each client computes the final updated operator via a convex blend of its current local evidence and the server reconstruction:

$$\boldsymbol{J}_c^{(t+1)} = w_c(t+1) \underbrace{\widetilde{\boldsymbol{J}}_c^{(t+1)}}_{\text{current local evidence}} + \big(1 - w_c(t+1)\big) \underbrace{\hat{\boldsymbol{J}}_s^{(t)}}_{\text{server reconstruction}} \tag{3.6}$$

where $w_c(t+1) \in [0, 1]$.

The plasticity weight is indexed by the round of the local batch it multiplies, consistent with the controller of Section 4.2 and with Eq. (4.29): the operator $\boldsymbol{J}_c^{(t+1)}$ fuses the batch of round $t+1$ with the server reconstruction $\hat{\boldsymbol{J}}_s^{(t)}$ of the previous round. We record here, once and for all, that the re-encoded server operator (3.3) and the consolidated operator $\boldsymbol{A}^{(t)}$ used in Section 4.2 denote the *same* matrix, $\boldsymbol{A}^{(t)} := \hat{\boldsymbol{J}}_s^{(t)}$; with this identification, replacing t by $t - 1$ in (3.6) reproduces (4.29) verbatim, and the two equations carry no conflicting timing convention. During the initial communication round, no adaptive estimate of the client quality is required since a server-side reconstruction is not yet available. The adaptive controller activates only after the first server reconstruction is broadcast back to the clients. From that point onward, the reconstructed archetypes serve as client-wise prototypes against which the quality of the current local batch can be dynamically evaluated (see Section 4.2).

(iv) **Iteration.** Steps (ii)–(iii) are repeated for $t = 0, \dots, T - 1$; at the last round the clients upload $\boldsymbol{J}_c^{(T)}$ and step (ii) is executed one final time on it, so that the reconstructed set $\{\hat{\boldsymbol{\xi}}_{s,\mu}^{(T)}\}_{\mu=1}^{\widehat{K}_{\text{rec}}(T)}$ returned by Algorithm 3 is well defined. The federation therefore consumes the $T + 1$ local batches indexed by $t = 0, 1, \dots, T$, the batch at $t = 0$ being the one used in the initialization step (i).

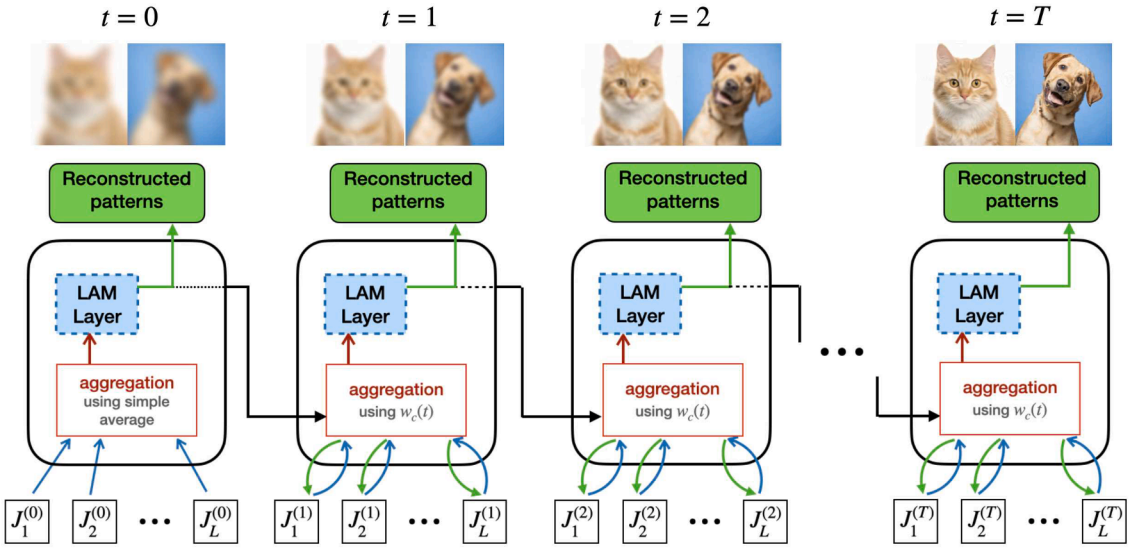


Fig. 3. Illustration of the federated pipeline. Each block represents a federation round t. At each round, the L client layers provide their synaptic matrices as input, each estimated from the batch of examples available at that round. These matrices are combined in the *aggregation* layer. For $t = 0$, the aggregation relies solely on the information received from the clients. For $t > 0$, the aggregation incorporates both (i) the client information from round t and (ii) the server-side information carried over from the previous round $t - 1$. The aggregated information can either be sent back to the clients as feedback to improve their local estimation of the ground-truth synaptic matrix, or forwarded to the *LAM* layer, where pattern reconstruction is performed. At round $t = 0$, there is no feedback from the federation to the clients. At each subsequent round, we can access both the reconstructed patterns outputted by the *LAM* layer and the updated client synaptic matrices. Panel (b) shows a zoom-in of the *LAM* layer, where the *pattern reconstruction* takes place. This layer takes the output of the *aggregation* layer and first applies an iterative sharpening procedure to estimate $\hat{\mathbf{J}}^{KS}$. It then generates a sufficiently large set of initial mixing states and feeds them into the LAM model to collect potential *pattern candidates*. These candidates are passed to the *pruning* layers, which remove duplicates and apply a quadratic acceptance criterion to discard spurious candidates. The final output is the set of $\hat{K}_{\text{rec}}(t)$ reconstructed patterns.

These steps are illustrated in Figs. 3 and 4. The full pseudo-code is detailed in Algorithm 3 in Appendix A.

Our pipeline significantly departs from standard federated learning frameworks: clients do not exchange gradients or model parameters, but instead share *synaptic operators* that capture archetypal structural information (Personnaz et al., 1986). Consistent with the federated paradigm, raw data remain strictly local and are never transmitted to the server; only aggregated correlation statistics are communicated. This design choice is not merely a practical convenience. By transmitting operator-level summaries, each client provides a concise representation of the underlying archetypal geometry. This enables the server to integrate a coherent global structure while strictly preserving raw-data locality. However, because these summaries are computed from local examples, they must be regarded as data-dependent aggregate statistics rather than as a formal privacy guarantee.

Remark 3 (Scope of the privacy claim). The privacy claim made in this work should be understood strictly as a *raw-data locality* statement, rather than a formal privacy guarantee. The uploaded Hebbian operators are inherently data-dependent, as they are computed directly from the local examples observed by each client. Consequently, they may reveal aggregate information about the local distributions—in particular, the specific archetypal directions that are sufficiently represented within a local batch. This aggregate structural information is precisely the signal exploited by the server-side spectral and LAM-based reconstruction procedure.

Crucially, however, the communicated object is a second-order, batch-averaged statistic,

$$\left(\widetilde{J}_c^{(t)}\right)_{ij} = \frac{1}{N M_c^t} \sum_{m=1}^{M_c^t} \left(\eta_c^t\right)_i^m \left(\eta_c^t\right)_j^m,$$

rather than the collection of raw examples $\{(\boldsymbol{\eta}_c^t)^m\}_{m=1}^{M_c^t}$. We stress that, for $t \geq 1$, the only data-dependent matrix leaving the client is precisely this fresh correlator $\widetilde{J}_c^{(t)}$ of (3.4): the blended operator $J_c^{(t)}$ of (3.6) differs from it only through the term $\hat{J}_s^{(t-1)}$, which the server has itself broadcast, so that the fusion step changes neither the nature nor the amount of local information disclosed. In the intended many-sample regime, this operator discards the ordering of the examples and averages out sample-specific noise fluctuations. The server's reconstruction target is therefore the set of shared archetypes $\{\hat{\boldsymbol{\xi}}_{s,\mu}\}$ encoded within the aggregate correlation structure, not the individual client realizations $\boldsymbol{\eta}$. Thus, while our method effectively avoids raw-data transmission and distills the reconstruction target down to archetypal statistics, it does not provide differential privacy or formally bound data leakage. Establishing such strict guarantees would require a dedicated privacy analysis, for instance by incorporating calibrated noise directly at the operator level.

A central component of this procedure is the client-wise convex combination defined in Eq. (3.6). The hyperparameter $w_c(t)$ acts as a *plasticity knob*: a small $w_c(t)$ favors consolidation (yielding lower variance but higher bias toward past global information), whereas a large $w_c(t)$ favors plasticity (reducing bias toward stale information but increasing variance due to finite-batch fluctuations). This operator-level mechanism intuitively mirrors the complementary learning systems (CLS) hypothesis in cognitive neuroscience—where fast sensory traces are integrated into a slower long-term memory store—providing an interpretable analogue for federated continual learning (Kumaran et al., 2016; O'Reilly et al., 2014).

Reconstruction metrics. To quantify the retrieval performance after each round, we monitor the alignment between the reconstructed server patterns and the ground-truth archetypes. Specifically, we report the *magnetization* of the reconstructed set:

$$m_t((\hat{\boldsymbol{\xi}}_s^{(t)}) \mid \boldsymbol{\xi}^\mu) := \max_{\nu \in \{1,\dots,\hat{K}_{\text{rec}}(t)\}} \frac{1}{N} \left| (\hat{\boldsymbol{\xi}}_s^{(t)})^\nu \cdot \boldsymbol{\xi}^\mu \right|, \tag{3.7}$$

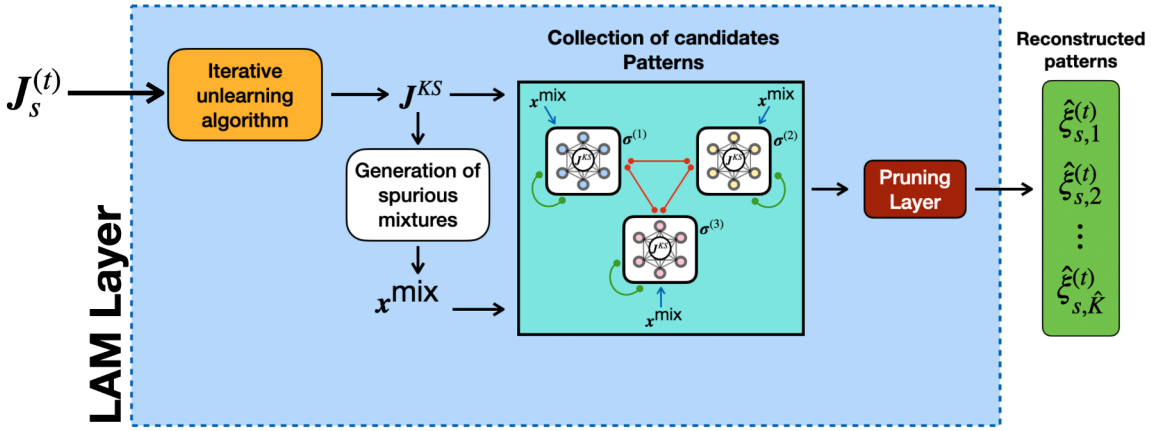


Fig. 4. Zoom-in of the *LAM* layer, where the *pattern reconstruction* is performed at each round. This layer takes the output of the *aggregation* layer as input and applies an iterative sharpening procedure to estimate $\widehat{\mathbf{J}}^{KS}$. It then generates a set of initial mixing states and feeds them into the LAM model to collect *pattern candidates*. These candidates are processed by the *pruning* layers, which remove duplicates and apply a quadratic acceptance criterion to filter out invalid configurations. The final output is the set of $\widehat{K}_{\text{rec}}(t)$ reconstructed patterns. The number of initial mixing states is fixed by the coverage heuristic of Eq. (D.11), which schedules a number of candidate blocks that grows with the estimated post-sharpening spectral rank $\widehat{K}_{\text{spec}}(t)$ and the number of layers, so that every detectable direction is explored before pruning. The spectral stage involves two conceptually distinct thresholds. The Marchenko–Pastur/BBP threshold $\tau_{\text{MP}}(t) = \lambda_+(q) + \delta_N$ is defined on the unsharpened aggregated operator and determines which signal directions are theoretically detectable. After the pseudo-inverse sharpening step, the separate threshold τ_{spec} is applied to the spectrum of $\widehat{\boldsymbol{J}}^{KS}$ for operational seed selection. In the well-separated regime, the two prescriptions identify the same signal subspace, while τ_{spec} should be regarded as a post-sharpening surrogate rather than as the MP edge itself. Duplicates are merged when their mutual overlap exceeds $\delta_{\text{dup}} = 0.5$ (Eq. (2.23)), and a candidate is accepted only if $N^{-1}(\tilde{\boldsymbol{\sigma}}^{\ell})^{\top}\widehat{\boldsymbol{J}}^{KS}\tilde{\boldsymbol{\sigma}}^{\ell} > \tau_{\text{acc}}$ (Eq. (2.27)). Each candidate is therefore either kept or discarded, with no soft scoring applied.

which measures the maximum normalized overlap between any reconstructed archetype and the true pattern set.

In synthetic experiments where the teacher operator is available, we additionally consider the normalized Frobenius error between the reconstructed kernel and the ground-truth Hebbian matrix:

$$\text{FRO}_t(\boldsymbol{J}) := \frac{\left\|\boldsymbol{J}^{(t)} - \boldsymbol{J}^{\star}\right\|_F}{\|\boldsymbol{J}^{\star}\|_F}, \tag{3.8}$$

where $\boldsymbol{J}^{\star}$ denotes the true Hebbian coupling matrix and $\|\cdot\|_F$ is the Frobenius norm. Here $\boldsymbol{J}^{\star} = \frac{1}{N}\sum_{\mu=1}^{K}\boldsymbol{\xi}^{\mu}(\boldsymbol{\xi}^{\mu})^{\top}$ is the clean teacher operator, assembled with unit weight per archetype. The server reconstruction $\hat{\boldsymbol{J}}_s^{(t)}$ of (3.3), being built from binary reconstructed archetypes $\hat{\boldsymbol{\xi}}_{s,\mu}^{(t)} \in \{-1,+1\}^N$ (so that $\|\hat{\boldsymbol{\xi}}_{s,\mu}^{(t)}\|^2 = N$, exactly as for the true archetypes) carried with unit weight and with the same $1/N$ prefactor, is on the same scale and can therefore approach $\boldsymbol{J}^{\star}$ as the reconstruction improves, exactly, when $\widehat{K}_{\text{rec}}(t) = K$ and each reconstructed archetype matches a distinct true one up to a global sign. A raw client correlator, by contrast, carries each archetype with weight $r^2\pi_{t,c}(\mu)$ together with an isotropic term of order σ^2/N (cf. the population form derived in (4.4)); even with unlimited local data it does not converge to $\boldsymbol{J}^{\star}$, and its normalized error retains a floor controlled by the channel quality r and the local exposure. This is the intended behavior: only the server-side factorization strips the noise scale and recovers the unit-weight archetypal operator, which is precisely the gap quantified in Fig. 11.

Now that the main components of our pipeline and the associated evaluation metrics have been introduced, we can proceed to the presentation of our core results, addressing both the theoretical and algorithmic facets of the proposed framework.

Remark 4 (Component-level characterization through prior work)**.** The present pipeline integrates four distinct building blocks: the low-rank Hebbian aggregation, the spectral sharpening and $\widehat{K}$ estimation, the LAM-based factorization, and the entropy-based controller. The first three components have been independently characterized in dedicated prior publications, which collectively provide an ablation-level analysis of each module:

- *LAM factorization (learning and retrieval):* The storage capacity, retrieval thresholds, robustness to noise, and critical operating regimes of the LAM architecture under both supervised and unsupervised protocols are characterized in Agliari et al. (2025a,b,c,d), Alessandrelli et al. (2025).
- *Spectral sharpening and acceptance criterion:* The sensitivity of the reconstruction pipeline to the spectral threshold and a formal derivation of the acceptance criterion are studied in Agliari et al. (2025c), Appendices B-C.
- *Hebbian aggregation under noisy observations:* The statistical properties of Hebbian correlation operators and the conditions under which they reliably encode archetypal structure under corrupted examples are analyzed in Agliari et al. (2025b,d).

The entropy-based controller is introduced in the present work; its theoretical grounding is developed in Section 4.2, and its advantage over a fixed-weight baseline is illustrated in numerical experiments in Section 5.

4. Theoretical findings

On the theory side, leveraging tools from random matrix theory, we characterize when and why *novelties* become detectable, namely when examples originating from archetypes that have never appeared before a given round start to surface in the federation. In particular, we show that *exposure* is the key quantity governing detectability: as exposure increases, Baik-Ben Arous-Péché (BBP)-type outliers split from the bulk spectrum and the corresponding leading eigenvectors align with archetypal directions, thereby providing reliable seeds for reconstruction.

On the algorithmic side, we introduce a data-driven, entropy-based controller that adapts the network plasticity over rounds. This mechanism allows the system to remain responsive to distributional drift and the arrival of new archetypes, while simultaneously protecting previously consolidated representations. As a by-product, the same controller provides a principled form of *preventive caution* against adverse conditions such as low-quality data batches or the presence of corrupted

(infected) clients, by automatically reducing over-reactivity when the incoming information is inconsistent.

Overall, our results substantiate two claims:

- **Exposure governs detectability.** When a class gains sufficient exposure, BBP outliers detach from the bulk and the leading eigenvectors align with archetypal directions, enabling reliable seeding and reconstruction. In the continual phase the relevant quantity is the *plasticity-weighted* exposure, and the feedback injected by the convex client update enters as a positive semidefinite, finite-rank, signal-aligned term that can reinforce consolidated directions but never create new ones (Proposition 1).
- **Entropy-driven stability–plasticity control.** An entropy-based controller yields robust schedule tracking without hand tuning, preserving consolidated attractors while remaining adaptive to drift. The system incorporates new archetypes with low detection latency and minimal disruption, expanding its effective representational subspace precisely when new directions become informative.

4.1. The role of exposure and BBP transitions

This section explains when and why archetypal structure becomes *spectrally visible* in the server-side estimator—a prerequisite for the seeding and counting steps in our pipeline. At communication round t, the server forms the aggregated Hebbian operator $\boldsymbol{J}_s^{(t)}$ by combining client-side correlators computed on heterogeneous, fragmented batches. Although the server never observes raw examples, it does observe data-dependent Hebbian operators. The key question is therefore spectral rather than sample-level: *under what conditions does $\boldsymbol{J}_s^{(t)}$ develop eigenvalue/eigenvector signatures that reveal which archetypes are currently present?*

Our answer is spectral. We show that the *population* operator associated with $\boldsymbol{J}_s^{(t)}$ has a spiked–Wishart structure in which each archetype contributes a rank-one spike with strength proportional to its round-level exposure. Moreover, $\boldsymbol{J}_s^{(t)}$ concentrates sharply around this population operator in operator norm (Theorem 1), while the precise localisation of the bulk inside the Marchenko–Pastur band and the detachment of the supercritical outliers are supplied separately by the isotropic local law and by the BBP analysis of Theorem 2; the two ingredients together yield the bulk/outlier geometry of the finite-sample spectrum up to controlled fluctuations. This yields explicit detectability thresholds (BBP transitions), predictions for outlier locations, and quantitative eigenvector alignment laws, which together justify the spectral steps used downstream (thresholding to estimate $\hat{K}(t)$ and using leading eigenvectors as reconstruction seeds).

Spiked decomposition and the meaning of exposure. With the data model of Section 2.2, let σ^2 denote the (per-coordinate) noise variance. In the Rademacher channel, this is given by the identity $\sigma^2 = 1 - r^2$, where for simplicity we set $r_c = r$ for all $c \in \{1, \dots, L\}$. To apply standard random matrix results, we analyze the spectrum of the *rescaled* operator $\tilde{\boldsymbol{J}}_s^{(t)} := N\boldsymbol{J}_s^{(t)}$. The population counterpart of this rescaled estimator decomposes into an isotropic noise part plus a finite-rank signal aligned with the true archetypes. Letting $u_\mu := \xi^\mu/\sqrt{N}$ denote the normalized archetype vectors, we have:

$$\mathbb{E}[\tilde{\boldsymbol{J}}_s^{(t)}] \;=\; \sigma^2\boldsymbol{I} \;+\; N\,r^2\sum_{\mu=1}^{K}\pi_t(\mu)\,u_\mu u_\mu^\top \;=\; \sigma^2\boldsymbol{I} \;+\; r^2\sum_{\mu=1}^{K}\pi_t(\mu)\,\xi^\mu(\xi^\mu)^\top. \tag{4.1}$$

The identity term $\sigma^2\boldsymbol{I}$ captures the isotropic contribution of random bit flips, while each archetype contributes a rank-one projector $u_\mu u_\mu^\top$ weighted by the global mixture mass $\pi_t(\mu)$, the signal intensity r^2, and the dimensional factor N produced by the rescaling $\tilde{\boldsymbol{J}}_s^{(t)} = N\boldsymbol{J}_s^{(t)}$, so that the population spike strength is $Nr^2\pi_t(\mu)$. The second equality, written in the unnormalized archetypes $\xi^\mu = \sqrt{N}\,u_\mu$, makes this scale manifest. Consequently, strictly identifying the spike strength in the spiked-covariance model yields:

$$\text{Spike strength of } \mu: \quad \theta_\mu^{(t)} = N\,r^2\,\pi_t(\mu)$$

$$\Longrightarrow \quad \kappa_\mu^{(t)} = \frac{\theta_\mu^{(t)}}{\sigma^2} = \frac{N\,r^2\,\pi_t(\mu)}{1-r^2}. \tag{4.2}$$

Intuitively, an archetype is invisible when it has (near-)zero exposure, and it becomes detectable only once its exposure makes the corresponding spike strong enough to separate from the noise bulk.

Derivation of the spiked structure from the generative model. We now obtain Eqs. (4.1)-(4.2) directly from the data model, keeping the rescaling explicit so that no scale factor is left implicit. By (2.13), each example can be written as $(\boldsymbol{\eta}_c^{t,m})_i = \xi_i^\mu\,\chi_i^{t,m}$ with $\chi_i^{t,m} \in \{-1,+1\}$, $\mathbb{E}[\chi_i^{t,m}] = r$ and $\mathrm{Var}(\chi_i^{t,m}) = 1 - r^2 = \sigma^2$. Since bit flips are independent across coordinates, $\mathbb{E}[\chi_i\chi_j] = r^2$ for $i \neq j$, while $\chi_i^2 = 1$. Averaging the client correlator (2.14) over a batch drawn from the local mixture $\pi_{t,c}$ therefore gives

$$\mathbb{E}\big[(\boldsymbol{J}_c^{(t)})_{ij}\big] = \frac{r^2}{N}\sum_{\mu=1}^{K}\pi_{t,c}(\mu)\,\xi_i^\mu\xi_j^\mu \quad (i \neq j), \qquad \mathbb{E}\big[(\boldsymbol{J}_c^{(t)})_{ii}\big] = \frac{r^2+\sigma^2}{N} = \frac{1}{N}. \tag{4.3}$$

Aggregating over clients through (3.2) and using $\pi_t(\mu) = L^{-1}\sum_c \pi_{t,c}(\mu)$ from (2.32), the round operator obeys, for all i, j,

$$\mathbb{E}\big[(\boldsymbol{J}_s^{(t)})_{ij}\big] = \frac{r^2}{N}\sum_{\mu=1}^{K}\pi_t(\mu)\,\xi_i^\mu\xi_j^\mu + \frac{\sigma^2}{N}\,\delta_{ij}. \tag{4.4}$$

The rescaled estimator $\tilde{\boldsymbol{J}}_s^{(t)} := N\boldsymbol{J}_s^{(t)} = M_{\text{round}}^{-1}\sum_{c,m}\boldsymbol{\eta}_c^{t,m}(\boldsymbol{\eta}_c^{t,m})^\top$ is thus a sample second-moment matrix over the M_{round} examples of the round. Multiplying (4.4) by N and using $\xi_i^\mu\xi_j^\mu = N\,(u_\mu u_\mu^\top)_{ij}$ recovers exactly the spiked decomposition (4.1), with population spike strength $\theta_\mu^{(t)} = Nr^2\pi_t(\mu)$ and signal-to-noise ratio $\kappa_\mu^{(t)} = \theta_\mu^{(t)}/\sigma^2$ as in (4.2). The factor N is the dimensional gain produced by the rescaling and cannot be omitted: at $r = 1$ and $\pi_t(\mu) = 1$ for a single archetype one has $\tilde{\boldsymbol{J}}_s^{(t)} = \xi(\xi)^\top$, whose leading eigenvalue is exactly $\|\xi\|^2 = N$, in agreement with $\theta^{(t)} = N$. The BBP analysis of Theorem 2 then applies to $\tilde{\boldsymbol{J}}_s^{(t)}$ as a spiked sample-covariance matrix with aspect ratio $q = N/M_{\text{round}}$, isotropic level σ^2, and spikes $\{\kappa_\mu^{(t)}\}_{\mu=1}^K$. Equivalently, archetype μ becomes spectrally detectable at round t precisely when its exposure exceeds the critical value

$$\pi_t^\star(\mu) = \frac{1-r^2}{r^2}\,\frac{1}{\sqrt{N\,M_{\text{round}}}}, \tag{4.5}$$

obtained by solving $\kappa_\mu^{(t)} = \sqrt{q}$ for $\pi_t(\mu)$. The independence assumptions used here ($\chi_i \perp \chi_j$ for $i \neq j$, and independence across archetypes and clients) are exactly the standing assumptions of Theorems 1 and 2.

The remaining task is to turn this intuition into a finite-sample statement: (a) show that $\boldsymbol{J}_s^{(t)}$ stays close to its expectation in operator norm at the relevant scales, and (b) characterize when the spikes of (4.1) produce outliers and aligned eigenvectors in the empirical spectrum.

Non-asymptotic stability of the round operator. We first collect concentration bounds ensuring that $\boldsymbol{J}_s^{(t)}$ is a small perturbation of its spiked population counterpart. Throughout, $\|\cdot\|_{\text{op}}$ denotes operator norm, $M_{\text{round}} = L\,M_c$[4], and σ^2 is as above.

Theorem 1 (Concentration). *Let $(\chi_j)_{j=1}^N$ be independent random variables. For a fixed communication round index t, let $\boldsymbol{J}_s^{(t)}$ be defined by (3.2). Then, for every deviation level $u > 0$:*

[4] While the general framework allows for time-varying and heterogeneous budgets M_c^t per client, for the sake of theoretical clarity we conduct the concentration analysis assuming a representative balanced round where $M_c^t \equiv M_c$. In this simplified setting, $M_{\text{round}} = LM_c$ acts as the effective aggregate sample volume; the results extend naturally to the general case by replacing M_{round} with the total round-level budget $M_{\text{round}}(t) = \sum_{c=1}^{L} M_c^t$.

(i) ***(Sub-Gaussian case)*** *If $\|\chi_j\|_{\psi_2} \le C$ for all j, there exists a constant $c_1 > 0$ (depending only on C) such that*

$$\mathbb{P}\Big(\Big\| \boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}] \Big\|_{\mathrm{op}} > u \Big) \;\le\; 2N \exp\left(- c_1\, M_{\mathrm{round}} \cdot \min\left\{ \frac{u^2}{\sigma^4}, \frac{u}{\sigma^2} \right\} \right), \tag{4.6}$$

where $\sigma^2 := \max_j \mathrm{Var}(\chi_j)$.

(ii) ***(Bounded / Rademacher channel)*** *If $\chi_j \in \{\pm 1\}$ with $\mathbb{E}[\chi_j] = r$, there exists a universal constant $c_* > 0$ such that*

$$\mathbb{P}\Big(\Big\| \boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}] \Big\|_{\mathrm{op}} > u \Big) \;\le\; 2N \exp\left(- c_*\, M_{\mathrm{round}} \cdot \min\left\{ \frac{u^2}{\sup_{\lambda\in[0,1]} \lambda(1-\lambda)}, u \right\} \right), \tag{4.7}$$

and, in particular,

$$\sup_{\lambda\in[0,1]} \lambda(1-\lambda) = \tfrac{1}{4}. \tag{4.8}$$

(iii) ***(Finite-N refined bound)*** *In the bounded/Rademacher setting of* (ii)*, for all N and M_c there exist universal constants $c_3, C > 0$ such that*

$$\mathbb{P}\Big(\Big\| \boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}] \Big\|_{\mathrm{op}} > u \Big) \;\le\; 2 \exp\left(- c_3\, M_{\mathrm{round}} \cdot \min\left\{ \frac{u^2}{\sup_{\lambda\in[0,1]} \lambda(1-\lambda)}, u \right\} \;+\; C \log N \right). \tag{4.9}$$

The bounds (4.6)–(4.9) are non-asymptotic and scale with the effective round size M_{round}. Operationally, they guarantee that $\boldsymbol{J}_s^{(t)}$ does not deviate appreciably from its spiked expectation in operator norm, unless the round is severely under-sampled. We stress, however, that operator-norm concentration alone is a coarse statement: it certifies the overall stability of the estimator, whereas the sharp localization of the bulk within the Marchenko–Pastur band and the detachment of the super-critical outliers are governed by the isotropic local law and the BBP analysis of Theorem 2, and not by the concentration bound in isolation. The two results are therefore complementary: Theorem 1 provides finite-sample stability of the aggregated operator, while Theorem 2 provides the precise detectability thresholds. (Proofs and auxiliary ingredients are deferred to the appendix.)

BBP transitions: when spikes separate and align. Given the stability above, we can locate the empirical spectrum relative to the Marchenko–Pastur (MP) bulk induced by the isotropic term, and describe precisely when an exposure-weighted spike produces a detectable outlier. The next theorem formalizes this in the standard spiked-covariance setting under near-orthogonality of the spike directions (which is natural for random $\{\pm 1\}$ archetypes).

Theorem 2 (Detection thresholds for the spiked covariance operator)**.** *Let $u_\mu := \xi^\mu/\sqrt{N}$ and consider the rank–K spiked population covariance*

$$C \;:=\; \sigma^2 I_N + \sum_{\mu=1}^{K} \theta_\mu\, u_\mu u_\mu^\top, \qquad \kappa_\mu := \frac{\theta_\mu}{\sigma^2},$$

with aspect ratio $q := N/M_{\mathrm{round}}$. Denote by $\lambda_\pm(q) := \sigma^2(1 \pm \sqrt{q})^2$ the Marchenko–Pastur edges (cf. (D.5)). Let $J := \tilde{\boldsymbol{J}}_s^{(t)} = N \boldsymbol{J}_s^{(t)}$ be the rescaled round–t operator *of (4.1), so that the isotropic level σ^2, the edges $\lambda_\pm(q)$ and the spikes $\theta_\mu = N r^2 \pi_t(\mu)$ all refer to this normalisation, which can be written as*

$$J = \Sigma^{1/2} S_0 \Sigma^{1/2}, \qquad S_0 := \frac{1}{M_{\mathrm{round}}} \sum_{k=1}^{M_{\mathrm{round}}} Z_k Z_k^\top,$$

with $\Sigma = C$ and whitened rows Z_k satisfying the standing assumptions under which the isotropic MP local law holds (Theorem 3). Assume that the number of spikes K is fixed (independent of N), that the spike directions u_μ have unit norm, and that their Gram matrix $\Gamma := U^\top U$ with $U := [u_1, \dots, u_K]$ satisfies

$$\varepsilon_{\mathrm{orth}} := \max_{\mu \ne \nu} |\langle u_\mu, u_\nu \rangle| \to 0 \quad \text{as } N \to \infty.$$

Assume additionally that the nonzero spike strengths $\{\kappa_\mu\}$ are pairwise distinct. Then, for every fixed $\delta \in (0,1)$ there exists a deterministic sequence $\varepsilon_N(\delta) \downarrow 0$ such that, with probability at least $1 - \delta$ for all N large enough, the following hold:

(D1) ***Bulk confinement.*** *All but at most K eigenvalues of J lie in*

$$\big[\lambda_-(q) - \varepsilon_N(\delta),\; \lambda_+(q) + \varepsilon_N(\delta)\big], \tag{4.10}$$

where

$$\varepsilon_N(\delta) \;=\; C \sqrt{\frac{\log(1/\delta) + \log N}{M_{\mathrm{round}}}} \;+\; C' \frac{\log^2 N}{N^{2/3}}, \tag{4.11}$$

for universal constants $C, C' > 0$ (depending only on q and the tail/regularity parameters).

(D2) ***BBP threshold (spike detection).*** *For each μ with $\kappa_\mu > 0$ there is at most one eigenvalue of J that can be asymptotically attributed to the spike κ_μ. In particular:*

- *If $\kappa_\mu > \sqrt{q}$, then there exists a unique outlier eigenvalue $\lambda_\mu(J) > \lambda_+(q)$ that converges to the population-dependent value in (D3).*
- *If $\kappa_\mu \le \sqrt{q}$, no eigenvalue separates from the bulk edge $\lambda_+(q)$ due to that spike.*

(D3) ***Outlier location (decoupled/orthogonal case).*** *In the idealized case where the spike directions are exactly orthonormal, $\langle u_\mu, u_\nu \rangle = \delta_{\mu\nu}$, the asymptotic location of the outlier associated with a spike of strength $\kappa > 0$ is*

$$\lambda_{\mathrm{out}}(\kappa) \;=\; \sigma^2 (1+\kappa)\Big(1 + \frac{q}{\kappa}\Big). \tag{4.12}$$

Moreover, $\lambda_{\mathrm{out}}(\kappa) > \lambda_+(q)$ if and only if $\kappa > \sqrt{q}$, and at $\kappa = \sqrt{q}$ the outlier merges with the upper MP edge.

In the nearly-orthogonal case, the non-orthogonality perturbs the population spikes by at most $O(\varepsilon_{\mathrm{orth}})$ (by standard eigenvalue perturbation of the Gram matrix $\Gamma = U^\top U$), hence the corresponding sample outliers concentrate around the same BBP locations up to this bias. More precisely, for each μ with $\kappa_\mu > \sqrt{q}$ and for every fixed $\delta \in (0,1)$, there exists $C_\delta > 0$ such that, for all N large enough,

$$\mathbb{P}\Big(\big| \lambda_\mu(J) - \lambda_{\mathrm{out}}(\kappa_\mu) \big| \le C_\delta \big(\varepsilon_{\mathrm{orth}} + N^{-1/2} \big) \Big) \;\ge\; 1 - \delta.$$

Equivalently, $\lambda_\mu(J) = \lambda_{\mathrm{out}}(\kappa_\mu) + O(\varepsilon_{\mathrm{orth}}) + O_p(N^{-1/2})$, and in particular $\lambda_\mu(J) \xrightarrow{P} \lambda_{\mathrm{out}}(\kappa_\mu)$ as $N \to \infty$ and $\varepsilon_{\mathrm{orth}} \to 0$.

Moreover, choosing $\delta = \delta_N$ summable (e.g. $\delta_N = N^{-D}$ for $D > 2$) and applying Borel–Cantelli yields the almost sure convergence $\lambda_\mu(J) \to \lambda_{\mathrm{out}}(\kappa_\mu)$.

(D4) ***Eigenvector alignment.*** *For each μ with $\kappa_\mu > \sqrt{q}$, let v_μ be the (random) unit eigenvector of J associated with the outlier eigenvalue near $\lambda_{\mathrm{out}}(\kappa_\mu)$, with sign chosen so that $\langle v_\mu, u_\mu \rangle \ge 0$. Then*

$$\big| \langle v_\mu, u_\mu \rangle \big|^2 \;\xrightarrow{P}\; \gamma(\kappa_\mu, q) := \frac{1 - \dfrac{q}{\kappa_\mu^2}}{1 + \dfrac{q}{\kappa_\mu}} \in (0,1), \tag{4.13}$$

as $N, M_{\mathrm{round}} \to \infty$ with $N/M_{\mathrm{round}} \to q$. In the nearly-orthogonal case the same limit holds, with finite-N deviations of order $O(\varepsilon_{\mathrm{orth}} + N^{-1/2})$.

Remark 5 (Effective single-population reduction)**.** In the federated i.n.i.d. setting the round operator $\tilde{\boldsymbol{J}}_s^{(t)}$ aggregates independent but *not* identically distributed samples: client c draws from its own mixture $\pi_{t,c}$, and in the continual phase the uploaded operators additionally carry the server feedback through the convex update (3.6). Strictly speaking, $\tilde{\boldsymbol{J}}_s^{(t)}$ is therefore not a single-population sample covariance, and the representation $J = \Sigma^{1/2} S_0 \Sigma^{1/2}$ assumed in Theorem 2 should be understood as that of the *effective* model whose population covariance equals the round-averaged expectation $\Sigma = \mathbb{E}[\tilde{\boldsymbol{J}}_s^{(t)}] = \sigma^2 \boldsymbol{I} + \sum_\mu \theta_\mu^{(t)} u_\mu u_\mu^\top$ of (4.1), with spikes built from the round-averaged exposure $\pi_t(\mu) = L^{-1} \sum_c \pi_{t,c}(\mu)$. Theorem 1 guarantees that the genuinely heterogeneous estimator concentrates around this same Σ, so that the bulk/outlier geometry predicted by the spiked model is the one observed empirically

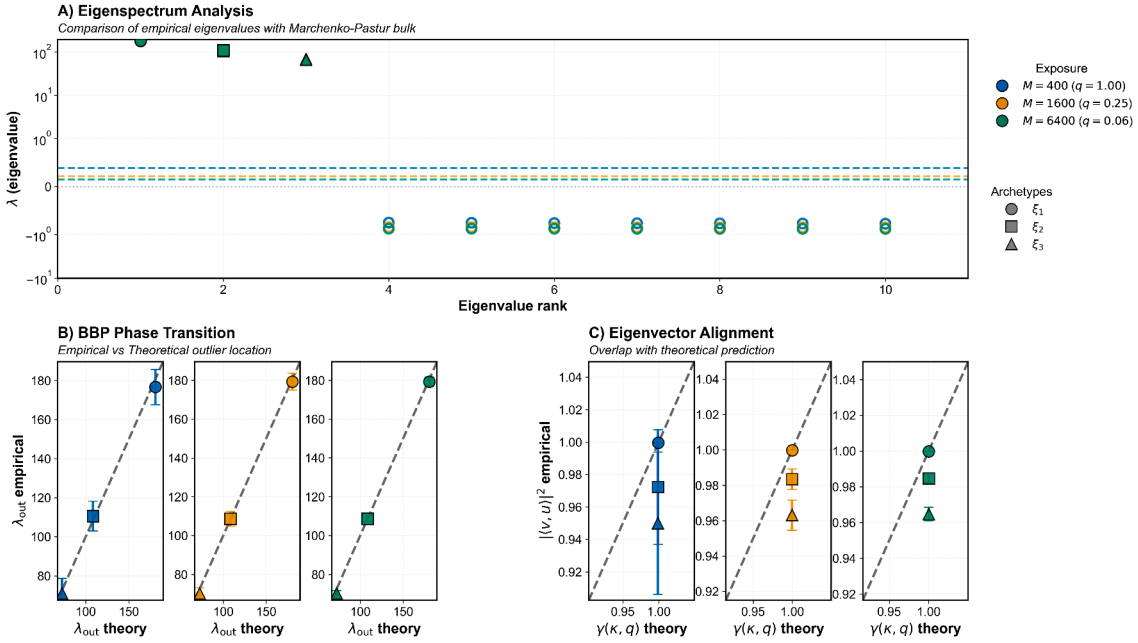


Fig. 5. Spectral validation of the BBP framework in federated Hebbian learning. *(Top)* Panel A displays the top 10 eigenvalues of the empirical Hebbian matrix for three representative exposure levels ($M \in \{400, 1600, 6400\}$), with the Marchenko–Pastur upper edge $\lambda_+(q)$ shown as a dashed line for each aspect ratio $\alpha_{\mathrm{MP}} := \frac{N}{M_{\mathrm{round}}}$. Filled markers denote eigenvalues above the MP threshold (detected spikes corresponding to the three strong archetypes), while empty markers indicate eigenvalues below threshold. The near-absence of empirical spikes for the weak archetypes validates that exposure below the BBP threshold $\kappa < \sqrt{\alpha_{\mathrm{MP}}}$ prevents spectral detection despite their theoretical presence. *(Bottom Left)* Panel B compares empirical spike eigenvalues (averaged over 50 trials) against the closed-form BBP prediction $\lambda_{\mathrm{out}}(\kappa, \alpha_{\mathrm{MP}})$ across all three archetypes and exposure levels. The near-diagonal scatter ($R^2 \in [0.955, 0.977]$) confirms that the spiked-Wishart population model, combined with concentration guarantees, accurately predicts empirical outlier locations at finite sample size ($N = 400$). *(Bottom Right)* Panel C shows eigenvector overlaps $|\langle v, u\rangle|^2$ compared against the theoretical alignment formula $\gamma(\kappa, \alpha_{\mathrm{MP}})$ from Theorem D4. The median gap between empirical and theoretical values is 0.0163, with the maximum gap of 0.0486 concentrated on the weakest archetype (ξ_3). This systematic gap for lower-exposure archetypes is consistent with the finite-size correction $O(N^{-1/2})$ predicted by the theory: the formula $\gamma(\kappa, \alpha_{\mathrm{MP}})$ is asymptotic ($N \to \infty$), and at $N = 400$ one observes deviations that scale inversely with exposure strength. The variance of empirical overlaps visibly decreases as M increases (moving left to right), reflecting the convergence of the sample estimator to its population limit.

(Fig. 5). A fully heterogeneous local law, tracking the client-wise covariances individually, would refine the finite-sample constants but leaves the detectability thresholds $\kappa_\mu > \sqrt{q}$ unchanged.

The feedback-dependent estimator for $t > 0$.. Remark 5 identifies the effective population operator, but it does not yet describe *how* the feedback produced by the convex client update (3.6) enters the round-t spectrum. Since this is exactly what separates the estimator actually used by the algorithm for $t > 0$ from the clean spiked-Wishart model of Theorem 2, we make the decomposition explicit here and delimit precisely which parts of the analysis remain exact and which are used as an approximation.

Let

$$\mathfrak{F}_{t-1} := \sigma\Big(\big\{\boldsymbol{\eta}_c^{s,m}\big\}_{s\le t-1,\,c,\,m},\ \big\{w_c(s)\big\}_{s\le t-1,\,c}\Big) \tag{4.14}$$

denote the σ-algebra generated by the entire history of the federation up to round $t-1$, and introduce the round-level plasticity level and the *plasticity-weighted exposure*

$$\bar{w}_t := \frac{1}{L}\sum_{c=1}^{L} w_c(t) \in [0,1], \qquad \pi_t^{w}(\mu) := \frac{1}{L}\sum_{c=1}^{L} w_c(t)\,\pi_{t,c}(\mu),$$
$$\pi_t^{\mathrm{eff}}(\mu) := \frac{\pi_t^{w}(\mu)}{\bar{w}_t}, \tag{4.15}$$

the latter being the convex combination of the local mixtures with weights $w_c(t)/\sum_{c'} w_{c'}(t)$. By construction $\pi_t^{\mathrm{eff}}(\mu)$ reduces to the round-level mixture $\pi_t(\mu)$ of (2.32) whenever the plasticity weights are homogeneous, $w_c(t) \equiv w(t)$.

Proposition 1 (Conditional decomposition of the feedback-dependent round operator)**.** *Fix $t \ge 1$, let the client operators be given by the convex update (3.6)–(4.29) and let $\boldsymbol{J}_s^{(t)}$ be the aggregate (3.2). Then the rescaled round operator splits* exactly *as*

$$\tilde{\boldsymbol{J}}_s^{(t)} = N\boldsymbol{J}_s^{(t)} = \underbrace{\frac{1}{L}\sum_{c=1}^{L} w_c(t)\,N\tilde{\boldsymbol{J}}_c^{(t)}}_{=:\ \boldsymbol{P}_t\ \text{(plastic part)}} + \underbrace{(1-\bar{w}_t)\sum_{v=1}^{\hat{K}_{\mathrm{rec}}(t-1)} \hat{\boldsymbol{\xi}}_{s,v}^{(t-1)}\big(\hat{\boldsymbol{\xi}}_{s,v}^{(t-1)}\big)^{\top}}_{=:\ \boldsymbol{F}_{t-1}\ \text{(feedback part)}}, \tag{4.16}$$

and the two terms enjoy the following properties.

(P1) ***The feedback term is positive semidefinite, of finite rank, and past-measurable.*** *One has $\boldsymbol{F}_{t-1} \succeq 0$ and* rank *$\boldsymbol{F}_{t-1} \le \hat{K}_{\mathrm{rec}}(t-1)$, the number of archetypes accepted at round $t-1$ (which we do* not *assume to be bounded by K: by item* (R4) *of Appendix E.3 the spectral stage may transiently accept residual directions); its eigendirections are $\mathfrak{F}_{t-1}$-measurable and its scalar prefactor is fixed by the round-t weights, so that $\boldsymbol{F}_{t-1}$ is measurable with respect to $\mathfrak{G}_t := \mathfrak{F}_{t-1} \vee \sigma\big(\{w_c(t)\}_{c=1}^{L}\big)$ and carries no fresh randomness. In particular $\boldsymbol{F}_{t-1}$ carries* no *bulk component: by Weyl's inequalities, for every $i > \hat{K}_{\mathrm{rec}}(t-1)$,*

$$\lambda_i(\boldsymbol{P}_t) \le \lambda_i\big(\tilde{\boldsymbol{J}}_s^{(t)}\big) \le \lambda_{i-\hat{K}_{\mathrm{rec}}(t-1)}(\boldsymbol{P}_t), \tag{4.17}$$

so that the bulk-confinement statement (D1) *of Theorem 2 transfers from $\boldsymbol{P}_t$ to $\tilde{\boldsymbol{J}}_s^{(t)}$, with K replaced by $K + \hat{K}_{\mathrm{rec}}(t-1)$. The relevant band is the one of the plastic part: for homogeneous weights $w_c(t) \equiv \bar{w}_t$ it is exactly $\bar{w}_t$ times the Marchenko–Pastur band (4.10), whereas for heterogeneous weights $\boldsymbol{P}_t$ obeys the generalized Marchenko–Pastur law of $H_t := L^{-1}\sum_c \delta_{w_c(t)}$ and the band must be inflated to $\big[w_{\min}(t)\big(\lambda_-(q) -$*

$\varepsilon_N(\delta)\big)$, $w_{\max}(t)\big(\lambda_+(q)+\varepsilon_N(\delta)\big)\big]$ *via the positive semidefinite sandwich established in the proof of Corollary 1.*

(P2) ***Conditionally on the past, the plastic part is a weighted sample second-moment matrix.*** *The round-t batches are generated by (2.13) independently of $\mathfrak{F}_{t-1}$. Hence, conditionally on $\mathfrak{G}_t$ and treating the weights as fixed before the current batch is drawn—the residual induced by their data dependence is quantified in Appendix E.3, item* (R1)*—one has*

$$\mathbb{E}\big[\boldsymbol{P}_t \;\big|\; \mathfrak{G}_t\big] = \bar{w}_t\,\sigma^2\,\boldsymbol{I} + Nr^2\sum_{\mu=1}^{K}\pi_t^{w}(\mu)\,u_\mu u_\mu^\top, \tag{4.18}$$

which is again of the spiked form (4.1), with the round-level exposure replaced by the plasticity-weighted exposure and the isotropic level rescaled by $\bar{w}_t$. The associated signal-to-noise ratios are therefore

$$\kappa_\mu^{\mathrm{plast}}(t) = \frac{Nr^2\,\pi_t^{\mathrm{eff}}(\mu)}{\sigma^2} = \frac{Nr^2\,\pi_t^{\mathrm{eff}}(\mu)}{1-r^2}, \tag{4.19}$$

so that the global plasticity level $\bar{w}_t$ cancels between spike and bulk and does not *displace the detection threshold: only the redistribution of the exposure across clients, through π_t^{eff}, does.*

(P3) ***The feedback term is signal-aligned and can only reinforce already consolidated directions.*** *Set $\hat{u}_\nu := \hat{\xi}_{s,\nu}^{(t-1)}/\sqrt{N}$ and let $m_{t-1}(\mu) := \max_{\nu\le\widehat{K}_{\mathrm{rec}}(t-1)}|\langle\hat{u}_\nu,u_\mu\rangle|$ be the magnetization (3.7) attained at the previous round. Then*

$$\begin{aligned} u_\mu^\top \boldsymbol{F}_{t-1}u_\mu &= (1-\bar{w}_t)\,N\sum_{\nu\le\widehat{K}_{\mathrm{rec}}(t-1)}\langle\hat{u}_\nu,u_\mu\rangle^2 \\ &= (1-\bar{w}_t)\,N\big[m_{t-1}(\mu)^2 + O(K/N)\big]. \end{aligned} \tag{4.20}$$

Consequently, in the large-spike regime $\theta_\mu^{(t)} = O(N) \gg \lambda_+(q) = O(1)$ and up to the near-orthogonality error $O(\varepsilon_{\mathrm{orth}})$ of Theorem 2, the effective signal-to-noise ratio carried by archetype μ at round t satisfies

$$\begin{aligned} \kappa_\mu^{\mathrm{eff}}(t) \;\ge\; &\frac{N}{\bar{w}_t\,\sigma^2}\Big[r^2\pi_t^{w}(\mu) \;+\; (1-\bar{w}_t)\,m_{t-1}(\mu)^2\Big] \\ &+\; O\big(\varepsilon_{\mathrm{orth}} + N^{-1/2}\big), \end{aligned} \tag{4.21}$$

the bound being an equality up to a relative *error $O\big(1-m_{t-1}(\mu)^2\big)$: the exact leading eigenvalue of the matched block (E.21) exceeds the bracketed expression by*

$$\frac{N(1-\bar{w}_t)^2\big(1-m_{t-1}(\mu)^2\big)}{r^2\pi_t^{w}(\mu)+(1-\bar{w}_t)} \;+\; O\Big(N\big(1-m_{t-1}(\mu)^2\big)^2\Big),$$

so that (4.21) is sharp only in the well-consolidated regime $m_{t-1}(\mu)\to 1$ and is, in every case, conservative for detection. Its right-hand side is nondecreasing in $m_{t-1}(\mu)$ and collapses onto $\kappa_\mu^{\mathrm{plast}}(t)$ of (4.19) whenever μ was not *represented in the previous reconstruction, i.e. whenever $m_{t-1}(\mu) = O(N^{-1/2})$.*

The proof, together with the explicit control of the residual terms, is given in Appendix E.3. Property (P3) isolates the operationally relevant consequence: *the feedback loop never manufactures spectral directions outside* $\mathrm{span}\{u_\mu\}_\mu \cup \mathrm{span}\{\hat{u}_\nu\}_\nu$. It can only push a direction that has already been reconstructed further away from the bulk—and, when that reconstruction is imperfect, populate the residual direction inside $\mathrm{span}\{u_\mu,\hat{u}_{\nu(\mu)}\}$ quantified in (4.27), cf. item (R4) of Appendix E.3—whereas a genuinely novel archetype—precisely the object that the detector $\widehat{K}_{\mathrm{MP}}(t)$ and the rank-expansion gate (4.39) are designed to reveal—receives no feedback contribution at all, and its detectability at round t is governed by the fresh, plasticity-weighted evidence alone. In this sense the use we make of the BBP threshold in the continual phase is *conservative*.

When the weights are heterogeneous, $\boldsymbol{P}_t$ is a sample second-moment matrix with a nonconstant scale profile, so its bulk is described by the generalized Marchenko–Pastur law associated with the empirical weight distribution $H_t := L^{-1}\sum_c \delta_{w_c(t)}$ rather than by (D.5). Instead of solving the corresponding Silverstein equation, the following two-sided sandwich turns the heterogeneity into an explicit inflation of the BBP threshold.

Corollary 1 (Detection threshold under heterogeneous plasticity)**.** *Let $w_{\min}(t) := \min_c w_c(t) > 0$ and $w_{\max}(t) := \max_c w_c(t)$, and define the* plasticity dispersion *of the round as*

$$\rho_t := \frac{w_{\max}(t)}{w_{\min}(t)} \;\ge\; 1. \tag{4.22}$$

Then, conditionally on $\mathfrak{F}_{t-1}$ and on $\{w_c(t)\}$, and on the high-probability event of Theorem 2 applied to the unweighted round design, archetype μ produces an eigenvalue of $\tilde{\boldsymbol{J}}_s^{(t)}$ lying strictly above the bulk of $\boldsymbol{P}_t$ as soon as

$$\kappa_\mu^{(t)} \;>\; \kappa_c(\rho_t,q) := \frac{b_t+\sqrt{b_t^2-4q}}{2}, \qquad b_t := \rho_t(1+\sqrt{q})^2-(1+q), \tag{4.23}$$

with $\kappa_\mu^{(t)}$ as in (4.2). The map $\rho\mapsto\kappa_c(\rho,q)$ is well defined ($b_t\ge 2\sqrt{q}$), increasing, and satisfies $\kappa_c(1,q)=\sqrt{q}$.

Corollary 1 makes the role of the adaptive controller transparent at the spectral level: for homogeneous weights the classical BBP threshold $\kappa_\mu > \sqrt{q}$ of Theorem 2 (D2) is recovered *exactly*, and client-wise heterogeneity of the plasticity weights is absorbed into the explicitly computable inflation factor $\kappa_c(\rho_t,q)/\sqrt{q}$. We stress that (4.23) is a *sufficient* condition obtained from a two-sided operator sandwich, not the exact threshold of the generalized Marchenko–Pastur law associated with H_t: it is tight only at $\rho_t = 1$, and since $\kappa_c(\rho_t,q)\sim\rho_t(1+\sqrt{q})^2$ the inflation factor grows linearly in the dispersion. Heterogeneity therefore need not raise the true detection threshold by this amount; (4.23) certifies detection, it does not certify non-detection. Since ρ_t is observable at the server, the inflated threshold (4.23) can be evaluated at run time and provides a conservative, self-certifying version of the detection rule.

Spectral reading of the stability–plasticity trade-off.. Eq. (4.21) also yields a purely spectral account of the two failure modes documented in Fig. 7. An archetype whose fresh evidence has vanished, $\pi_t^{w}(\mu) = 0$, remains supercritical provided that

$$\frac{N\,(1-\bar{w}_t)}{\bar{w}_t\,\sigma^2}\,m_{t-1}(\mu)^2 \;>\; \sqrt{q} = \sqrt{\frac{N}{M_{\mathrm{round}}}}. \tag{4.24}$$

The left-hand side exceeds the right-hand side by a factor of order $\sqrt{N\,M_{\mathrm{round}}}$, so a consolidated direction stays spectrally visible for every $\bar{w}_t$ bounded away from 1: this is the memory-protection mechanism of the convex update, read on the spectrum. In the purely data-driven limit $\bar{w}_t\to 1$ the feedback term disappears from (4.16), an unexposed archetype instantaneously falls below the detection threshold, and the reconstruction is lost—exactly the catastrophic forgetting displayed by the $w_c(t)\equiv 1$ baseline in Fig. 7-D. In the opposite, memory-driven limit $\bar{w}_t\to 0$ one has $\boldsymbol{P}_t\to\boldsymbol{0}$ and the spectrum of $\tilde{\boldsymbol{J}}_s^{(t)}$ freezes onto that of $\boldsymbol{F}_{t-1}$, which is the rigidity displayed in Fig. 7-C. The mechanism here is *not* a change of the ratio $\kappa_\mu^{\mathrm{plast}}(t)$, which by (4.19) is invariant under the global rescaling $\boldsymbol{P}_t\mapsto\bar{w}_t\boldsymbol{P}_t$: it is the collapse of the *absolute* separation $\bar{w}_t\sigma^2\big[\lambda_{\mathrm{out}}(\kappa_\mu^{\mathrm{plast}})-\lambda_+(q)\big] = O(\bar{w}_t)$ of the plastic outliers, which is swamped both by the $O(N)$ feedback spikes of $\boldsymbol{F}_{t-1}$ and by the finite-size cushion δ_N of the detector (4.25), both fixed on an absolute scale. A fresh direction therefore ceases to be resolvable at finite N, even though in the idealized $N\to\infty$ model it would remain supercritical for every $\bar{w}_t > 0$. The adaptive controller of Section 4.2 is precisely the mechanism that keeps $\bar{w}_t$ in the interior of $(0,1)$, where both terms of (4.16) remain spectrally active.

Remark 6 (Scope of the BBP analysis across rounds: exact at $t=0$, controlled approximation for $t>0$)**.** We state explicitly the status of Theorem 2 along the federated trajectory.

(*i*) *Initialization ($t=0$).* No server reconstruction is available yet, so $\boldsymbol{F}_{-1} = \boldsymbol{0}$ and $w_c(0)\equiv 1$. The operator $\tilde{\boldsymbol{J}}_s^{(0)} = M_{\mathrm{round}}^{-1}\sum_{c,m}\boldsymbol{\eta}_c^{0,m}(\boldsymbol{\eta}_c^{0,m})^\top$ is a sample second-moment matrix built from M_{round} genuinely independent examples, and Theorem 2 applies to it *without any approximation*. This is the regime in which the quantitative predictions (D2)–(D4) are tested in Fig. 5.

(*ii*) *Continual phase ($t \geq 1$), exact part.* The decomposition (4.16) is an algebraic identity; (P1) and hence bulk confinement (D1) are exact, and the conditional spiked structure (4.18) of the plastic part is exact as well. Under homogeneous weights, the threshold $\kappa_\mu > \sqrt{q}$ is exact conditionally on $\mathfrak{F}_{t-1}$; under heterogeneous weights it is replaced by the explicit conservative threshold of Corollary 1.

(*iii*) *Continual phase ($t \geq 1$), approximate part.* Three ingredients are *not* established rigorously and are used as an approximation. First, $w_c(t)$ is itself a functional of the round-t batch, so conditioning on $\mathfrak{F}_{t-1}$ does not make the weights deterministic; we replace them by their conditional means, an operation whose residual we control only heuristically (Appendix E.3, item (R1)). Second, all statements are *quenched*, i.e. conditional on the realized past: the effective population operator in (4.18) is itself random and $\mathfrak{F}_{t-1}$-measurable, so an annealed description joint across rounds would require a local law for a recursively defined, time-dependent ensemble (item (R2)). Third, $\tilde{\boldsymbol{J}}_s^{(t)}$ mixes a multiplicative finite-rank perturbation (the plastic spikes) with an additive one (the feedback); we use the leading-order superposition (4.21), legitimate because the spikes are $O(N)$ while the bulk is $O(1)$, rather than the exact joint transform (item (R5)).

Accordingly, for $t > 0$ Theorem 2 should be read as a *controlled approximation*—structurally justified by Proposition 1, conservative for the novelty-detection task by (P3), and validated a posteriori by the numerical experiments of Section 5, in particular by the rank-expansion dynamics of Fig. 9, which occurs at the round predicted by the exposure threshold (4.5) even though the estimator is feedback-dependent throughout. A fully rigorous treatment of item (*iii*) is an interesting problem in its own right and is left to future work.

From theory to an operational detector. Theorem 2 yields a direct detection rule on the unsharpened aggregated operator $\tilde{\boldsymbol{J}}_s^{(t)} = N\boldsymbol{J}_s^{(t)}$. Defining

$$\tau_{\mathrm{MP}}(t) := \lambda_+(q) + \delta_N, \tag{4.25}$$

where δ_N is a finite-size cushion consistent with Eq. (4.11), we introduce the BBP/MP effective rank

$$\widehat{K}_{\mathrm{MP}}(t) = \#\big\{i : \lambda_i\big(\tilde{\boldsymbol{J}}_s^{(t)}\big) > \tau_{\mathrm{MP}}(t)\big\}. \tag{4.26}$$

This quantity estimates the number of empirically supercritical directions. For $t = 0$ the threshold (4.25) is the exact BBP cut of Theorem 2. For $t > 0$ it must be evaluated on the effective quantities supplied by Proposition 1: the aspect ratio q and the isotropic level are those of the plastic part $\boldsymbol{P}_t$, and the conservative variant (4.23) should be used whenever the plasticity dispersion ρ_t is appreciably larger than one. Two further consequences of the decomposition (4.16) are worth recording, because they explain why the pipeline does not rely on $\widehat{K}_{\mathrm{MP}}(t)$ alone. On the one hand, by (P1) the feedback can displace at most $\widehat{K}_{\mathrm{rec}}(t-1)$ additional eigenvalues, so $\widehat{K}_{\mathrm{MP}}(t)$ can never undercount the supercritical fresh directions. On the other hand, an imperfectly reconstructed archetype, $m_{t-1}(\mu) < 1$, contributes a residual direction orthogonal to $\hat{u}_\mu$ inside $\mathrm{span}\{u_\mu, \hat{u}_\mu\}$, whose population eigenvalue is

$$\lambda_\mu^{\mathrm{res}}(t) \;\simeq\; N\,\frac{r^2\pi_t^w(\mu)(1-\bar{w}_t)}{r^2\pi_t^w(\mu)+(1-\bar{w}_t)}\,\big(1-m_{t-1}(\mu)^2\big), \tag{4.27}$$

and which may itself become supercritical. Such residual outliers are reconstruction-error directions rather than new archetypes; they are removed downstream by the duplicate-merging rule (2.23) and the acceptance criterion (2.27), which is one of the reasons why the operational counts $\widehat{K}_{\mathrm{spec}}(t)$ and $\widehat{K}_{\mathrm{rec}}(t)$, rather than $\widehat{K}_{\mathrm{MP}}(t)$, drive the reconstruction stage.

For the subsequent LAM initialization, the aggregated operator is first sharpened into $\hat{\boldsymbol{J}}^{KS,(t)}$. We then define the operational post-sharpening rank

$$\widehat{K}_{\mathrm{spec}}(t) = \#\Big\{i : \lambda_i\Big(\hat{\boldsymbol{J}}^{KS,(t)}\Big) > \tau_{\mathrm{spec}}\Big\}. \tag{4.28}$$

The associated eigenvectors are used to construct the LAM seeds. In the well-separated regime, $\widehat{K}_{\mathrm{spec}}(t) \simeq \widehat{K}_{\mathrm{MP}}(t)$, since the sharpening map sends supercritical signal directions close to unity while leaving the bulk close to zero. Outside this regime, $\widehat{K}_{\mathrm{MP}}(t)$ retains the direct BBP interpretation, whereas $\widehat{K}_{\mathrm{spec}}(t)$ is an operational post-sharpening quantity.

Interpretation for federated dynamics. Eq. (4.1) makes the organizing principle explicit: exposure determines spike strength, spike strength determines spectral separation, and separation determines whether an archetype can be discovered from $\boldsymbol{J}_s^{(t)}$ at that round. In particular, archetypes with negligible exposure cannot generate outliers and remain invisible until their exposure increases; conversely, once exposure crosses the BBP threshold, their eigenvalues detach and their eigenvectors begin to align, enabling reliable reconstruction. Fig. 5 provides a numerical validation of the outlier and alignment predictions at finite N.

Taken together, these results close the loop between the federated sampling process and the spectral heuristics used in the pipeline: a principled BBP/MP thresholding step yields $\widehat{K}_{\mathrm{MP}}(t)$, while the corresponding sharpened spectrum provides the operational seed count $\widehat{K}_{\mathrm{spec}}(t)$ for the subsequent heteroassociative refinement. In the next sections we exploit this lens to study non-stationary regimes, where π_t drifts and new archetypes enter the federation, and to quantify the resulting detection latency and interference effects.

Remark 7 (Scope of Theoretical Guarantees)**.** The concentration bounds and spectral thresholds derived in Theorems 1 and 2 are established under the assumption of a fixed effective noise level characterizing the aggregated operator. While our numerical simulations explore a more complex regime involving fully adaptive, client-specific weights $w_c(t)$ and the presence of pure-noise attackers, the theoretical analysis provides the fundamental detectability guarantees for the *effective* federation. Essentially, the theory predicts *when* archetypes become spectrally detectable provided that the adaptive controller successfully suppresses high-variance clients, thereby validating the consistency of the reconstruction limits in the asymptotic regime. The effect of the client-specific weights on the detection threshold is now quantified explicitly by Proposition 1 and Corollary 1, and the exact/approximate status of each statement along the federated trajectory is summarized in Remark 6.

4.2. Plasticity and stability: The role of $w_c(t)$

In this subsection, we detail the construction of the adaptive client-wise weight $w_c(t)$ introduced in the blending rule (3.6). Recall that, at round t, the effective operator associated with client c is obtained by combining the server reconstruction from the previous round with the correlator computed from its current local batch:

$$\boldsymbol{J}_c^{(t)} \;=\; \big(1 - w_c(t)\big)\,\boldsymbol{A}^{(t-1)} \;+\; w_c(t)\,\tilde{\boldsymbol{J}}_c^{(t)}. \tag{4.29}$$

Here,

$$\boldsymbol{A}^{(t-1)} \;=\; \frac{1}{N}\sum_{\mu=1}^{\widehat{K}_{\mathrm{rec}}(t-1)} \hat{\boldsymbol{\xi}}_{s,\mu}^{(t-1)}\big(\hat{\boldsymbol{\xi}}_{s,\mu}^{(t-1)}\big)^\top,$$

$$\tilde{\boldsymbol{J}}_c^{(t)} \;=\; \frac{1}{N\,M_c^t}\sum_{m=1}^{M_c^t} (\boldsymbol{\eta}_c^{(t)})^m\big[(\boldsymbol{\eta}_c^{(t)})^m\big]^\top. \tag{4.30}$$

Thus, $\boldsymbol{A}^{(t-1)}$ encodes the archetypal structure reconstructed at the server up to round $t-1$, whereas $\tilde{\boldsymbol{J}}_c^{(t)}$ captures the statistics of the new batch observed by client c at round t.

As detailed in Section 3, the fresh correlator $\tilde{\boldsymbol{J}}_c^{(t)}$ is uploaded before the adaptive blend is assembled. The server first forms the unweighted aggregate $\boldsymbol{J}_{\mathrm{fresh}}^{(t)} = 1/L\sum_{c=1}^{L}\tilde{\boldsymbol{J}}_c^{(t)}$ and evaluates the preliminary spectral-rank diagnostic. After receiving the resulting rank-expansion indicator, each client computes and returns the scalar weight $w_c(t)$, and the effective operator (4.29) can be assembled directly at the server.

An effective adaptive rule for $w_c(t)$ must distinguish two qualitatively different sources of discrepancy: (i) degradation of the example-to-archetype channel and (ii) genuine distributional shifts, such as the emergence of previously unseen archetypes. We use two complementary diagnostics for this purpose. First, the fresh spectral-rank test introduced in Section 3 determines whether the current federation contains additional structured directions that are not represented by the previous server reconstruction. Second, the sign-agreement statistic defined below quantifies how strongly the complete current-batch correlator differs from the consolidated operator. The spectral diagnostic therefore distinguishes structured rank expansion from an unsupported discrepancy, whereas the entropy controller determines how strongly the incoming evidence should affect the memory.

To keep notation light, we set $\boldsymbol{A} = \boldsymbol{A}^{(t-1)}$ and $\boldsymbol{B} = \tilde{\boldsymbol{J}}_c^{(t)}$, and define the corresponding sign matrices:

$$s_{ij}^A = \operatorname{sign}(A_{ij}),$$
$$s_{ij}^B = \operatorname{sign}(B_{ij}).$$

Since the theoretical sign-matching baseline is defined over non-degenerate, clean Hebbian entries, we restrict our sign comparison to the off-diagonal elements where the consolidated sign is genuinely nonzero. Specifically, letting

$$D_A^{(t)} := \{(i,j) : i \neq j,\ s_{ij}^A \neq 0\}, \tag{4.31}$$

we define the empirical agreement probability as

$$p_c(t) := \frac{1}{|D_A^{(t)}|} \sum_{(i,j)\in D_A^{(t)}} \mathbf{1}\{s_{ij}^A = s_{ij}^B\}, \tag{4.32}$$

and map it to a scalar uncertainty score via the binary entropy:

$$H_{AB,c}(t) := h_2\big(p_c(t)\big) = -p_c(t)\log_2 p_c(t) - \big(1-p_c(t)\big)\log_2\big(1-p_c(t)\big). \tag{4.33}$$

By construction, $H_{AB,c}(t) \in [0,1]$. It reaches its maximum value $H_{AB,c}(t) = 1$ when $p_c(t) = 1/2$, namely when the signs of the current and consolidated operators agree at approximately chance level. In the relevant regime $p_c(t) \gtrsim 1/2$, smaller entropy corresponds to stronger sign consistency, whereas $H_{AB,c}(t) \to 1$ indicates the loss of a preferred sign agreement with the consolidated memory.

Crucially, $H_{AB,c}(t)$ cannot be interpreted in absolute terms without accounting for the intrinsic corruption noise in the data. Indeed, $\boldsymbol{B}$ is a Hebbian correlator built from noisy examples, whereas $\boldsymbol{A}$ is derived from reconstructed archetypes. Even in the ideal scenario where the reconstruction is perfect (i.e., $\boldsymbol{A}$ exactly matches the true archetypal Hebbian matrix), the noisy example channel inherently introduces a baseline disagreement.

Under the generative model (2.13), the parameter r quantifies the correlation between an example and its generating archetype. Because the learning problem is unsupervised, the client observes only the examples $(\boldsymbol{\eta}_c^{(t)})^m$ for $m = 1, \dots, M_c^t$, without access to their true generating archetypes. For analytical purposes only, let $\boldsymbol{\eta}^{\nu,a}$ denote the ath example generated from archetype $\boldsymbol{\xi}^\nu$. This is equivalent to relabelling the sample index m as a pair (ν, a) whenever the latent generator is exposed in the teacher model. In a theoretically balanced batch with M examples per archetype, we would have $M_c^t = KM$. In the general heterogeneous case, however, the number of examples associated with each archetype is a random variable governed by the local mixture distribution $\pi_{t,c}$.

Using this analytical notation, if $\boldsymbol{\eta}^{\nu,a}$ is generated from $\boldsymbol{\xi}^\nu$, we have

$$\mathbb{E}\left[\frac{1}{N}\sum_{i=1}^N \xi_i^\mu \eta_i^{\nu,a}\right] = r\,\delta_{\mu\nu}, \tag{4.34}$$

up to finite-size fluctuations. In practice, however, r is rarely known a priori. We therefore estimate an effective client-wise quality after the first communication round, using the archetypes reconstructed and broadcast by the server as reference prototypes.

More precisely, for $t \geq 1$, we define the overlap between the mth current example of client c and the μth archetype reconstructed at the previous round as

$$m_{\mu m}^{(c,t)} := \left|\frac{1}{N}\big(\hat{\boldsymbol{\xi}}_{s,\mu}^{(t-1)}\big)^\top (\boldsymbol{\eta}_c^{(t)})^m\right|, \qquad \mu = 1, \dots, \hat{K}_{\mathrm{rec}}(t-1). \tag{4.35}$$

The absolute value accounts for the global sign ambiguity inherent in each reconstructed archetype. For every example, we define its best alignment with the currently represented archetypal set as

$$a_{c,m}(t) := \max_{\mu \leq \hat{K}_{\mathrm{rec}}(t-1)} m_{\mu m}^{(c,t)}. \tag{4.36}$$

Let $r_{\mathrm{floor}} > 0$ be a small quality threshold separating meaningful prototype overlaps from the finite-size random-overlap floor. If the current batch contains only examples generated from previously reconstructed archetypes, the natural all-batch estimator is

$$\hat{r}_{c,\mathrm{all}}(t) := \frac{1}{M_c^t}\sum_{m=1}^{M_c^t} a_{c,m}(t). \tag{4.37}$$

In the ideal regime where the server prototypes are accurate and the current examples are generated from already represented archetypes, $\hat{r}_{c,\mathrm{all}}(t)$ concentrates around the true channel correlation r.

This interpretation no longer holds when the same batch also contains examples generated from genuinely new archetypes. Let $\gamma_c(t)$ denote the fraction of current examples originating from archetypes already represented by the server. If the remaining examples are generated from novel archetypes that are asymptotically orthogonal to the old archetypal subspace, then, up to finite-size fluctuations we have

$$\hat{r}_{c,\mathrm{all}}(t) \simeq \gamma_c(t)\, r + \big(1-\gamma_c(t)\big)\,\varepsilon_N, \qquad \varepsilon_N = \Theta\left(\sqrt{2\log \hat{K}_{\mathrm{rec}}(t-1)/N}\right). \tag{4.38}$$

The all-batch estimate therefore mixes the intrinsic channel quality with the fraction of examples already explained by the current memory. It may fall below r_{floor} even when the previously represented and the newly emerging examples are both generated through a high-quality channel.

To prevent this dilution, the quality estimator is conditioned on the spectral-rank diagnostic applied to the unweighted fresh aggregate $\boldsymbol{J}_{\mathrm{fresh}}^{(t)}$ (3.5). In the additive archetype-emergence regime studied in Section 5.2, we define the number of additional spectrally detectable directions as

$$\hat{K}_{\mathrm{new}}(t) := \left[\hat{K}_{\mathrm{spec}}^{\mathrm{fresh}}(t) - \hat{K}_{\mathrm{rec}}(t-1)\right]_+, \qquad [x]_+ := \max\{0, x\}. \tag{4.39}$$

If $\hat{K}_{\mathrm{new}}(t) = 0$, no statistically detectable expansion of the represented archetypal subspace is present and the original all-batch estimate is retained:

$$\hat{r}_c(t) = \hat{r}_{c,\mathrm{all}}(t). \tag{4.40}$$

If instead $\hat{K}_{\mathrm{new}}(t) > 0$, low overlap with the previously reconstructed archetypes is no longer interpreted automatically as poor channel quality. We define the local reference set

$$\mathcal{I}_{c,\mathrm{old}}(t) := \big\{m \in \{1, \dots, M_c^t\} : a_{c,m}(t) > r_{\mathrm{floor}}\big\}, \tag{4.41}$$

which contains the current examples that remain compatible with at least one archetype reconstructed at round $t-1$. Whenever this set is nonempty, the reference quality is estimated only from these examples:

$$\hat{r}_{c,\mathrm{old}}(t) := \frac{1}{|\mathcal{I}_{c,\mathrm{old}}(t)|}\sum_{m\in\mathcal{I}_{c,\mathrm{old}}(t)} a_{c,m}(t). \tag{4.42}$$

The final rank-conditioned estimate used by the adaptive controller is therefore

$$\hat{r}_c(t) = \begin{cases} \hat{r}_{c,\mathrm{all}}(t), & \hat{K}_{\mathrm{new}}(t) = 0, \\ \hat{r}_{c,\mathrm{old}}(t), & \hat{K}_{\mathrm{new}}(t) > 0 \text{ and } \|\mathcal{I}_{c,\mathrm{old}}(t)\| > 0, \\ 0, & \hat{K}_{\mathrm{new}}(t) > 0 \text{ and } \|\mathcal{I}_{c,\mathrm{old}}(t)\| = 0. \end{cases} \tag{4.43}$$

Under the shared-channel assumption used in the additive-emergence experiment of Section 5.2, the corruption parameter is independent of the archetype identity. The examples generated from previously represented archetypes can therefore provide a non-degenerate reference estimate of the same observation quality without dilution by the low-overlap novel examples. If no previously represented example is available at the client, the quality cannot be inferred through this reference construction and the contribution is conservatively suppressed. The present mechanism therefore directly addresses the additive-emergence regime studied in this work, where old and new archetypes coexist after the transition.

Crucially, the examples outside $\mathcal{I}_{c,\mathrm{old}}(t)$ are excluded only from the estimation of $\hat{r}_c(t)$. They are not removed from the complete current-batch correlator $\tilde{\boldsymbol{J}}_c^{(t)}$ in (4.30). Consequently, examples generated by newly emerging archetypes continue to modify the local operator and contribute to the sign-discrepancy entropy $H_{AB,c}(t)$. The reference subset estimates the reliability of the local observation channel, whereas the complete batch measures the structural discrepancy between the incoming evidence and the consolidated memory.

The rank-conditioned construction cannot be applied at the first communication round, since neither a previously reconstructed archetypal set nor a reference spectral rank is available yet. Consistently with Remark 6(*i*), we set $w_c(0) \equiv 1$, so that $\boldsymbol{J}_c^{(0)} = \tilde{\boldsymbol{J}}_c^{(0)}$ and the round-0 aggregate is built from purely local evidence; this value also initializes the recursion (4.48), whose first adaptive output $w_c(1) = \alpha\, w_{c,\mathrm{inst}}(1) + (1-\alpha)$ is therefore biased toward plasticity during the first rounds.

The entropy baseline is then evaluated using the rank-conditioned client-quality estimate $\hat{r}_c(t)$ defined in Eq. (4.43). It is useful to distinguish between two levels of approximation. A simple pairwise proxy follows from the probability that a single noisy product $\eta_i\eta_j$ preserves the sign of the corresponding clean product $\xi_i^\mu \xi_j^\mu$:

$$H_{\min,c}^{\mathrm{pair}}(t) = h_2\!\left(\frac{1+\hat{r}_c(t)^2}{2}\right). \tag{4.44}$$

This expression is transparent and computationally cheap, serving as an intuitive proxy for the intrinsic uncertainty induced by the example channel. However, it remains a single-pair approximation. The actual object evaluated by the controller is the sign of a finite-sample Hebbian correlator, which involves summing over both archetypal contributions and local examples.

Let $p_{K,M}(r)$ denote the probability that the sign of the noisy empirical Hebbian element agrees with that of the clean archetypal Hebbian element (conditioned on the latter being non-degenerate):

$$p_{K,M}(r) := \mathbb{P}\left(\mathrm{sign}\left(\sum_{\mu=1}^{K} \xi_i^\mu \xi_j^\mu\right) = \mathrm{sign}\left(\sum_{m=1}^{M_c^t} \eta_i^m \eta_j^m\right) \,\middle|\, \sum_{\mu=1}^{K} \xi_i^\mu \xi_j^\mu \neq 0\right). \tag{4.45}$$

We can then define

$$H_{\min,c}(t) = h_2\!\left(\widetilde{p}^{\star}_{K,M}(\hat{r}_c(t))\right), \tag{4.46}$$

where $\widetilde{p}^{\star}_{K,M}(\hat{r})$ is the large-M Gaussian approximation of Eq. (4.45) (detailed in Appendix C). This provides a mathematically consistent and numerically efficient substitute for the exact finite-sample expression.

Thus, Eq. (4.44) should be viewed as a lightweight single-product proxy useful for intuition or simple ablations, whereas Eq. (4.46) serves as the finite-sample baseline for the sign-agreement statistic used by the adaptive controller. In both cases, $H_{\min,c}(t)$ estimates the uncertainty expected from the reference channel quality before any additional discrepancy is attributed to a change in the incoming archetypal structure. A large residual discrepancy is interpreted as evidence of novelty only when the independent spectral-rank diagnostic indicates that additional structured directions are present.

We now define the instantaneous adaptive weight through the same regularized entropy rule, using the rank-conditioned estimate $\hat{r}_c(t)$ introduced above. Since r_{floor} has already been defined in the quality-estimation step, we only introduce the clipping operator

$$[x]_{[0,1]} := \min\{1, \max\{0, x\}\}.$$

Then,

$$w_{c,\mathrm{inst}}(t) = \begin{cases} 0, & \hat{r}_c(t) \le r_{\mathrm{floor}}, \\ \left[\dfrac{H_{AB,c}(t) - H_{\min,c}(t)}{1 - H_{\min,c}(t)}\right]_{[0,1]}, & \hat{r}_c(t) > r_{\mathrm{floor}}. \end{cases} \tag{4.47}$$

The revised construction separates the estimation of the observation-channel quality from the measurement of the structural change. The examples retained in $\mathcal{I}_{c,\mathrm{old}}(t)$ determine whether the client provides a non-degenerate quality reference, whereas the complete batch correlator $\tilde{\boldsymbol{J}}_c^{(t)}$ determines how strongly the incoming evidence differs from the consolidated memory.

The qualitative behavior of the rule can be summarized as follows:

- If no rank expansion is detected, $\widehat{K}_{\mathrm{new}}(t) = 0$, the estimator reduces to the original all-batch construction. If $\hat{r}_c(t) \le r_{\mathrm{floor}}$, the current contribution is treated conservatively as non-informative and its plastic component is suppressed.
- If a rank expansion is detected, $\widehat{K}_{\mathrm{new}}(t) > 0$, the examples having low overlap with the previously reconstructed archetypes are excluded only from the quality estimate. They remain present in $\tilde{\boldsymbol{J}}_c^{(t)}$ and can therefore increase the discrepancy between the current operator and the consolidated memory without artificially driving $\hat{r}_c(t)$ toward zero.
- If $\hat{r}_c(t) > r_{\mathrm{floor}}$ and $H_{AB,c}(t) \approx H_{\min,c}(t)$, the observed discrepancy is largely explained by the intrinsic corruption noise associated with the estimated channel quality. The rule consequently yields $w_{c,\mathrm{inst}}(t) \ll 1$ and favors consolidation.
- If $\hat{r}_c(t) > r_{\mathrm{floor}}$ and $p_c(t) \to 1/2$, equivalently $H_{AB,c}(t) \to 1$, the complete current-batch correlator has approximately chance-level sign agreement with the consolidated operator. Conditional on the independent rank-expansion test $\widehat{K}_{\mathrm{new}}(t) > 0$, this discrepancy is interpreted as evidence of structured novelty rather than unstructured noise. In this limit $w_{c,\mathrm{inst}}(t) \longrightarrow 1$, and the controller strongly increases plasticity.
- If rank expansion is detected but $\mathcal{I}_{c,\mathrm{old}}(t) = \varnothing$, no reference example is available to estimate the current channel quality. The client contribution is therefore suppressed conservatively in the present formulation rather than assigning a large weight from spectral evidence alone.

Below the spectral detectability threshold, a genuinely new direction cannot yet be distinguished reliably from finite-sample fluctuations. The controller therefore remains conservative until sufficient exposure produces a detectable rank expansion, consistently with the BBP mechanism developed in Section 4.1.

Remark 8. We state explicitly the consequence of this conservative choice. If the batch of client c consists entirely of examples generated by new archetypes, then $\mathcal{I}_{c,\mathrm{old}}(t) = \varnothing$, leading to $\hat{r}_c(t) = 0$ and $w_{c,\mathrm{inst}}(t) = 0$. The client is thus completely suppressed, and its plasticity weight decays geometrically. While this prevents the consolidated memory from expanding if no client retains examples compatible with the old set, it acts as a strictly necessary safeguard. Locally, a purely novel batch is statistically indistinguishable from a severe noise injection or an adversarial attack. Relaxing this restriction would require external guarantees, such as verifying cross-client consensus among local synaptic matrices to distinguish structured novelty from independent noise, or exploiting within-batch statistics $N^{-1}(\boldsymbol{\eta}_c^{(t)})^m \cdot (\boldsymbol{\eta}_c^{(t)})^{m'}$, which we leave to future work.

Finally, to prevent abrupt fluctuations in $w_c(t)$ across successive rounds, we apply an exponential moving average to the instantaneous

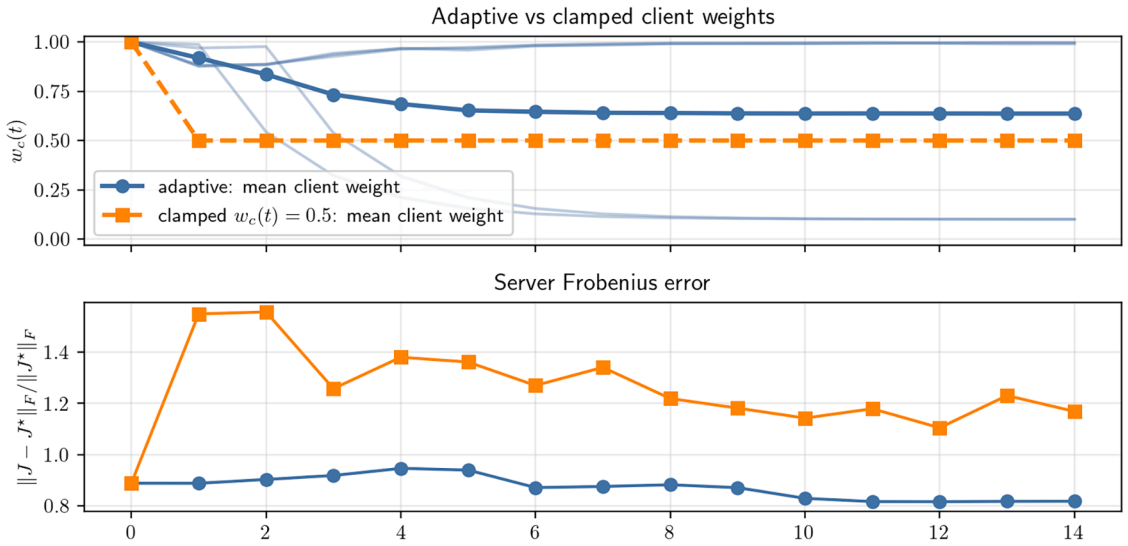


Fig. 6. We compare the proposed adaptive plasticity controller with a simple FedAvg-like baseline in which all clients use a fixed, clamped weight $w_c(t) = 1/2$ throughout the federation. The upper panel reports the evolution of the client weights: pale blue curves show the individual adaptive weights, the dark blue curve their mean, and the orange dashed curve the fixed clamped baseline. The adaptive rule produces heterogeneous client-dependent weights, suppressing client contributions that are judged unreliable while retaining larger weights for informative evidence and for structurally novel contributions supported by the spectral-rank diagnostic. The lower panel reports the corresponding server-side normalized Frobenius error $\|\boldsymbol{J} - \boldsymbol{J}^\star\|_F / \|\boldsymbol{J}^\star\|_F$. The fixed-weight baseline rapidly accumulates a larger reconstruction error, showing that a uniform FedAvg-like blending of local and server information is not sufficient under heterogeneous or corrupted client contributions. In contrast, the adaptive controller maintains a lower server reconstruction error over the whole trajectory, demonstrating that client-aware weighting improves the robustness of the federated Hebbian consolidation mechanism.

value $w_{c,\mathrm{inst}}(t)$ computed via Eq. (4.47):

$$w_c(t) = \alpha\, w_{c,\mathrm{inst}}(t) + (1-\alpha)\, w_c(t-1), \qquad \alpha \in [0,1]. \tag{4.48}$$

To immediately isolate the role and necessity of this adaptive weighting mechanism, we compare it against a simple FedAvg-like fixed-weight baseline. In this baseline, the client plasticity weights are clamped to a constant value, $w_c(t) \equiv w_{\mathrm{fix}} = 1/2$ for all clients and all rounds, so that each client combines its current local Hebbian estimate and the previous server reconstruction with a time-independent, client-independent coefficient.

This comparison highlights a fundamental limitation of uniform averaging: the fixed-weight baseline treats all clients as equally reliable throughout the federation, independently of whether their local operators are aligned with the server-side reconstruction or corrupted by heterogeneous exposure. By contrast, the proposed controller combines channel-quality estimation with the spectral-rank diagnostic: it suppresses unreliable local evidence while preserving strongly discrepant contributions when they are supported by a detectable expansion of the archetypal subspace.

As shown in Fig. 6, the FedAvg-like fixed-weight rule yields a substantially larger server-side Frobenius reconstruction error. After the first few rounds, the clamped baseline remains around a high-error plateau, whereas the adaptive protocol keeps the server reconstruction consistently closer to the ground-truth operator $\boldsymbol{J}^\star$. This demonstrates that, in the present heterogeneous Hebbian setting, uniform averaging is not sufficient: the gain comes from conditioning plasticity jointly on client quality and on whether the incoming discrepancy is compatible with structured rank expansion rather than uninformative noise.

This smoothly adaptive and robust behavior is further explored and practically illustrated under various heterogeneous conditions in the subsequent numerical experiments (Figs. 7 and 8).

5. Numerical simulations

We now empirically validate the proposed federated associative-memory pipeline and the theoretical mechanisms developed in Sections 4.1–4.2. Unless otherwise stated, clients follow the pipeline of Section 3: at each round they upload local correlators, the server aggregates them into $\boldsymbol{J}_s^{(t)}$, applies spectral sharpening and LAM-based factorization (Section 2.1.1), and broadcasts back the reconstructed operator $\hat{\boldsymbol{J}}_s^{(t)}$ used by clients in the convex fusion update (3.6). Reconstruction quality is assessed via the magnetization (3.7).

We organize the numerical analysis into five complementary parts: (i) non-stationary *exposure drift* and the stability–plasticity trade-off controlled by $w_c(t)$; (ii) *novelty emergence*, where new archetypes enter the federation mid-training; (iii) a *spectral-only ablation* isolating the contribution of the LAM factorization; (iv) strong Dirichlet client heterogeneity and local-only consolidation baselines; and (v) structured-data environments, where the adaptive aggregation principle is evaluated on a federated Boolean–Gaussian RBM.

Remark 9 (Theory-to-experiment cross-references)**.** The experiments of this section are organized to validate, one by one, the theoretical results of Section 4. Fig. 5 is the direct empirical counterpart of Theorem 2: Panel A tests bulk confinement (D1) and the BBP detection threshold $\kappa_\mu > \sqrt{q}$ (D2), Panel B the outlier-location law (4.12) (D3), and Panel C the eigenvector-alignment law (4.13) (D4). These three panels are evaluated on the aggregated correlator assembled from fresh, mutually independent client batches, that is, in the feedback-free setting of Remark 6(*i*), where Theorem 2 holds without approximation; this is why they can be used as a quantitative test of the closed-form laws (D3)–(D4). The feedback-dependent estimator of the continual phase is instead probed by Fig. 9, which is run with the full convex update (3.6) active at every round $t \geq 1$ and therefore constitutes the a posteriori validation of the controlled approximation discussed in Remark 6(*iii*). The entropy-controller construction of Section 4.2 is exercised in Figs. 6–8, where the failure of the fixed-weight limits (memory-driven and data-driven) makes the necessity of the adaptive rule explicit. Fig. 9 probes the exposure-driven detection mechanism of Theorem 2 (D2) in a continual regime with genuine class emergence, while Fig. 11 isolates the benefit of federated consolidation over local-only accumulation, which is the operational content of the concentration guarantee of Theorem 1. Fig. 10 isolates the contribution of the LAM factorization by comparing the full reconstruction with direct binarization of the dominant eigen-

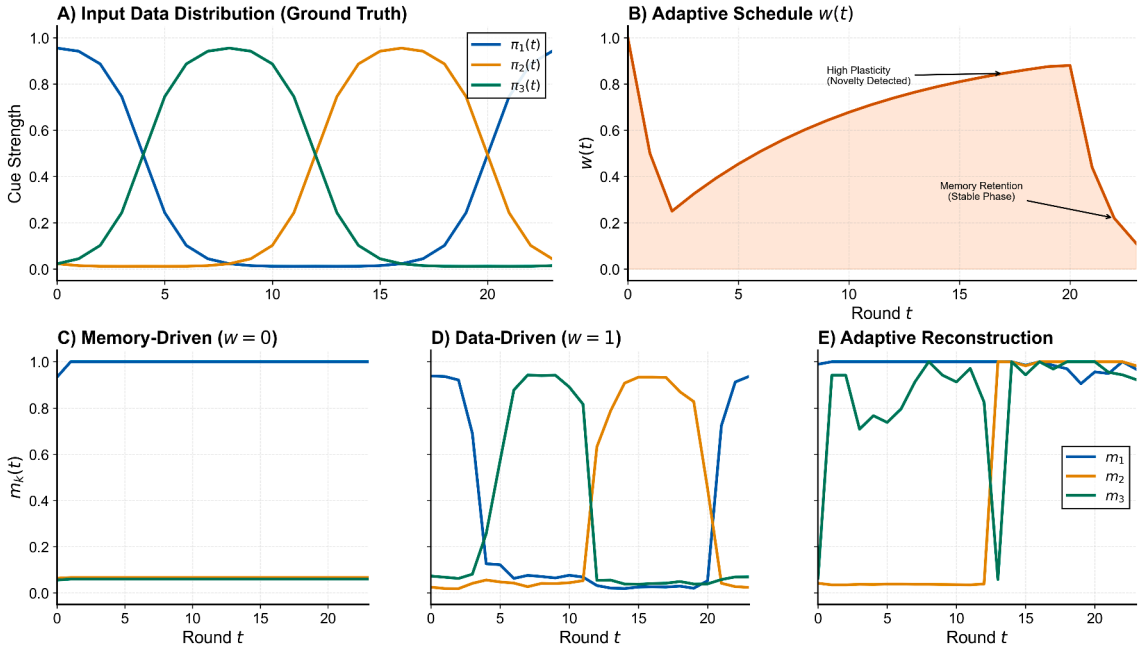


Fig. 7. Adaptive plasticity resolves the stability–plasticity dilemma under exposure drift. *A*) Ground-truth round-level mixture $\pi_t(\mu)$ (cf. (2.32)) in a sequential schedule where archetypes become dominant one after another. *B*) Adaptive client weight $w_c(t)$ computed from the entropy controller in (4.47) and smoothed via (4.48). Peaks align with mixture transitions (novelty/shift), while plateaus yield small $w_c(t)$ (consolidation). *C*) Memory-driven limit ($w_c(t) \equiv 0$): the operator update ignores new evidence and the system fails to acquire later archetypes ($m_2, m_3 \simeq 0$). *D*) Data-driven limit ($w_c(t) \equiv 1$): the update is overly plastic and magnetizations track the instantaneous mixture, leading to rapid forgetting when exposures decrease. *E*) Adaptive rule: newly exposed archetypes are acquired while previously consolidated ones remain retrievable, yielding sustained high magnetization across all modes.

vectors of the same final sharpened operator. It thereby tests whether the spectrally detected signal subspace is already sufficient to recover the individual Boolean archetypes. Finally, Fig. 12 transfers the same adaptive-aggregation principle to the structured-data (RBM) setting of Section 5.5.

Remark 10. The hyperparameters used in the numerical simulations are fixed according to the theoretically optimal operating regimes of the associative-memory blocks employed in the model. In particular, the structural parameters of the LAM modules are chosen as prescribed by the theoretical analysis developed in the works where these architectures were introduced and characterized (Agliari et al., 2025b,c). These choices correspond to the retrieval/disentangling regime in which the associative-memory blocks operate optimally.

The entropy-controller weights and the spectral detection threshold are likewise fixed according to the stability and detectability criteria described in the theoretical part of the manuscript. Thus, the numerical experiments are performed using the theoretically prescribed hyperparameter values of the constituent modules, rather than treating them as free empirical tuning parameters.

5.1. Adaptive plasticity under exposure drift and varying data quality

We first study continual learning under *exposure drift*, i.e. a time-varying round-level class mixture $\pi_t \in \Delta^{K-1}$ (Section 2.2). In this initial experiment, we use $L = 3$ clients and $K = 3$ archetypes to illustrate the temporal dynamics. At each round, clients upload correlators and the server forms $\boldsymbol{J}_s^{(t)}$; at the population level, $\mathbb{E}[\boldsymbol{J}_s^{(t)}]$ follows the spiked decomposition (4.1) with spike strengths controlled by exposure. Hence, as the mixture π_t shifts, the spectral evidence supporting each archetype shifts accordingly: when an archetype gains exposure, its spike becomes supercritical and produces an outlier/eigenvector alignment as described by the BBP theory (Theorem 2); when exposure wanes, the same direction becomes weakly supported by fresh data and risks being overwritten by overly plastic updates.

This is precisely the role of the convex fusion update (3.6). For clarity, we denote by $\boldsymbol{A}^{(t-1)}$ the operator broadcast by the server at the previous round (constructed from reconstructed archetypes as in (3.3)), and by $\tilde{\boldsymbol{J}}_c^{(t)}$ the client correlator from the current batch. The weight $w_c(t)$ trades consolidation versus adaptation (Section 4.2). In Fig. 7A, exposures shift in stages, inducing bursts of non-stationarity separated by plateaus.

The two fixed-w extremes illustrate complementary failure modes. In the *memory-driven* regime ($w_c(t) \equiv 0$), early consolidated structure dominates and subsequent exposures fail to reshape the operator, preventing acquisition of later archetypes (Fig. 7-C). Conversely, in the *data-driven* regime ($w_c(t) \equiv 1$), the system becomes overly reactive to the current batch: magnetizations quickly decay when an archetype stops being prevalent, producing catastrophic forgetting (Fig. 7-D).

The adaptive entropy controller (4.47) resolves this trade-off. As shown in Fig. 7-B, $w_c(t)$ spikes during transitions (large sign inconsistency between consolidated memory and incoming data) and relaxes during stable phases (consistency up to the intrinsic noise floor $H_{\min}(r)$). The resulting dynamics (Fig. 7-E) retains previously learned archetypes while still incorporating newly dominant ones. In other words, exposure governs *when* an archetype becomes spectrally learnable (via BBP outliers), while adaptive $w_c(t)$ governs *how strongly* the update should trust current data versus consolidated memory to prevent overwrite. Proposition 1 makes this statement quantitative and accounts for both fixed-w failure modes at the level of the spectrum: in the data-driven limit $\bar{w}_t \to 1$ the feedback term $\boldsymbol{F}_{t-1}$ of (4.16) vanishes, the retention condition (4.24) fails as soon as the fresh exposure of an archetype drops to zero, and the corresponding outlier is reabsorbed into the bulk (Fig. 7-D); in the memory-driven limit $\bar{w}_t \to 0$ the plastic term $\boldsymbol{P}_t$ vanishes, the spectrum freezes and no new supercritical direction can ever appear (Fig. 7-C). The adaptive rule keeps $\bar{w}_t$ in the interior of $(0, 1)$, where both mechanisms remain active simultaneously (Fig. 7-E).

Having established the global temporal dynamics, we now move to a more complex scenario to validate the per-client adaptive weight mecha-

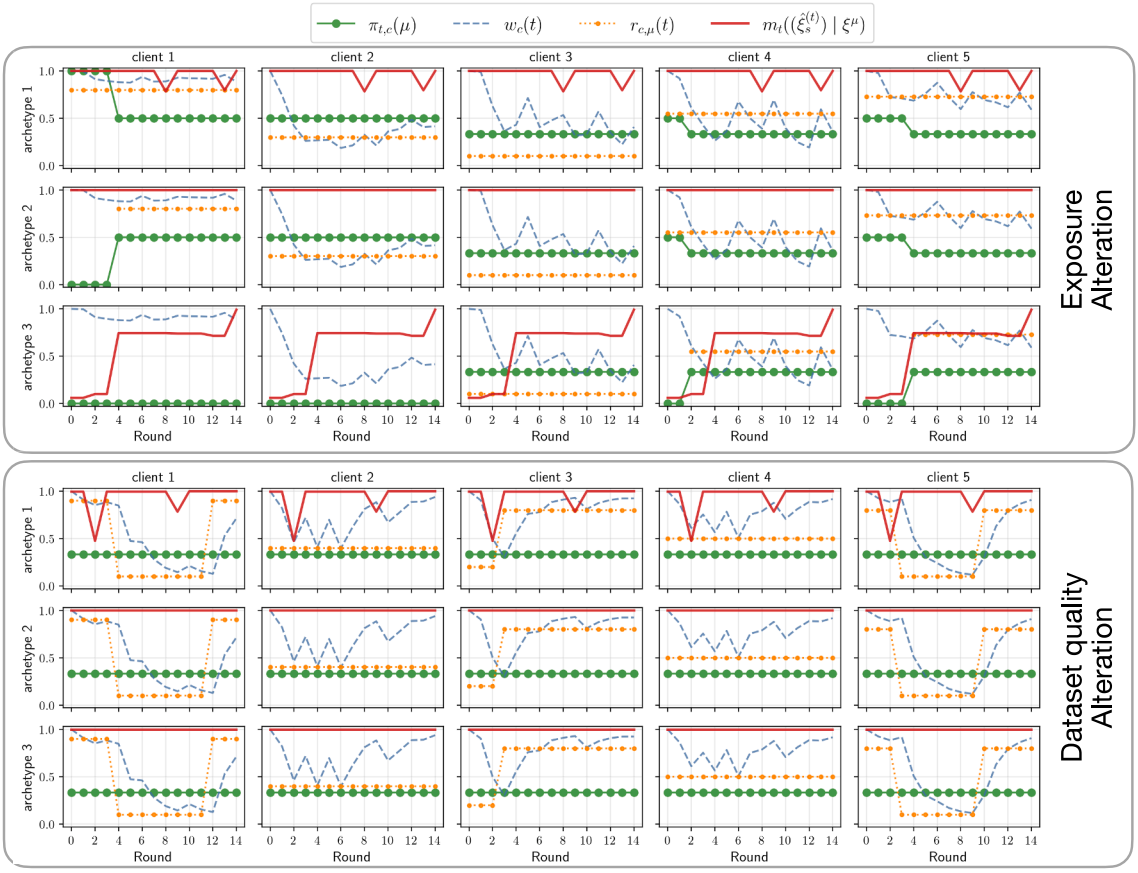


Fig. 8. Fine-grained dynamics of adaptive weighting and server-side archetype reconstruction under two complementary sources of heterogeneity. Each block is organized as a grid of archetype–client panels: rows correspond to archetypes $\mu = 1, 2, 3$, while columns correspond to clients $c = 1, \dots, 5$. In each panel, the green curve denotes the local exposure $\pi_{t,c}(\mu)$, the orange dotted curve denotes the local dataset-quality parameter $r_{c,\mu}(t)$, the blue dashed curve reports the adaptive plasticity weight $w_c(t)$ assigned to client c, and the red curve reports the server-side reconstruction quality of archetype μ, measured by the magnetization $m_t((\hat{\xi}_s^{(t)}) \mid \xi^\mu)$. Since $w_c(t)$ is client-specific but not archetype-specific, the same weight profile is repeated along each column; conversely, the magnetization is archetype-specific but not client-specific, and is therefore repeated along each row. **Top block: exposure alteration.** Here the dataset quality is fixed in time and depends only on the client, while the class exposure $\pi_{t,c}(\mu)$ varies across clients, archetypes, and rounds. This regime models heterogeneous client sampling, where different clients observe different archetypes with different frequencies even when their intrinsic data quality is stable. **Bottom block: dataset-quality alteration.** Here the exposure is kept fixed and non-zero for all archetypes and all clients, so that each client continuously observes all classes. The heterogeneity is instead introduced through time-dependent variations of $r_{c,\mu}(t)$, mimicking transient corruptions or adversarial attacks affecting the quality of the local batches. The comparison between the two blocks shows that the adaptive controller reacts to qualitatively different perturbations: it tracks exposure-driven changes when class availability varies, while suppressing unreliable contributions when dataset quality deteriorates, thereby preserving high server-side reconstruction magnetization for the archetypes that remain sufficiently informative. Parameters: $K = 3$ archetypes, $L = 5$ clients/layers, $T = 15$ communication rounds and the other hyperparameters set as in Agliari et al. (2025b,c).

nism described in Section 4.2. Specifically, we conduct two complementary federated unsupervised learning experiments designed to disentangle the effect of local class exposure from that of local dataset quality. The results are reported in Fig. 8. Each block is organized as a grid of archetype–client panels: rows correspond to the $K = 3$ archetypes, while columns correspond to the $L = 5$ clients. In each panel, we show the local exposure $\pi_{t,c}(\mu)$, the dataset-quality parameter $r_{c,\mu}(t)$, the adaptive plasticity weight $w_c(t)$, and the server-side reconstruction magnetization $m_t((\hat{\xi}_s^{(t)}) \mid \xi^\mu)$.

The top block of Fig. 8 considers an *exposure-alteration* regime. Here, the quality parameter is kept fixed in time for each client, while the class exposure $\pi_{t,c}(\mu)$ varies across clients, archetypes, and rounds. This setting models heterogeneous client sampling: different clients observe different archetypes with different frequencies, even though their intrinsic data quality is stable. The adaptive weights remain primarily controlled by the quality of the local Hebbian evidence, while the server-side magnetizations increase once the corresponding archetypes receive sufficient exposure from informative clients. This confirms that reconstruction depends not only on the nominal presence of an archetype in the federation, but also on its effective exposure across reliable clients.

The bottom block considers the complementary *dataset-quality alteration* regime. Here, the exposure is kept fixed and non-zero for all archetypes and clients, so that all clients continuously observe all archetypal classes. The perturbation is instead introduced through time-dependent variations of $r_{c,\mu}(t)$, which mimic transient corruptions or adversarial attacks affecting the quality of the local batches. In this case, drops in $r_{c,\mu}(t)$ induce a corresponding reduction of the adaptive plasticity weights, thereby limiting the influence of unreliable local correlators on the server aggregation. At the same time, the server-side magnetizations remain stable whenever enough high-quality information is still available from the remaining clients.

Overall, Fig. 8 shows that the entropy-based controller reacts to two distinct sources of heterogeneity in the intended way. When class availability changes, the reconstruction dynamics tracks the effective exposure of each archetype. When data quality deteriorates, the controller suppresses unreliable client contributions without requiring labels or direct access to the ground-truth archetypes. This demonstrates that local sign-agreement entropy, computed independently by each client from its Hebbian operator and the server reconstruction, provides a useful unsupervised signal for balancing plasticity and stability in decen-

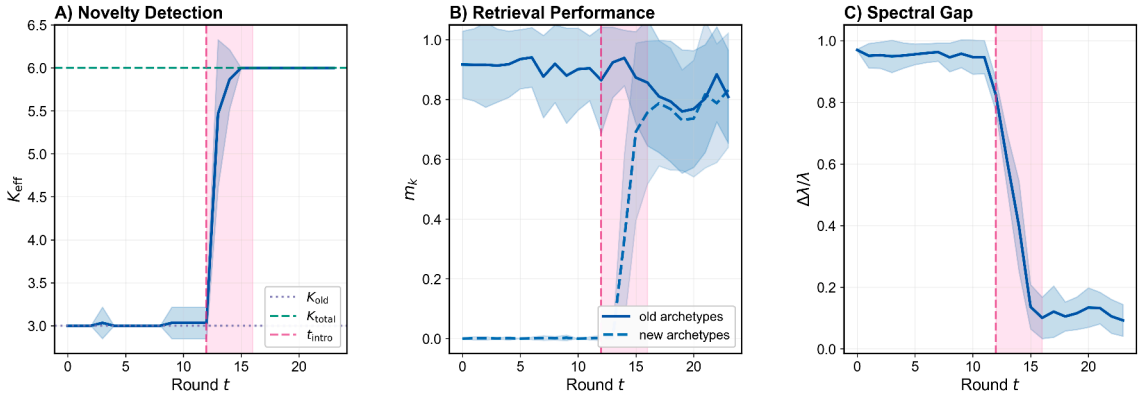


Fig. 9. Detecting and integrating novel archetypes in a federated setting. Setup: $L = 3$ clients, $N = 400$, $T = 24$, example quality $r = 0.8$, and M_c^t fixed across rounds. The federation starts with $K_{\text{old}} = 3$ archetypes; $K_{\text{new}} = 3$ additional archetypes are introduced at $t_{\text{intro}} = 12$ via a four-round ramp (vertical dashed line and shaded interval). Curves are averaged over 30 random seeds (shaded regions: standard deviation). *(Left)* Post-sharpening effective dimensionality $\hat{K}_{\text{spec}}(t)$ estimated by eigenvalue thresholding (4.28). The system rapidly transitions from $\hat{K} \simeq 3$ to $\hat{K} \simeq 6$ within a few rounds after the introduction, indicating low-latency novelty detection. *(Center)* Magnetizations (overlaps (3.7)) for old archetypes (solid) and newly introduced ones (dashed). New archetypes are acquired during/shortly after the ramp, while old archetypes exhibit only a mild transient dip and recover, consistent with targeted (rather than indiscriminate) plasticity. *(Right)* Relative spectral gap at the K_{old} boundary, $(\lambda_{K_{\text{old}}} - \lambda_{K_{\text{old}}+1})/\lambda_{K_{\text{old}}}$, computed from the (sharpened) server operator. The gap collapses during the ramp, signaling the breakdown of the K_{old}-dimensional hypothesis, and stabilizes once the representation has expanded to $K_{\text{tot}} = 6$ directions.

tralized learning. The mechanism therefore extends beyond the simple good-client versus attacker scenario, and naturally applies to heterogeneous federations with partial observations, time-dependent noise, client-specific attacks, and distributional drift.

We next move from drift in prevalences to genuine *class emergence*, where the representational subspace must expand when new archetypes become exposed.

5.2. Continual learning with archetype emergence

We now intensify non-stationarity by letting new archetypes enter the federation mid-training. The system starts with $K_{\text{old}} = 3$ archetypes. At round $t_{\text{intro}} = 12$, $K_{\text{new}} = 3$ additional archetypes are introduced through a four-round ramp in the mixture, after which all $K_{\text{tot}} = K_{\text{old}} + K_{\text{new}} = 6$ modes have non-zero exposure. Clients follow the same pipeline without replay; therefore, successful adaptation hinges on engaging plasticity when novel directions become informative, while maintaining stability on previously consolidated attractors.

Fig. 9 summarizes the key dynamics. The post-sharpening effective-rank estimate $\hat{K}_{\text{spec}}(t)$ increases from 3 to 6 within a few rounds after t_{intro}, reflecting the exposure-driven BBP mechanism: as the new archetypes gain exposure, their spikes become detectable and contribute additional outliers/eigenvectors (Theorem 2). The overlap curves (center) show that newly introduced archetypes are learned quickly, while previously consolidated ones remain retrievable, exhibiting only a transient decrease around the transition. Finally, the spectral-gap diagnostic (right) provides an architecture-agnostic signature of novelty: the gap shrinks when the old subspace ceases to be sufficient and stabilizes again once the representation has expanded.

This experiment is also the empirical test of the scope statement of Remark 6. From $t = 1$ onward the server operator is feedback-dependent: by Proposition 1 it carries, in addition to the fresh plastic evidence, the finite-rank term $\boldsymbol{F}_{t-1}$ built from the $\hat{K}_{\text{rec}}(t-1) = 3$ archetypes consolidated before t_{intro}. Two predictions of the decomposition are visible in the data. First, by (P3) the newly introduced archetypes receive no feedback contribution, so their detection is governed by the fresh plasticity-weighted exposure alone: the rank estimate indeed expands only after the ramp has pushed π_t^{eff} above the threshold (4.5), and not at the nominal introduction round. Second, by the retention condition (4.24) the three consolidated archetypes remain supercritical throughout the transition, which is what the mild and transient dip of the solid magnetization curves shows: plasticity is engaged on the new directions without dissolving the old outliers. What we regard as the a posteriori justification of the use of Theorem 2 as an approximation for $t > 0$ is therefore the qualitative pattern displayed by the data: the expansion of $\hat{K}_{\text{spec}}(t)$ is systematically delayed with respect to the nominal introduction round t_{intro}, as required by a threshold crossing in the plasticity-weighted exposure, and it is not accompanied by the loss of the previously consolidated outliers.

Taken together, these results support the central claim of the paper: the system expands its representational subspace precisely when new directions become informative (exposure-driven detectability), and it does so with limited interference thanks to the stability–plasticity control implemented by $w_c(t)$.

5.3. Spectral-only ablation of the LAM factorization

We next isolate the contribution of the LAM factorization stage from that of the preceding spectral extraction. As discussed in Section 2.1.1, the dominant eigenspace of the sharpened operator $\hat{\boldsymbol{J}}^{KS}$ spans the detectable archetypal subspace, but its orthonormal eigenvectors generally correspond to linear combinations of the latent patterns rather than to the individual Boolean archetypes.

To assess whether this spectral representation is sufficient by itself, we introduce a spectral-only control without LAM refinement. At the final communication round, let

$$\boldsymbol{v}_1^{(T)}, \dots, \boldsymbol{v}_K^{(T)}$$

be the K dominant eigenvectors of the sharpened server operator $\hat{\boldsymbol{J}}^{KS,(T)}$. We directly convert them into Boolean candidates through component-wise sign binarization:

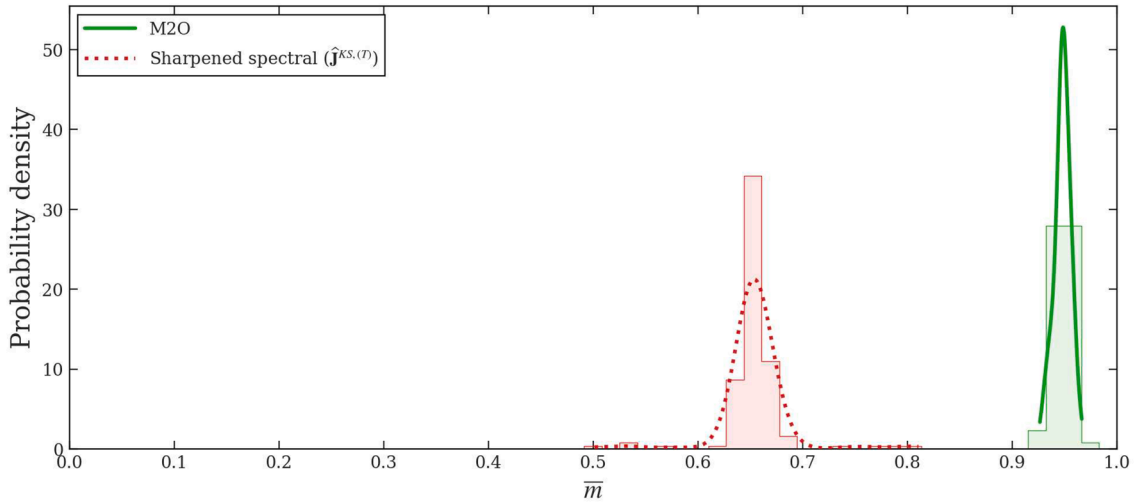


Fig. 10. Distribution over 150 independent federated realizations of the final mean best-match magnetization $\overline{m}$ defined in Eq. (5.3). The green distribution reports the archetypes reconstructed by the complete M2O pipeline, including synthetic-mixture generation and LAM refinement. The red distribution reports the spectral-only control, obtained by directly binarizing the K dominant eigenvectors of the final sharpened operator $\hat{J}^{KS,(T)}$, without applying the LAM dynamics. Histograms are normalized to probability density and the superimposed curves are kernel-density estimates. The marked shift of the full M2O distribution toward larger magnetizations shows that spectral subspace identification alone is not sufficient to recover the individual Boolean archetypes.

$$\hat{\xi}_k^{\mathrm{spec}} = \mathrm{sign}\left(v_k^{(T)}\right), \qquad k = 1, \dots, K. \tag{5.1}$$

This control uses neither the synthetic-mixture generation nor the subsequent LAM dynamics. It therefore shares the same final sharpened spectral input as the complete pipeline, while removing the archetype-factorization stage.

For a generic set of reconstructed candidates $\{\hat{\xi}^{\nu}\}_{\nu=1}^{K}$, we form the magnetization matrix

$$M_{\mu\nu} = \frac{1}{N}\left|(\xi^{\mu})^{\top}\hat{\xi}^{\nu}\right|, \qquad \mu, \nu = 1, \dots, K. \tag{5.2}$$

Consistently with the reconstruction metric in Eq. (3.7), each ground-truth archetype is associated with its best reconstructed candidate, and the final score for one realization is

$$\overline{m} = \frac{1}{K}\sum_{\mu=1}^{K} \max_{\nu=1,\dots,K} M_{\mu\nu}. \tag{5.3}$$

We compare the spectral-only candidates of Eq. (5.1) with the final archetypes produced by the complete M2O pipeline. The experiment is repeated over 150 independent federated realizations, using the same data-generation parameters and 5 examples per archetype, per client, and per communication round. Within each realization, the spectral-only and full reconstructions are evaluated on the same latent archetypes and the same final server operator.

As shown in Fig. 10, direct binarization of the dominant spectral directions yields substantially lower pattern-level magnetizations than the complete M2O reconstruction. The spectral-only baseline identifies informative directions, but those directions need not coincide with the individual archetypes because eigenvectors provide an orthogonal basis of the signal subspace and may represent rotations or linear combinations of the planted patterns. The LAM dynamics instead explores nonlinear mixtures within this subspace and disentangles them into distinct Boolean configurations. The comparison therefore provides a direct ablation of the LAM stage and confirms that its role is not merely to improve spectral detection, but to perform the actual archetype-level factorization.

5.4. Baselines under dirichlet client heterogeneity.

To assess the benefit of federated consolidation under strong statistical heterogeneity, we compare the proposed protocol with a local-only/no-feedback baseline. We consider a Dirichlet-partitioned setting in which clients have different archetype exposure probabilities $\pi_{t,c}(\mu)$ and therefore observe only partial and imbalanced views of the latent dataset. The first row of Fig. 11 reports the corresponding client–archetype exposure maps.

In the federated protocol, each client sends to the server the operator

$$J_c^{\mathrm{send}}(t) = w_c(t)\widetilde{J}_c(t) + \left(1 - w_c(t)\right)\hat{J}_s(t-1),$$

where $\widetilde{J}_c(t)$ is the Hebbian operator estimated from the current local batch and $\hat{J}_s(t-1)$ is the operator reconstructed by the server at the previous round. The middle panel of Fig. 11 shows the normalized Frobenius error of the client-side operators and of the server-side reconstruction with respect to the ground-truth Hebbian matrix $J^{\star}$. The black curve always denotes the federated server reconstruction obtained after aggregation and LAM-based factorization.

As a baseline, we remove the feedback channel from the server to the clients. In this local-only setting, each client accumulates only its own local Hebbian evidence across rounds and never receives the server reconstruction. The bottom panel of Fig. 11 reports the resulting client-side errors, while keeping the same federated server curve as a reference.

The results show that no isolated client can match the server-side reconstruction. This holds both within the federated protocol and, more markedly, in the local-only baseline. In the absence of server feedback, clients remain strongly biased by their restricted archetype exposure and the gap with respect to the federated server becomes substantially larger. Thus, the improvement is not merely due to accumulating more local Hebbian samples over time, but to the many-to-one federated consolidation mechanism, which transfers archetypal information across clients through the aggregated server reconstruction.

5.5. Structured-data extension: Federated Boolean–Gaussian RBM on MNIST

To complement the associative-memory experiments discussed above, we also consider a structured-data benchmark based on the MNIST dataset. The goal of this experiment is twofold: first, it provides an external federated baseline based on an established unsupervised energy-based model; second, it allows us to test whether our adaptive aggregation principle remains beneficial when latent prototypical representations are not explicitly prescribed by a teacher model, but must instead be learned directly from structured data.

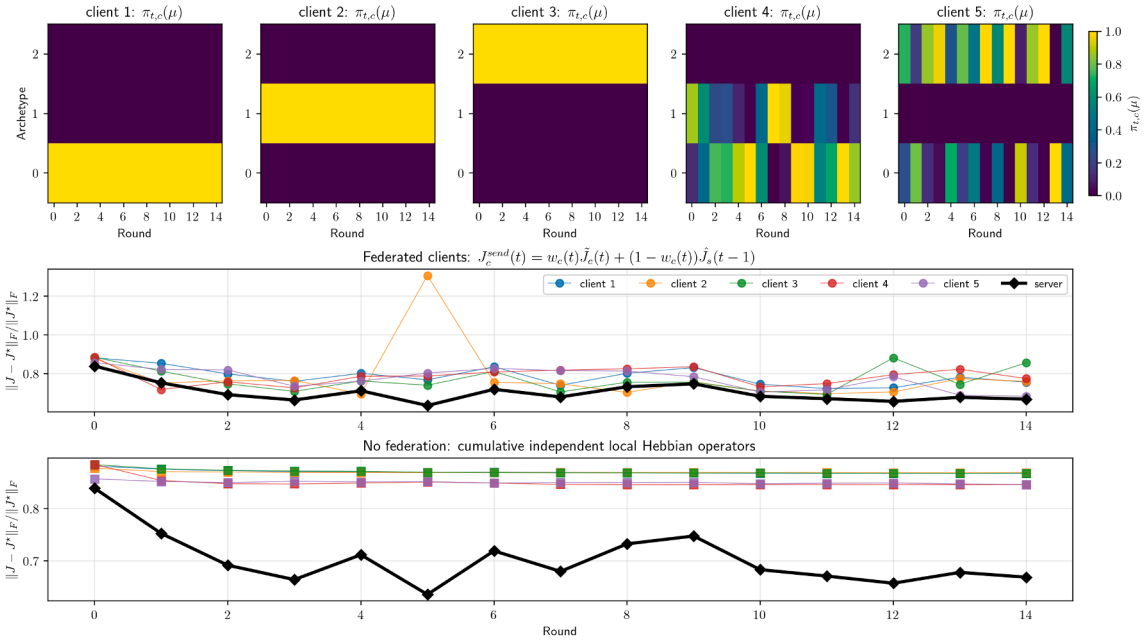


Fig. 11. The first row reports the archetype exposure probabilities $\pi_{t,c}(\mu)$ for each client c across communication rounds. The experiment is designed to stress the federation under strong non-i.i.d. conditions: some clients observe only one archetype, while others receive mixed, unstable, or corrupted batches. The middle panel shows the normalized Frobenius reconstruction error $\|J - J^\star\|_F / \|J^\star\|_F$ for the proposed federated protocol, in which each client sends to the server the convex combination $J_c^{\text{send}}(t) = w_c(t)\widetilde{J}_c(t) + \big(1 - w_c(t)\big)\widehat{J}_s(t-1)$. Colored curves correspond to the client-side operators generated within the federated protocol, while the black curve reports the server-side reconstruction obtained after aggregation and LAM-based factorization. The bottom panel reports a local-only/no-feedback baseline for the clients: each client independently accumulates only its own local Hebbian evidence and never receives the reconstructed server operator. The black curve is kept fixed as the same federated server reference, not as a local-only server. This comparison shows that no individual client matches the federated server reconstruction, either when clients participate in the federated protocol or when they are left isolated. Moreover, in the local-only setting the gap with respect to the federated server is substantially larger, confirming that the gain is due to the many-to-one federated consolidation mechanism rather than to local accumulation alone.

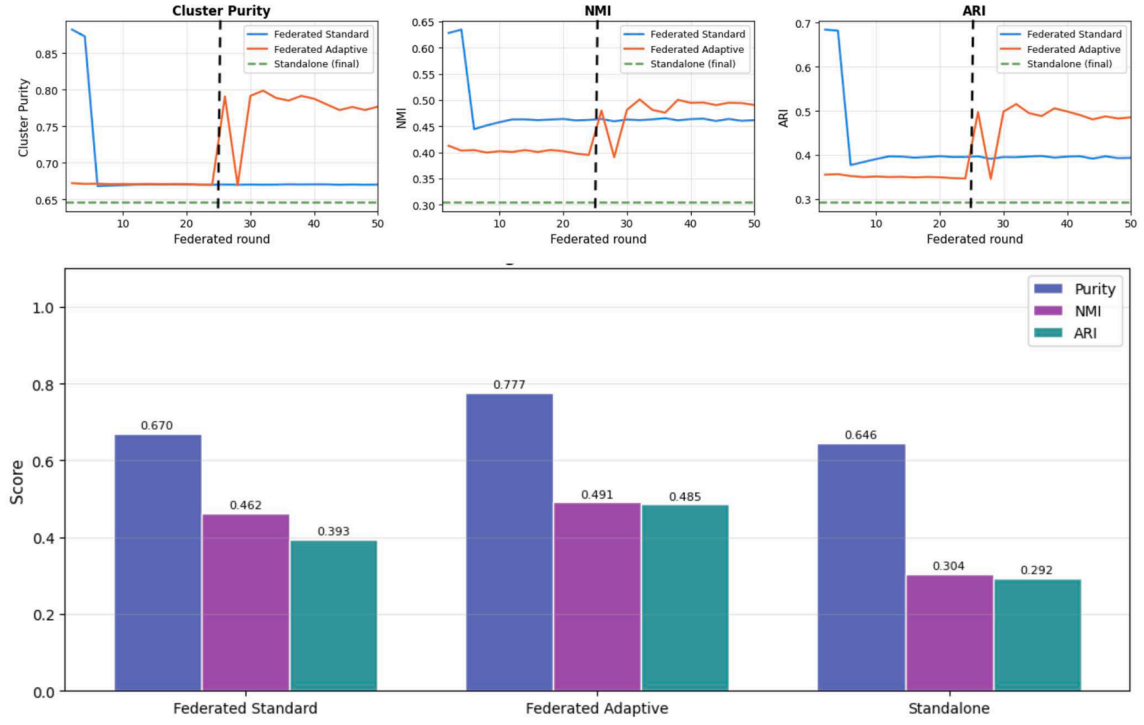


Fig. 12. The top row reports the evolution of cluster purity, normalized mutual information (NMI), and adjusted Rand index (ARI) across communication rounds for standard FedAvg aggregation and for the adaptive quality-aware variant. The dashed vertical line marks the onset of the streaming/novelty phase. The bottom panel compares the final scores of three regimes: standard federated RBM training, adaptive federated RBM training, and a standalone RBM trained without federation. Adaptive aggregation yields the best final performance, reaching purity 0.777, NMI 0.491, and ARI 0.485, compared with 0.670, 0.462, and 0.393 for standard federated training, and 0.646, 0.304, and 0.292 for the standalone baseline. This indicates that quality-aware aggregation is especially beneficial under heterogeneous and streaming client updates, improving the latent clustering structure without using labels during training.

The connection with our main framework is conceptual and statistical-mechanical. As recalled in Section 2, there is a well-known duality between Hopfield networks and Boolean–Gaussian Restricted Boltzmann Machines (RBMs). By marginalizing over the Gaussian hidden variables, the visible marginal of the RBM becomes equivalent to a Hopfield-type model with an effective low-rank interaction matrix.

We exploit precisely this dual representation to extend our federated protocol. The energy function of the Boolean–Gaussian RBM is defined as:

$$\mathcal{H}_{\mathrm{RBM}}(\mathbf{v},\mathbf{h}\mid W,\beta=1)=\frac{1}{2}\sum_{\mu=1}^{K}h_\mu^2-\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\sum_{\mu=1}^{K}W_{i\mu}v_ih_\mu$$

where the visible layer $\mathbf{v}\in\{\pm1\}^N$ contains binarized MNIST images ($N=784$), the continuous Gaussian hidden layer $\mathbf{h}\in\mathbb{R}^K$ learns latent directions acting as data-driven archetypal features, and $W\in\mathbb{R}^{N\times K}$ is the weight matrix trained via Contrastive Divergence. Consequently, this experiment does not replace the LAM-based factorization mechanism, but rather operationalizes the same federated adaptive-learning principle in a scenario where archetypes are implicit.

True dataset labels are never used during RBM training. They are employed solely to construct controlled non-i.i.d. client partitions and to evaluate the post-hoc clustering quality of the learned latent representations. For this evaluation, we compare three regimes: *i*) *Standalone*: A Boolean–Gaussian RBM trained locally without federation. *ii*) *Standard Federated (FedAvg)*: An RBM trained with traditional sample-size proportional aggregation. *iii*) *Adaptive Federated*: Our proposed approach, where client updates are reweighted according to local quality diagnostics.

Full implementation details regarding the local training procedure, the definition of the adaptive quality scores, and the complete hyperparameter configuration are deferred to Appendix B.

Fig. 12 demonstrates that the adaptive federated Boolean–Gaussian RBM consistently improves final clustering quality compared to both baselines. The performance gap becomes particularly evident after the onset of the streaming/novelty phase. While the standard FedAvg trajectory plateaus after the initial transient, the adaptive protocol continues to increase in purity and ARI. This indicates that quality-aware aggregation successfully exploits informative local updates while suppressing less coherent ones.

By the final round, adaptive aggregation improves over standard FedAvg across all metrics (from 0.670 to 0.777 in purity, from 0.462 to 0.491 in NMI, and from 0.393 to 0.485 in ARI). The standalone RBM baseline performs significantly worse, confirming both the necessity of federation in this partitioned setting and the distinct advantage of the adaptive mechanism over naive sample-size averaging.

Overall, this experiment reinforces the main claim of the paper from a complementary perspective. Even in a conventional unsupervised energy-based setting, adaptive client weighting improves representation learning when faced with heterogeneous federated data. Consistent with our associative-memory results, this confirms that in streaming non-i.i.d. regimes, client quality cannot be treated as uniform; aggregation rules that dynamically assess the reliability of local updates yield significantly more stable and informative global representations.

6. Conclusion

We proposed a federated associative-memory framework for archetype learning under heterogeneous data, client drift, and streaming novelty, in regimes that are naturally modeled as *independent but non-identically distributed* (i.n.i.d.) across clients. By communicating compact Hebbian operators rather than raw samples, clients contribute data-local aggregate correlation statistics that the server can aggregate and factorize to reconstruct global archetypes. This communication protocol avoids raw-example transmission, but the transmitted operators are still data-dependent and should not be interpreted as providing a formal privacy guarantee.

On the theoretical side, framing aggregation as a low-rank-plus-noise spectral problem enables random-matrix predictions of detectability and robustness: global archetypes emerge reliably once signal eigenstructure separates from heterogeneity-induced noise, providing a principled view of when federated consolidation is feasible under independence and heterogeneity. Empirically, our method exhibits improved archetype reconstruction and retrieval stability across heterogeneous clients and drifting regimes, and the proposed entropy-based controller provides a practical mechanism to balance stability and plasticity without centralized replay. In addition to the controlled associative-memory experiments, the federated Boolean–Gaussian RBM experiment on MNIST (Section 5.5 and Appendix B) illustrates how the same federated principle can be transferred to structured datasets where archetypes are not specified by a teacher model, but must be inferred as latent representations from data. Although MNIST is originally designed as a classification benchmark rather than as an archetype-reconstruction task, the RBM results provide a first indication that adaptive federated energy-based learning can extract meaningful prototypical structure from real, structured data.

Despite its theoretical grounding and empirical validation, the proposed framework entails several limitations that we discuss openly:

- **Behavior under extreme non-i.i.d. distributions.** The spectral detectability of an archetype is governed by its round-level exposure $\pi_t(\mu)$ relative to the BBP threshold $\kappa_\mu>\sqrt{q}$ (Theorem 2). In strongly imbalanced regimes where certain archetypes have very low global exposure, the corresponding spikes remain below the detection threshold for many rounds, leading to delayed or failed reconstruction. The global coverage assumption $\bigcup_c K_c=\{1,\dots,K\}$ guarantees eventual detectability but does not bound the detection latency as a function of the imbalance ratio. The entropy-based controller partially mitigates this by protecting already consolidated archetypes against forgetting, but it cannot accelerate the initial detection of chronically under-exposed archetypes. We regard this as a genuine failure mode of the approach in the extreme non-i.i.d. regime.
- **Status of the spectral theory in the feedback-driven phase.** The random-matrix analysis is exact only at initialization, where the aggregated operator is built from genuinely independent client batches. From the second round onward the estimator carries the server feedback through the convex update (3.6), and the samples entering successive rounds are temporally correlated. Proposition 1 shows that this feedback enters as a positive semidefinite, finite-rank, past-measurable and signal-aligned term, so that bulk confinement remains exact, the detection threshold is unchanged for homogeneous plasticity weights and inflated by the explicit factor of Corollary 1 otherwise, and a genuinely novel archetype receives no feedback contribution at all. The residual ingredients—the data dependence of the weights $w_c(t)$, the quenched nature of the conditional statements, and the leading-order superposition of the additive and multiplicative finite-rank perturbations—are quantified in Appendix E.3 but not proved. For $t>0$ the BBP predictions should therefore be read as a controlled approximation, validated a posteriori by the experiments of Section 5. Establishing them rigorously would require a local law for a recursively defined, time-dependent random-matrix ensemble, presumably through free-probability control of the cross terms between the fresh and the consolidated components; we regard this as a natural and non-trivial continuation of the present work.
- **Computational overhead of server-side factorization.** The dominant computational cost at the server is the eigendecomposition of the sharpened operator $\hat{\boldsymbol{J}}^{KS}$, which scales as $O(N^3)$ per communication round. The iterative unlearning procedure (2.26) contributes $O(N^2)$ per iteration, and the subsequent LAM dynamics scale as $O(m\cdot T_{\mathrm{LAM}}\cdot N^2)$, where m is the number of initial mixing states and T_{LAM} is the number of synchronous updates. In the numerical experiments presented here ($N\lesssim 1000$), all these operations are man-

ageable on standard hardware. For high-dimensional feature spaces (large N) or many archetypes (large K), however, the eigendecomposition becomes a bottleneck. Approximate spectral methods such as randomized SVD could alleviate this cost, though their interaction with the LAM dynamics and the detectability thresholds would need to be analyzed carefully.

- **Communication cost.** At each round every client uploads one symmetric operator $\boldsymbol{J}_c^{(t)} \in \mathbb{R}^{N\times N}$, carrying $N(N+1)/2$ independent entries, and receives back one operator $\widehat{\boldsymbol{J}}_s^{(t)}$ of the same size, for a total of $O(LN^2)$ scalars exchanged per round. This cost is *independent of the number of archetypes* K and of the number of local examples, and is comparable to that of gradient-based federated methods when N matches the model dimension. For large N it can be reduced by transmitting a low-rank or sparsified encoding of the operators, at the price of a controlled spectral degradation that interacts with the detectability thresholds of Theorem 2; a quantitative study of this communication–accuracy trade-off is left to future work.

These limitations naturally point to several open directions. First, extending the reconstruction step beyond linear Hebbian operators to kernelized, energy-based, or deep-feature variants could enlarge the class of representable archetypes while preserving the operator-communication interface. In this direction, the RBM experiment suggests a promising path toward federated representation-learning models in which the archetypal variables are not fixed in advance, but learned jointly with the latent feature space. Second, integrating formal privacy guarantees (e.g., differential privacy noise calibrated at the operator level) and analyzing the induced spectral degradation would clarify the privacy–utility frontier. Third, asynchronous and partial participation regimes may be studied through time-varying random-matrix models, connecting communication constraints to phase transitions in archetype detectability.

We hope that this work stimulates further connections between federated learning, associative memories, energy-based models, and spectral theory for continual, data-local representation learning.

Data and code availability

No new experimental data were collected for this work. The code is available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Andrea Alessandrelli: Writing – review & editing, Writing – original draft, Supervision, Conceptualization; **Fabrizio Durante:** Writing – review & editing, Supervision; **Andrea Ladiana:** Writing – review & editing, Software, Formal analysis, Data curation; **Andrea Lepre:** Writing – review & editing, Software, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Algorithms and pseudocode

See Algorithms 1–3.

Algorithm 1: Pattern reconstruction in LAM: pre-processing and candidate generation.

Parameters: *Number of layers L, neurons per layer N; number of nonlinear mixtures m; number of parallel dynamic steps N_p; inverse thermal noise β; eigenvalue threshold τ_{spec}; KS tolerance $\Delta_{\min} = 10^{-4}$; KS step ϵ as in (2.26).*
Input: Hebbian coupling matrix $\boldsymbol{J}^{\text{Hebb}} = \{J_{ij}^{\text{Hebb}}\}_{i,j=1,\dots,N}$.
Output: Candidate reconstructed patterns $\mathcal{V} = \{\boldsymbol{\zeta}\}$.

1. Spectral pre-processing and initialization
Initialize KS iteration:;
$\boldsymbol{J} \leftarrow \boldsymbol{J}^{\text{Hebb}}$, $\Delta \leftarrow 10^4$, $k \leftarrow 0$;
while $\Delta \geq \Delta_{\min}$ **do**
  Update KS-renormalized coupling:
  $\boldsymbol{J}_{\text{new}} \leftarrow \left(1 + \frac{\epsilon}{1+\epsilon k}\right)\boldsymbol{J} - \left(\frac{\epsilon}{1+\epsilon k}\right)\boldsymbol{J}^2$;
  Compute update magnitude: $\Delta \leftarrow \|\boldsymbol{J}_{\text{new}} - \boldsymbol{J}\|$;
  Set $\boldsymbol{J} \leftarrow \boldsymbol{J}_{\text{new}}$ and increment $k \leftarrow k+1$;
Set $\widehat{\boldsymbol{J}}^{KS} \leftarrow \boldsymbol{J}$;
Compute $\widehat{\boldsymbol{J}}^{KS}$ eigendecomposition;
Retain the $\widehat{K}_{\text{spec}}$ eigenvectors $\{\tilde{\boldsymbol{x}}^{\delta}\}_{\delta=1}^{\widehat{K}_{\text{spec}}}$ associated with eigenvalues $\lambda_\delta > \tau_{\text{spec}}$ (see (4.28));
Generate m nonlinear initial configurations:;
for $\alpha \leftarrow 1$ **to** m **do**
  Sample coefficients $c_\delta^\alpha \sim \mathcal{N}(0,1)$ or Rademacher for $\delta = 1, \dots, \widehat{K}_{\text{spec}}$;
  Construct mixture and binarize: $\boldsymbol{x}^\alpha \leftarrow \text{sign}\Big(\sum_{\delta=1}^{\widehat{K}_{\text{spec}}} c_\delta^\alpha\, \tilde{\boldsymbol{x}}^\delta\Big)$;

2. LAM dynamics and candidate collection
Initialize candidate set $\mathcal{V} \leftarrow \emptyset$;
for $\alpha \leftarrow 1$ **to** m **do**
  Initialize all layers/modules of the LAM model with input $\boldsymbol{x}^\alpha$ and set $t \leftarrow 0$;
  while $t < N_p$ **do**
    Update network state according to (2.21) with thermal noise β^{-1};
    $t \leftarrow t+1$;
  Extract final-layer configuration(s) as reconstructed candidate(s) $\boldsymbol{\zeta}$;
  Append candidate(s): $\mathcal{V} \leftarrow \mathcal{V} \cup \{\boldsymbol{\zeta}\}$;
return $\mathcal{V}$;

Appendix B. Implementation details for the federated Boolean–Gaussian RBM baseline

This appendix reports the technical details of the federated Boolean–Gaussian RBM baseline introduced in Section 5.5. The Hopfield–RBM duality has already been recalled in Section 2; here we specify the experimental parametrization, the local learning rule, the federated ag-

Algorithm 2: Pattern reconstruction in LAM: pruning and acceptance filtering.

Parameters: *Neurons per layer N; overlap threshold δ_{dup} (default 0.5); acceptance threshold τ_{acc}.*

Input: Candidate set $\mathcal{V} = \{\boldsymbol{\zeta}\}$; KS-renormalized coupling matrix $\hat{\boldsymbol{J}}^{KS}$.

Output: Accepted reconstructed patterns $\mathcal{Z} = \{\hat{\boldsymbol{\xi}}\}$.

1. Duplicate removal via overlap threshold

Initialize pruned set $\widetilde{\mathcal{V}} \leftarrow \emptyset$;

foreach $\boldsymbol{\zeta} \in \mathcal{V}$ **do**

  `is_duplicate` $\leftarrow$ `false`;

  foreach $\boldsymbol{\zeta}' \in \widetilde{\mathcal{V}}$ **do**

    Compute overlap:

$$q(\boldsymbol{\zeta}, \boldsymbol{\zeta}') \leftarrow \frac{1}{N} \sum_{i=1}^{N} \zeta_i \, \zeta_i'$$

    if $q(\boldsymbol{\zeta}, \boldsymbol{\zeta}') \geq \delta_{\text{dup}}$ **then**

      `is_duplicate` $\leftarrow$ `true`;

      break;

  if $\neg$ *is_duplicate* **then**

    $\widetilde{\mathcal{V}} \leftarrow \widetilde{\mathcal{V}} \cup \{\boldsymbol{\zeta}\}$;

2. Acceptance test via quadratic score

Initialize accepted set $\mathcal{Z} \leftarrow \emptyset$;

foreach $\boldsymbol{\zeta} \in \widetilde{\mathcal{V}}$ **do**

  Compute pattern score under $\hat{\boldsymbol{J}}^{KS}$:

$$p(\boldsymbol{\zeta}) \leftarrow \frac{1}{N} \, \boldsymbol{\zeta}^T \, \hat{\boldsymbol{J}}^{KS} \, \boldsymbol{\zeta}$$

  if $p(\boldsymbol{\zeta}) \geq \tau_{\text{acc}}$ **then**

    $\hat{\boldsymbol{\xi}} \leftarrow \boldsymbol{\zeta}$;

    $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{\hat{\boldsymbol{\xi}}\}$;

return $\mathcal{Z}$;

Algorithm 3: Federated M2O pipeline.

Parameters: *Number of clients L; neurons N; communication rounds T; batch sizes $\{M_c^t\}$; blending weights $\{w_c(t) \in [0,1]\}$; server factorization routine* LAM$(\cdot)$ *(Sec.2.1.1).*

Input: For each client c, a stream of *local* batches $\{\boldsymbol{\eta}_c^{(t)}\}_{t=0}^{T}$, with $\boldsymbol{\eta}_c^{(t)} = \{\boldsymbol{\eta}_c^{(t),m}\}_{m=1}^{M_c^t}$ and $\boldsymbol{\eta}_c^{(t),m} \in \{-1,+1\}^N$ (never shared).

Output: Server reconstructed patterns $\{\hat{\boldsymbol{\xi}}_{s,\mu}^{(T)}\}_{\mu=1}^{\hat{K}_{\text{rec}}(T)}$.

Initialization (client-side, round $t = 0$)

foreach $c \leftarrow 1$ **to** L ***in parallel*** **do**

  Compute local synaptic matrix from the first batch:;

$$\left(J_c^{(0)}\right)_{ij} \leftarrow \frac{1}{N \, M_c^0} \sum_{m=1}^{M_c^0} \left(\eta_c^{(0),m}\right)_i \left(\eta_c^{(0),m}\right)_j$$

  Send $\boldsymbol{J}_c^{(0)}$ to server;

for $t \leftarrow 0$ **to** $T-1$ **do**

  Server aggregation

  Receive $\{\boldsymbol{J}_c^{(t)}\}_{c=1}^{L}$ and compute:

$$\left(J_s^{(t)}\right)_{ij} \leftarrow \frac{1}{L} \sum_{c=1}^{L} \left(J_c^{(t)}\right)_{ij}$$

  Server factorization and re-encoding

  Using LAM model to reconstruct patterns underlied the $\boldsymbol{J}_s^{(t)}$:

$$\{\hat{\boldsymbol{\xi}}_{s,\mu}^{(t)}\}_{\mu=1}^{\hat{K}_{\text{rec}}(t)} \leftarrow \text{LAM}\left(\boldsymbol{J}_s^{(t)}\right)$$

  Compute server synaptic matrix

$$\left(\hat{J}_s^{(t)}\right)_{ij} \leftarrow \frac{1}{N} \sum_{\mu=1}^{\hat{K}_{\text{rec}}(t)} \left(\hat{\xi}_{s,\mu}^{(t)}\right)_i \left(\hat{\xi}_{s,\mu}^{(t)}\right)_j;$$

  Broadcast $\hat{\boldsymbol{J}}_s^{(t)}$ to all clients;

  Client fusion (new batch at round t+1) and upload

  foreach $c \leftarrow 1$ **to** L ***in parallel*** **do**

    Compute operator from the *new* local batch:

$$\left(J_c^{\text{loc}}\right)_{ij} \leftarrow \frac{1}{N \, M_c^{t+1}} \sum_{m=1}^{M_c^{t+1}} \left(\eta_c^{(t+1),m}\right)_i \left(\eta_c^{(t+1),m}\right)_j$$

    Fuse local evidence with server reconstruction after server preliminary spectral-rank diagnostic:

$$\boldsymbol{J}_c^{(t+1)} \leftarrow w_c(t{+}1) \, \boldsymbol{J}_c^{\text{loc}} + \left(1 - w_c(t{+}1)\right) \hat{\boldsymbol{J}}_s^{(t)}$$

    Upload $\boldsymbol{J}_c^{(t+1)}$ to server;

return $\{\hat{\boldsymbol{\xi}}_{s,\mu}^{(T)}\}_{\mu=1}^{\hat{K}_{\text{rec}}(T)}$;

gregation protocol, and the post-hoc clustering evaluation used for the MNIST experiment.

Data preprocessing. Each MNIST image is normalized to $[0,1]$ and flattened into a vector of dimension $N = 784$. A pixel value $x_i \in [0,1]$ is mapped to a Boolean visible spin by

$$\sigma_i = \text{sign}(x_i - \tau_{\text{bin}}) \qquad \tau_{\text{bin}} = 0.5.$$

Labels are never used during RBM training. They are used only to construct controlled non-i.i.d. client partitions and to evaluate the learned latent representations after training.

Local contrastive divergence (CD-k) training. Consider a mixed Bernoulli-Gaussian RBM, where the visible layer consists of bipolar Boolean units $\mathbf{v} \in \{\pm 1\}^N$ and the hidden layer consists of continuous Gaussian units $\mathbf{h} \in \mathbb{R}^K$. The network is thus parameterized by the weight matrix $W \in \mathbb{R}^{N \times K}$ with no visible or hidden bias presence.

The exact gradient update rule for the weights aims to align the data distribution with the model distribution. In the federated setup, each client approximates this gradient locally using the Contrastive Divergence algorithm with k steps (CD-k) evaluated on mini-batches.

Let $V_0 \in \{\pm 1\}^{B \times N}$ be a mini-batch of size B containing the empirical training data. In the *positive phase*, the network computes the matrix of conditional hidden means $M_0 \in \mathbb{R}^{B \times K}$, which captures the data-driven correlations:

$$M_0 = \sqrt{\frac{\beta}{N}} V_0 W. \tag{B.1}$$

The *negative phase* initializes the Gibbs chain at the data (V_0) and alternates conditional sampling across the layers to estimate the model's equilibrium distribution. For $\ell = 1, \ldots, k$, the continuous hidden states $H_\ell \in \mathbb{R}^{B \times K}$ and the bipolar visible states $V_\ell \in \{\pm 1\}^{B \times N}$ are sampled as:

$$(H_\ell)_{b,\mu} \sim \mathcal{N}\left(\left(\sqrt{\frac{\beta}{N}} V_{\ell-1} W\right)_{b,\mu}, 1\right), \tag{B.2}$$

$$P\left((V_\ell)_{b,i} = +1 \mid H_\ell\right) = \sigma\left(2\left(\sqrt{\frac{\beta}{N}} H_\ell W^\top\right)_{b,i}\right), \tag{B.3}$$

where $\sigma(\cdot)$ is the logistic sigmoid function, $b \in [1, B]$ is the batch index, $\mu \in [1, K]$, and $i \in [1, N]$.

After k Gibbs steps, we obtain the reconstructed visible batch V_k. To reduce the variance of the gradient estimator, the corresponding nega-

tive phase hidden states are replaced by their conditional means:

$$M_k = \sqrt{\frac{\beta}{N}} V_k W. \tag{B.4}$$

The CD-k approximation of the log-likelihood gradient over the mini-batch is then computed in a fully vectorized form as the difference between the positive phase (data correlations) and the negative phase (model hallucinations):

$$\nabla_W = \sqrt{\frac{\beta}{N}} \frac{1}{B} \left(V_0^\top M_0 - V_k^\top M_k \right). \tag{B.5}$$

Finally, the local model weights are updated using stochastic gradient ascent equipped with momentum and weight decay. Let $U_W \in \mathbb{R}^{N \times K}$ be the momentum buffer; the local update equations are:

$$G_W = \nabla_W - \lambda_{\text{wd}} W, \tag{B.6}$$

$$U_W = \mu_{\text{mom}} U_W + \varepsilon G_W, \tag{B.7}$$

$$W = W + U_W, \tag{B.8}$$

where λ_{wd} is the weight decay coefficient, μ_{mom} is the momentum factor, and ε is the learning rate.

Federated aggregation. At communication round t, the server stores a global weight matrix

$$W_s^{(t)} \in \mathbb{R}^{N \times K}. \tag{B.9}$$

It samples without replacement a subset S_t of participating clients. Each client $c \in S_t$ receives $W_s^{(t)}$, initializes its local RBM with this matrix, performs E local epochs of CD-k on its private dataset D_c, and returns the local weight update delta:

$$\Delta W_c^{(t)} = W_c^{(t)} - W_s^{(t)}. \tag{B.10}$$

The client also returns its local sample count n_c and the diagnostic quantities used by the adaptive rule.

For the standard federated baseline, client deltas are aggregated using traditional sample-size proportional weights:

$$\omega_{c,\text{FedAvg}}^{(t)} = \frac{n_c}{\sum_{j \in S_t} n_j}. \tag{B.11}$$

The server update is given by:

$$W_s^{(t+1)} = W_s^{(t)} + \sum_{c \in S_t} \omega_{c,\text{FedAvg}}^{(t)} \Delta W_c^{(t)}. \tag{B.12}$$

Since $\sum_{c \in S_t} \omega_{c,\text{FedAvg}}^{(t)} = 1$, Eq. (B.12) is algebraically equivalent to a direct convex combination of the locally updated weights:

$$W_s^{(t+1)} = \sum_{c \in S_t} \omega_{c,\text{FedAvg}}^{(t)} W_c^{(t)}. \tag{B.13}$$

Adaptive quality-aware aggregation. The adaptive variant replaces the pure sample-size weight with a convex combination of the sample-size prior and a local empirical quality score. For each participating client c, we compute two diagnostic quantities over its local dataset D_c:

- the mean absolute free-energy magnitude $\phi_c^{(t)}$;
- the hidden-response participation entropy $h_c^{(t)}$.

For any scalar diagnostic x_c computed over the active clients $c \in S_t$, we define the batch extrema:

$$x_{\min}^{(t)} = \min_{j \in S_t} x_j, \qquad x_{\max}^{(t)} = \max_{j \in S_t} x_j. \tag{B.14}$$

If $x_{\max}^{(t)} > x_{\min}^{(t)}$, we define the min-max normalization mapping to $[0, 1]$:

$$\mathcal{N}_\uparrow(x_c) = \frac{x_c - x_{\min}^{(t)}}{x_{\max}^{(t)} - x_{\min}^{(t)}}. \tag{B.15}$$

If $x_{\max}^{(t)} = x_{\min}^{(t)}$, all clients are assigned the same normalized value of 1.

Both the free-energy magnitude and the hidden-response entropy are normalized with $\mathcal{N}_\uparrow$. The final composite quality score is the unweighted average of the two normalized metrics:

$$q_c^{(t)} = \frac{1}{2} \left[\mathcal{N}_\uparrow(\phi_c^{(t)}) + \mathcal{N}_\uparrow(h_c^{(t)}) \right]. \tag{B.16}$$

The diagnostics are formally defined as follows. The mean absolute free-energy magnitude measures the overall scale of the energy landscape:

$$\phi_c^{(t)} = \frac{1}{|D_c|} \sum_{\mathbf{v} \in D_c} |F_W(\mathbf{v})|. \tag{B.17}$$

The hidden-response participation entropy measures the diversity of hidden unit activations. Given a visible configuration $\mathbf{v} \in D_c$, the conditional hidden mean vector is:

$$\mathbf{m}(\mathbf{v}) = \sqrt{\frac{\beta}{N}} W^\top \mathbf{v}. \tag{B.18}$$

We compute the unnormalized response energy for each hidden unit $\mu \in \{1, \dots, K\}$:

$$R_{c,\mu} = \sum_{\mathbf{v} \in D_c} m_\mu(\mathbf{v})^2, \qquad R_c = \sum_{\nu=1}^{K} R_{c,\nu}. \tag{B.19}$$

If $R_c > 0$, we define the empirical activity distribution $A_{c,\mu} = R_{c,\mu}/R_c$, such that $\sum_{\mu=1}^{K} A_{c,\mu} = 1$. If $R_c = 0$ (dead network), we set $A_{c,\mu} = 1/K$ for all μ to ensure a uniform baseline. The normalized participation entropy is:

$$h_c^{(t)} = -\frac{1}{\log K} \sum_{\mu=1}^{K} A_{c,\mu} \log(A_{c,\mu}). \tag{B.20}$$

Finally, the adaptive aggregation weight merges the sample-size prior with the quality score via a hyperparameter $\alpha \in [0, 1]$:

$$\omega_c^{(t)} = \frac{(1-\alpha)\omega_{c,\text{FedAvg}}^{(t)} + \alpha q_c^{(t)}}{\sum_{j \in S_t} \left[(1-\alpha)\omega_{j,\text{FedAvg}}^{(t)} + \alpha q_j^{(t)} \right]}. \tag{B.21}$$

The adaptive server update structurally mirrors standard FedAvg, applying the normalized adjusted weights:

$$W_s^{(t+1)} = W_s^{(t)} + \sum_{c \in S_t} \omega_c^{(t)} \Delta W_c^{(t)}. \tag{B.22}$$

Setting $\alpha = 0$ recovers the standard FedAvg baseline, while $\alpha = 1$ yields a fully quality-driven aggregation.

Latent codes and clustering evaluation. After training, each test image $\mathbf{v}$ is mapped to its hidden conditional mean:

$$\mathbf{m}(\mathbf{v}) = \sqrt{\frac{\beta}{N}} W_s^\top \mathbf{v} \in \mathbb{R}^K. \tag{B.23}$$

These continuous vectors are extracted as latent representations (codes) and clustered using the standard k-means algorithm with ten clusters. Clustering quality is evaluated through cluster purity, normalized mutual information (NMI), and the adjusted Rand index (ARI). True dataset labels are strictly reserved for this post-hoc evaluation stage and are never observed by the model during the unsupervised federated training.

Default hyperparameters. The default network configuration used in the reported MNIST experiments is:

$$N = 784, \qquad K = 256, \qquad \beta = 1. \tag{B.24}$$

The federated setup consists of 10 total clients, with a subset of $|S_t| = 5$ clients sampled per communication round, for a total of 50 rounds. Each participating client performs $E = 1$ local epoch of CD-1 on its private dataset with a mini-batch size of $B = 64$. The local optimizer uses a learning rate $\varepsilon = 0.01$, a momentum coefficient $\mu_{\text{mom}} = 0.5$, and a weight decay factor $\lambda_{\text{wd}} = 10^{-4}$.

Appendix C. Sign-reconstruction probability

In this appendix we compute the probability that a noisy empirical Hebbian element preserves the sign of the corresponding clean archetypal Hebbian element. This quantity provides the finite-sample baseline used in Section 4.2 to normalize the sign-agreement entropy.

We work in a balanced teacher setting, used only for the analytical calculation. For each archetype ξ^μ, $\mu = 1, \dots, K$, we consider M noisy examples, indexed by $a = 1, \dots, M$. Thus the total number of examples in the balanced batch is KM. This notation should be distinguished from the general client-side batch size M_c^t used in the main text: in the balanced analytical case one has $M_c^t = KM$, whereas in heterogeneous client batches the number of examples associated with each archetype is random and governed by the local mixture distribution.

For a fixed off-diagonal pair $i \neq j$, define the clean and noisy Hebbian sums

$$S_{ij} := \sum_{\mu=1}^{K} \xi_i^\mu \xi_j^\mu, \qquad T_{ij} := \sum_{\mu=1}^{K} \sum_{a=1}^{M} \eta_i^{\mu,a} \eta_j^{\mu,a}. \tag{C.1}$$

The clean sign is informative only when $S_{ij} \neq 0$. For odd K, this condition is always satisfied, whereas for even K the clean sum vanishes with positive probability. We therefore define the empirical non-degenerate sign-overlap as

$$p_N^\star := \frac{\sum_{i\neq j} \mathbf{1}\{S_{ij} \neq 0\} \mathbf{1}\{\operatorname{sign}(T_{ij}) = \operatorname{sign}(S_{ij})\}}{\sum_{i\neq j} \mathbf{1}\{S_{ij} \neq 0\}}. \tag{C.2}$$

For odd K, $p_N^\star$ coincides with the usual sign-overlap. For even K, the pairs with $S_{ij} = 0$ are excluded because their clean sign is intrinsically undefined.

In the regime $N \gg 1$, with K fixed, the empirical quantity $p_N^\star$ is described by the corresponding single-pair probability

$$\widetilde{p}_{K,M}^\star(r) := \mathbb{P}\big(\operatorname{sign}(T_{ij}) = \operatorname{sign}(S_{ij}) \,\big|\, S_{ij} \neq 0\big). \tag{C.3}$$

The goal is to compute this probability explicitly.

Each noisy example is generated as

$$\eta_i^{\mu,a} = \xi_i^\mu \chi_i^{\mu,a}, \tag{C.4}$$

where the independent noise variables satisfy

$$\mathbb{P}(\chi_i^{\mu,a} = +1) = \frac{1+r}{2}, \qquad \mathbb{P}(\chi_i^{\mu,a} = -1) = \frac{1-r}{2}. \tag{C.5}$$

For $i \neq j$, introduce

$$s_\mu := \xi_i^\mu \xi_j^\mu \in \{-1, +1\}, \qquad z_{\mu,a} := \chi_i^{\mu,a} \chi_j^{\mu,a}. \tag{C.6}$$

Then

$$S_{ij} = \sum_{\mu=1}^{K} s_\mu, \qquad T_{ij} = \sum_{\mu=1}^{K} \sum_{a=1}^{M} s_\mu z_{\mu,a}. \tag{C.7}$$

Since the noises at different sites are independent,

$$\mathbb{P}(z_{\mu,a} = +1) = q := \frac{1+r^2}{2}, \qquad \mathbb{P}(z_{\mu,a} = -1) = 1 - q = \frac{1-r^2}{2}. \tag{C.8}$$

Thus, at the level of a single pairwise product, the sign is preserved with probability $(1 + r^2)/2$. The full Hebbian sign-reconstruction problem is, however, different: it concerns the sign of a sum over archetypes and noisy examples.

Let

$$K_+ := \#\{\mu : s_\mu = +1\}. \tag{C.9}$$

Since the variables s_μ are i.i.d. Rademacher random variables,

$$K_+ \sim \text{Binomial}\left(K, \frac{1}{2}\right). \tag{C.10}$$

Conditioning on $K_+ = k$, define

$$d_k := 2k - K. \tag{C.11}$$

Then

$$S_{ij} = d_k. \tag{C.12}$$

Moreover, if

$$U_+ \sim \text{Binomial}(Mk, q), \qquad U_- \sim \text{Binomial}(M(K-k), q), \tag{C.13}$$

with U_+ and U_- independent, then

$$T_{ij} = 2(U_+ - U_-) - M d_k. \tag{C.14}$$

Indeed, the k positive clean contributions produce $2U_+ - Mk$, whereas the $K - k$ negative clean contributions produce $-\big(2U_- - M(K-k)\big)$.

For $d_k \neq 0$, correct sign reconstruction is equivalent to

$$\operatorname{sign}(T_{ij}) = \operatorname{sign}(S_{ij}) \iff d_k\big[2(U_+ - U_-) - M d_k\big] > 0. \tag{C.15}$$

Possible ties $T_{ij} = 0$ are not counted as correct sign reconstructions, consistently with the convention that the sign is then uninformative.

We now write a formula valid for both odd and even K. Define the set of non-degenerate values

$$\mathcal{K}_\star := \{k \in \{0, \dots, K\} : 2k - K \neq 0\}. \tag{C.16}$$

For odd K, $\mathcal{K}_\star = \{0, \dots, K\}$. For even K, the value $k = K/2$ is excluded. The total probability mass of the non-degenerate sector is

$$Z_K := \sum_{k\in\mathcal{K}_\star} \binom{K}{k} 2^{-K} = \mathbb{P}(S_{ij} \neq 0). \tag{C.17}$$

Equivalently,

$$Z_K = \begin{cases} 1, & K \text{ odd}, \\ 1 - \binom{K}{K/2} 2^{-K}, & K \text{ even}. \end{cases} \tag{C.18}$$

The exact non-degenerate sign-reconstruction probability is therefore

$$\widetilde{p}_{K,M}^\star(r) = \frac{1}{Z_K} \sum_{k\in\mathcal{K}_\star} \binom{K}{k} 2^{-K} \sum_{u=0}^{Mk} \sum_{v=0}^{M(K-k)} \mathbf{1}\{d_k\big[2(u-v) - M d_k\big] > 0\} \times \binom{Mk}{u} \binom{M(K-k)}{v} q^{u+v} (1-q)^{MK-u-v}. \tag{C.19}$$

For odd K, this reduces to the usual sign-reconstruction probability. For even K, it removes the degenerate sector $S_{ij} = 0$ and renormalizes the remaining probability mass.

Gaussian approximation for large M. For large M, the conditional law of T_{ij} given $K_+ = k$ can be approximated by a Gaussian random variable. From (C.14), using $\mathbb{E}[U_+] = Mkq$, $\mathbb{E}[U_-] = M(K-k)q$, and $\operatorname{Var}(U_\pm) = M(\cdot)q(1-q)$, one obtains

$$\mathbb{E}[T_{ij} \mid K_+ = k] = Mr^2 d_k, \qquad \operatorname{Var}(T_{ij} \mid K_+ = k) = MK(1 - r^4). \tag{C.20}$$

Hence

$$T_{ij} \mid K_+ = k \approx \mathcal{N}\big(Mr^2 d_k,\, MK(1-r^4)\big). \tag{C.21}$$

For $d_k \neq 0$, this gives

$$\mathbb{P}\big(\operatorname{sign}(T_{ij}) = \operatorname{sign}(S_{ij}) \,\big|\, K_+ = k\big) \approx \Phi\left(\frac{|d_k| r^2 \sqrt{M}}{\sqrt{(1-r^4)K}}\right), \tag{C.22}$$

where Φ denotes the cumulative distribution function of a standard Gaussian random variable. The limit $r = 1$ is understood by continuity: in that case $T_{ij} = MS_{ij}$ and sign reconstruction is exact.

Averaging over the non-degenerate values of k yields

$$\widetilde{p}_{K,M}^\star(r) \approx \frac{1}{Z_K} \sum_{k\in\mathcal{K}_\star} \binom{K}{k} 2^{-K} \Phi\left(\frac{|2k-K| r^2 \sqrt{M}}{\sqrt{(1-r^4)K}}\right). \tag{C.23}$$

For odd K, $Z_K = 1$. For even K, Eq. (C.23) becomes

$$\widetilde{p}_{K,M}^\star(r) \approx \frac{1}{1 - \binom{K}{K/2} 2^{-K}} \sum_{\substack{k=0 \\ k\neq K/2}}^{K} \binom{K}{k} 2^{-K} \Phi\left(\frac{|2k-K| r^2 \sqrt{M}}{\sqrt{(1-r^4)K}}\right). \tag{C.24}$$

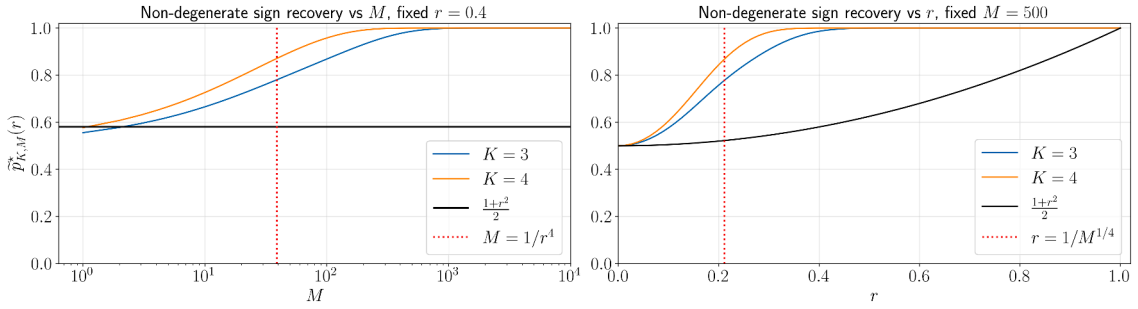


Fig. 13. Non-degenerate sign-reconstruction probability for noisy Hebbian correlators. Left: $\widetilde{p}^{\star}_{K,M}(r)$ as a function of the number M of noisy examples per archetype, at fixed channel quality $r = 0.4$. Right: $\widetilde{p}^{\star}_{K,M}(r)$ as a function of the channel quality r, at fixed $M = 500$. The blue and orange curves correspond respectively to $K = 3$ and $K = 4$ archetypes, after conditioning on the non-degenerate sector $S_{ij} \neq 0$. The black curve shows the single-product pairwise proxy $(1 + r^2)/2$, which does not account for the averaging effect of the Hebbian sum. The red dotted lines mark the scaling crossover $M \simeq r^{-4}$, equivalently $r \simeq M^{-1/4}$, predicted by the Gaussian approximation. The plot shows that sign recovery improves sharply once Mr^4 becomes of order one or larger, and that the finite-sample Hebbian sign-reconstruction probability can be substantially higher than the pairwise proxy because of averaging over repeated noisy examples.

Scaling with the channel quality. Eq. (C.23) shows that the relevant scaling variable is Mr^4. Indeed, setting

$$M = \frac{c}{r^4}, \tag{C.25}$$

one obtains

$$\widetilde{p}^{\star}_{K,M}(r) \approx \frac{1}{Z_K} \sum_{k \in \mathcal{K}_\star} \binom{K}{k} 2^{-K} \Phi\left(\frac{|2k - K|\sqrt{c}}{\sqrt{(1 - r^4)K}} \right). \tag{C.26}$$

In the small-r regime this reduces to

$$\widetilde{p}^{\star}_{K,M}(r) \approx \frac{1}{Z_K} \sum_{k \in \mathcal{K}_\star} \binom{K}{k} 2^{-K} \Phi\left(|2k - K| \sqrt{\frac{c}{K}} \right). \tag{C.27}$$

Thus, after removing the degenerate pairs with $S_{ij} = 0$, odd and even values of K are governed by the same scaling law: if

$$Mr^4 \gg 1, \tag{C.28}$$

then

$$\widetilde{p}^{\star}_{K,M}(r) \longrightarrow 1. \tag{C.29}$$

Equivalently, in terms of the total balanced batch size $M_{\text{tot}} = KM$, the condition can be written as $M_{\text{tot}} r^4 / K \gg 1$.

Relation with the pairwise proxy. The simple pairwise expression

$$q = \frac{1 + r^2}{2} \tag{C.30}$$

is recovered from the present calculation in the elementary case $K = 1$ and $M = 1$. In that case there is only one clean product and one noisy product, so sign reconstruction is correct exactly when $z_{\mu,a} = +1$. For general K and M, however, the controller compares the signs of Hebbian sums rather than individual products. The appropriate baseline is therefore the non-degenerate probability $\widetilde{p}^{\star}_{K,M}(r)$ in (C.19), or its Gaussian approximation (C.23) when M is large (see Fig. 13).

Appendix D. Random Matrix theory digression

The blended correlator $\boldsymbol{J}^{(t)}_{\text{rec}}(w_t)$ is informative but still entangles signal with bulk noise. We recall that the true archetypes $\{\boldsymbol{\xi}^\mu\}_{\mu=1}^K$ are eigenvectors, with eigenvalue equal to 1, of the pseudo–inverse coupling matrix

$$\left(J^{KS}\right)_{ij} = \frac{1}{N} \sum_{\mu,\nu=1}^{K} \xi_i^\mu \left(C^{-1}\right)_{\mu\nu} \xi_j^\nu,$$

where

$$C_{\mu\nu} = \frac{1}{N} \sum_{i=1}^{N} \xi_i^\mu \, \xi_i^\nu$$

is the pattern correlation matrix.

We can approximate the latter coupling matrix as the fixed point of the following algorithm (Fachechi et al., 2019):

$$\Phi_\varepsilon(\boldsymbol{J}) \;:=\; \boldsymbol{J} + \varepsilon\big(\boldsymbol{J} - \boldsymbol{J}^2\big), \qquad \varepsilon \in (0, 1]. \tag{D.1}$$

We compose a finite number of micro–steps with a decaying step size,

$$\widehat{\boldsymbol{J}}^{KS,(t)} \;:=\; \Phi_{\varepsilon_{n_{\text{prop}}-1}} \circ \cdots \circ \Phi_{\varepsilon_1} \circ \Phi_{\varepsilon_0}\left(\boldsymbol{J}^{(t)}_{\text{rec}}(w_t)\right),$$

$$\varepsilon_k = \frac{\varepsilon_0}{1 + k\varepsilon_0}, \quad k = 0, \ldots, n_{\text{prop}} - 1. \tag{D.2}$$

In the eigenbasis of $\boldsymbol{J}$, the map (D.1) updates each eigenvalue through a forward–Euler step of a logistic flow,

$$\lambda \;\longmapsto\; \lambda + \varepsilon\, \lambda(1 - \lambda). \tag{D.3}$$

Eigenvalues in $(0, 1)$ are pushed upward toward 1, eigenvalues larger than 1 are pulled downward, and values very close to 0 undergo only small absolute changes. After a finite number of iterations, the gap between the leading signal eigenvalues and the small bulk modes becomes more pronounced, making the subsequent thresholding step more robust.

Once per round, we also enforce symmetry and remove diagonal terms,

$$\widehat{\boldsymbol{J}}^{KS,(t)} \;\leftarrow\; \frac{1}{2}\left(\widehat{\boldsymbol{J}}^{KS,(t)} + \widehat{\boldsymbol{J}}^{KS,(t)\top}\right), \qquad \widehat{\boldsymbol{J}}^{KS,(t)} \;\leftarrow\; \mathrm{Off}\left(\widehat{\boldsymbol{J}}^{KS,(t)}\right), \tag{D.4}$$

so that the operator remains symmetric and focused on cross–feature interactions. The map (D.1) is a matrix polynomial and is therefore cheap to evaluate; when an eigendecomposition is already available, one can equivalently apply (D.3) directly to the eigenvalues.

Marchenko–Pastur/BBP threshold on the unsharpened operator. The detectability results of Theorem 2 apply to the rescaled, unsharpened aggregated operator

$$\widetilde{\boldsymbol{J}}^{(t)}_s \;:=\; N \boldsymbol{J}^{(t)}_s.$$

Let

$$q \;:=\; \frac{N}{M_{\text{round}}}$$

be the corresponding aspect ratio. We estimate the effective bulk variance σ_n^2 from the lower part of the unsharpened spectrum and define the upper Marchenko–Pastur edge as

$$\lambda_+(q) \;=\; \sigma_n^2\big(1 + \sqrt{q}\big)^2. \tag{D.5}$$

Optionally including a finite–sample correction $\delta_N > 0$, we define the BBP/MP threshold

$$\tau_{\text{MP}}(t) \;:=\; \lambda_+(q) + \delta_N. \tag{D.6}$$

The corresponding MP–based effective rank is

$$\widehat{K}_{\mathrm{MP}}(t) \;:=\; \#\Big\{ i \,:\, \lambda_i\Big(\widetilde{\boldsymbol{J}}_s^{(t)}\Big) > \tau_{\mathrm{MP}}(t) \Big\}. \tag{D.7}$$

Within the spiked-covariance model of Theorem 2, $\widehat{K}_{\mathrm{MP}}(t)$ coincides, up to finite–sample fluctuations, with the number of supercritical spikes satisfyingec

$$\kappa_\mu > \sqrt{q}$$

with high probability. Thus, $\widehat{K}_{\mathrm{MP}}(t)$ has a direct BBP interpretation and estimates the number of archetypal directions that are spectrally distinguishable from the bulk at round t.

For $t > 0$, the actual estimator also contains the feedback-dependent contribution introduced by the convex client update and the adaptive weights $w_c(t)$. Consequently, unless the additional reduction assumptions discussed in Remark 5 are imposed, the MP/BBP rule above should be interpreted as the reference criterion supplied by the corresponding effective spiked-covariance approximation, rather than as an exact threshold for the full adaptive estimator. The precise form taken by this effective model is derived in Proposition 1: the aspect ratio and the isotropic level in (D.5) are those of the plastic part $\boldsymbol{P}_t$ of the decomposition (4.16), the exposure entering the spike is the plasticity-weighted exposure π_t^{eff} of (4.15), and the conservative threshold of Corollary 1 replaces $\kappa_\mu > \sqrt{q}$ when the plasticity weights are heterogeneous. The residual terms that are not controlled rigorously are listed in Appendix E.3.

Post-sharpening spectral threshold. The pseudo–inverse propagation (D.2) is a separate conditioning step applied after aggregation. The operational spectral rank used to construct the LAM initial states is defined from the sharpened operator as

$$\widehat{K}_{\mathrm{spec}}(t) \;:=\; \#\Big\{ i \,:\, \lambda_i\Big(\widehat{\boldsymbol{J}}^{KS,(t)}\Big) > \tau_{\mathrm{spec}} \Big\}, \tag{D.8}$$

where $\tau_{\mathrm{spec}} \in (0,1)$ is a post-sharpening eigenvalue threshold.

The two thresholds therefore act on different operators and serve different purposes:

$$\tau_{\mathrm{MP}}(t) \quad \text{acts on} \quad \widetilde{\boldsymbol{J}}_s^{(t)}$$

and determines theoretical BBP/MP detectability, whereas

$$\tau_{\mathrm{spec}} \quad \text{acts on} \quad \widehat{\boldsymbol{J}}^{KS,(t)}$$

and determines which sharpened eigendirections are retained for operational seed generation.

In the well-separated regime, the two prescriptions typically identify the same signal subspace,

$$\widehat{K}_{\mathrm{spec}}(t) \simeq \widehat{K}_{\mathrm{MP}}(t),$$

because the finite sharpening procedure moves the separated signal eigenvalues toward 1, while the small bulk modes remain below the operational cut. Outside this regime, however, only $\widehat{K}_{\mathrm{MP}}(t)$ retains a direct BBP interpretation; $\widehat{K}_{\mathrm{spec}}(t)$ should instead be regarded as a post-sharpening operational estimate.

Shuffle threshold. As a data-driven alternative to a fixed τ_{spec}, we may construct a null model directly on the sharpened operator. We randomly reshuffle its off-diagonal entries while preserving symmetry and leaving the diagonal unchanged, compute the empirical distribution of $\lambda_{\max}$ over several reshuffles, and define $\tau_{\mathrm{shuf}}(t)$ as its $(1-\alpha)$ quantile. The corresponding operational rank is

$$\widehat{K}_{\mathrm{shuf}}(t) \;:=\; \#\Big\{ i \,:\, \lambda_i\Big(\widehat{\boldsymbol{J}}^{KS,(t)}\Big) > \tau_{\mathrm{shuf}}(t) \Big\}. \tag{D.9}$$

This choice is useful when the post-sharpening noise distribution is uncertain. It is an empirical alternative to τ_{spec} and should not be identified with the MP edge $\tau_{\mathrm{MP}}(t)$, which is defined on the unsharpened operator.

Effective spike-strength estimate. For each outlier eigenvalue

$$\lambda = \lambda_i\Big(\widetilde{\boldsymbol{J}}_s^{(t)}\Big) > \tau_{\mathrm{MP}}(t),$$

we estimate its effective spike strength through the closed–form inversion of the outlier-location map in Eq. (4.12):

$$\widehat{\kappa}(\lambda) \;=\; \frac{\big(\lambda/\sigma_n^2 - 1 - q\big) + \sqrt{\big(\lambda/\sigma_n^2 - 1 - q\big)^2 - 4q}}{2}. \tag{D.10}$$

This quantity provides a continuous diagnostic of exposure intensity consistent with the outlier-location law in Theorem 2, item (D3).

Coverage-aware initialization. Initialization does *not* run only once per retained eigendirection. Instead, we schedule a number of blocks according to a coverage heuristic. Specifically, for a block size P—with $P = L$ in the present L-layer implementation—we set

$$s(t) = 10 \cdot \left\lfloor \frac{\widehat{K}_{\mathrm{spec}}(t)}{P} \log\!\left(\frac{\widehat{K}_{\mathrm{spec}}(t)}{0.01} \right) \right\rfloor, \tag{D.11}$$

so that a total of $s(t) \times P$ candidate states are explored before pruning. This choice follows the standard balls–into–bins estimate

$$m_{\min} \approx \frac{K}{P} \log\!\Big(\frac{K}{\varepsilon}\Big), \qquad \varepsilon = 0.01.$$

Each block $b = 1, \dots, s(t)$ selects a multiset S_b of P seeds generated from the $\widehat{K}_{\mathrm{spec}}(t)$ retained sharpened eigendirections, for instance uniformly or according to their spectral gaps, and runs the corresponding P-layer refinement. In the present architecture, $P = L$. The resulting raw candidate pool is

$$\Xi_{\mathrm{raw}}^{(t)} \;:=\; \bigcup_{b=1}^{s(t)} \mathrm{Refine}\big(S_b; P\big), \tag{D.12}$$

and is subsequently processed by the two acceptance stages introduced in Section 2.1.1: duplicate removal through δ_{dup} and quadratic candidate acceptance through τ_{acc}.

The eigenvector-to-seed step is justified by BBP alignment. Once an archetypal spike becomes supercritical, $\kappa_\mu > \sqrt{q}$, its sample eigenvalue separates from the MP bulk and the corresponding eigenvector acquires a non-vanishing alignment with the latent archetypal direction. Because the sharpening transformation is a matrix polynomial, it preserves these eigendirections before the final symmetry and diagonal corrections while improving their eigenvalue separation.

The retained eigenvectors therefore provide informative directions spanning the detectable signal subspace. They need not, however, coincide with the individual archetypes, since they may represent orthogonalized linear combinations of the underlying patterns. The nonlinear initialization and subsequent LAM refinement perform the corresponding archetype-level factorization, while the coverage schedule ensures sufficient exploration of the retained subspace.

Appendix E. Detailed Proofs of Theorems

E.1. Proof of Theorem 1

Proof. Fix a communication round t. Throughout the proof, t denotes this fixed round index, while we use $u > 0$ (or $s > 0$) for generic deviation levels in the matrix Bernstein bounds. Recall that the round-wise Hebbian estimator is

$$\boldsymbol{J}_s^{(t)} := \frac{1}{N M_{\mathrm{round}}} \sum_{\ell=1}^{L} \sum_{m=1}^{M_c} \eta_{\ell,t,m}\, \eta_{\ell,t,m}^{\top}, \qquad M_{\mathrm{round}} := L M_c,$$

where $\eta_{\ell,t,m} \in \mathbb{R}^N$ denotes the mth local sample at client ℓ and round t. For each pair (ℓ, m) set

$$B_{\ell,m} := \eta_{\ell,t,m}\, \eta_{\ell,t,m}^{\top} \in \mathbb{R}^{N \times N}, \qquad \bar{B}_\ell := \mathbb{E}[B_{\ell,m}],$$

where the expectation is over the sampling of $\eta_{\ell,t,m}$ under the generative model of Section 2. By construction, $\bar{B}_\ell$ does not depend on m but may

depend on the client ℓ, reflecting statistical non-i.i.d. behavior across the federation.

Define the centered and rescaled increments

$$Z_{\ell,m} := \frac{B_{\ell,m} - \bar{B}_\ell}{N M_{\text{round}}}, \qquad S := \sum_{\ell=1}^{L} \sum_{m=1}^{M_c} Z_{\ell,m}.$$

Each $Z_{\ell,m}$ is self-adjoint, mean-zero, and the family $\{Z_{\ell,m}\}$ is independent over (ℓ, m). Moreover,

$$S = \frac{1}{N M_{\text{round}}} \sum_{\ell,m} B_{\ell,m} - \frac{1}{N M_{\text{round}}} \sum_{\ell,m} \bar{B}_\ell = \boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}],$$

so that $\|S\|_{\text{op}} = \|\boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}]\|_{\text{op}}$.

For each client ℓ, the population covariance at round t can be written as

$$\Theta_{t,\ell} := \frac{1}{N} \bar{B}_\ell = \frac{\sigma^2}{N} I + r^2 \sum_{\mu=1}^{K} \pi_{t,\ell}(\mu)\, u_\mu u_\mu^\top, \qquad u_\mu := \xi^\mu / \sqrt{N},$$

where $\pi_{t,\ell}$ is the client-specific class mixture and (u_μ) are the normalized archetypes. The round-level population operator is the average

$$\Theta_t := \mathbb{E}[\boldsymbol{J}_s^{(t)}] = \frac{1}{N M_{\text{round}}} \sum_{\ell,m} \bar{B}_\ell = \frac{1}{L} \sum_{\ell=1}^{L} \Theta_{t,\ell} = \frac{\sigma^2}{N} I + r^2 \sum_{\mu=1}^{K} \pi_t(\mu)\, u_\mu u_\mu^\top,$$

with $\pi_t := L^{-1} \sum_\ell \pi_{t,\ell}$. In particular, each $\Theta_{t,\ell}$ and Θ_t is positive semidefinite, with spectrum contained in some compact interval $[0, \Lambda_\star]$, and

$$\Lambda_t := \max_\ell \lambda_{\max}(\Theta_{t,\ell}) < \infty.$$

We first control $\|Z_{\ell,m}\|_{\text{op}}$. Using $\|B_{\ell,m}\|_{\text{op}} = \|\eta_{\ell,t,m}\|_2^2$ and $\|\bar{B}_\ell\|_{\text{op}} = N\, \lambda_{\max}(\Theta_{t,\ell})$, we obtain

$$\|Z_{\ell,m}\|_{\text{op}} = \frac{\|B_{\ell,m} - \bar{B}_\ell\|_{\text{op}}}{N M_{\text{round}}} \le \frac{\|B_{\ell,m}\|_{\text{op}} + \|\bar{B}_\ell\|_{\text{op}}}{N M_{\text{round}}} \le \frac{\|\eta_{\ell,t,m}\|_2^2 / N + \lambda_{\max}(\Theta_{t,\ell})}{M_{\text{round}}}.$$

In the bounded/Rademacher channel (items (ii)–(iii) in the theorem), we have $\|\eta_{\ell,t,m}\|_2^2 \equiv N$ almost surely, hence

$$\|Z_{\ell,m}\|_{\text{op}} \le \frac{1 + \Lambda_t}{M_{\text{round}}} =: R_t.$$

In the sub-Gaussian channel (item (i)), the same scaling holds up to a universal constant: standard sub-Gaussian moment bounds imply $\|\eta_{\ell,t,m}\|_2^2 \le CN$ with high probability, and a truncation argument (or a sub-exponential matrix Bernstein/Freedman inequality, see e.g. Tropp (2012, 2015)) allows one to absorb C into R_t. This only affects absolute numerical constants and leaves the functional dependence on M_{round} unchanged.

We now compute the variance parameter

$$v_t := \Big\| \sum_{\ell,m} \mathbb{E}[Z_{\ell,m}^2] \Big\|_{\text{op}}.$$

We first treat the bounded/Rademacher channel. Since $B_{\ell,m} = \eta_{\ell,t,m} \eta_{\ell,t,m}^\top$ and

$$B_{\ell,m}^2 = (\eta_{\ell,t,m} \eta_{\ell,t,m}^\top)^2 = (\eta_{\ell,t,m}^\top \eta_{\ell,t,m})\, \eta_{\ell,t,m} \eta_{\ell,t,m}^\top,$$

the assumption $\|\eta_{\ell,t,m}\|_2^2 \equiv N$ implies

$$B_{\ell,m}^2 = N B_{\ell,m} \quad \Longrightarrow \quad \mathbb{E}[B_{\ell,m}^2] = N \bar{B}_\ell.$$

Therefore

$$\mathbb{E}\big[(B_{\ell,m} - \bar{B}_\ell)^2\big] = \mathbb{E}[B_{\ell,m}^2] - \bar{B}_\ell^2 = N \bar{B}_\ell - \bar{B}_\ell^2 = N^2 \big(\Theta_{t,\ell} - \Theta_{t,\ell}^2\big) = N^2\, \Theta_{t,\ell}(I - \Theta_{t,\ell}).$$

Using $Z_{\ell,m} = (B_{\ell,m} - \bar{B}_\ell)/(N M_{\text{round}})$, we get

$$\mathbb{E}[Z_{\ell,m}^2] = \frac{1}{N^2 M_{\text{round}}^2} \mathbb{E}\big[(B_{\ell,m} - \bar{B}_\ell)^2\big] = \frac{1}{M_{\text{round}}^2} \Theta_{t,\ell}(I - \Theta_{t,\ell}),$$

and hence

$$\sum_{\ell,m} \mathbb{E}[Z_{\ell,m}^2] = \frac{1}{M_{\text{round}}^2} \sum_{\ell,m} \Theta_{t,\ell}(I - \Theta_{t,\ell}).$$

Taking operator norm and using that the spectrum of each $\Theta_{t,\ell}$ is contained in $[0, \Lambda_\star]$, we obtain

$$\begin{aligned} v_t &:= \Big\| \sum_{\ell,m} \mathbb{E}[Z_{\ell,m}^2] \Big\|_{\text{op}} \\ &\le \frac{1}{M_{\text{round}}^2} \sum_{\ell,m} \|\Theta_{t,\ell}(I - \Theta_{t,\ell})\|_{\text{op}} \\ &= \frac{1}{M_{\text{round}}^2} \sum_{\ell,m} \max_{\lambda \in \text{spec}(\Theta_{t,\ell})} \lambda(1-\lambda) \\ &\le \frac{M_{\text{round}}}{M_{\text{round}}^2} \sup_{\lambda \in [0,1]} \lambda(1-\lambda) = \frac{1}{4 M_{\text{round}}}, \end{aligned}$$

since in the present model each $\Theta_{t,\ell}$ is positive semidefinite with $\lambda_{\max}(\Theta_{t,\ell}) \le \frac{\sigma^2}{N} + r^2 \le 1$, so that $\text{spec}(\Theta_{t,\ell}) \subseteq [0,1]$, where $\lambda \mapsto \lambda(1-\lambda)$ is nonnegative and attains its maximum $1/4$ at $\lambda = 1/2$. The restriction to $[0,1]$ is essential: $|\lambda(1-\lambda)|$ is unbounded on $\lambda \ge 0$, and the bound would fail for a population operator with $\lambda_{\max} > 1$.

In the homogeneous case $\Theta_{t,\ell} \equiv \Theta_t$ for all ℓ, the computation simplifies and yields the exact identity

$$\sum_{\ell,m} \mathbb{E}[Z_{\ell,m}^2] = \frac{1}{M_{\text{round}}} \Theta_t(I - \Theta_t), \qquad v_t = \frac{1}{M_{\text{round}}} \Big\| \Theta_t(I - \Theta_t) \Big\|_{\text{op}},$$

so that the bound $v_t \le 1/(4M_{\text{round}})$ follows as a special case.

In the sub-Gaussian channel, an analogous argument applies. The identity $B_{\ell,m}^2 = (\eta_{\ell,t,m}^\top \eta_{\ell,t,m})\, B_{\ell,m}$ still holds, and standard sub-Gaussian moment bounds give $\mathbb{E}[\|\eta_{\ell,t,m}\|_2^4] \le CN^2$ for some universal C. This implies a uniform bound

$$\|\mathbb{E}[B_{\ell,m}^2]\|_{\text{op}} \le CN^2,$$

and hence

$$\Big\| \mathbb{E}[(B_{\ell,m} - \bar{B}_\ell)^2] \Big\|_{\text{op}} \le CN^2.$$

It follows that $v_t \le C/M_{\text{round}}$, which again has the same $1/M_{\text{round}}$ dependence and can be absorbed into a universal constant in the final bound.

We now apply a standard matrix Bernstein inequality for sums of independent self-adjoint matrices (see, e.g., Tropp, 2012, 2015). Let $d := N$ be the dimension. For all $u > 0$,

$$\mathbb{P}\big(\|S\|_{\text{op}} > u\big) \le 2d \exp\Big(- \frac{u^2/2}{v_t + (R_t u)/3} \Big).$$

Substituting the bounds on v_t and R_t, and using the elementary inequality

$$\frac{x^2}{a + bx} \ge \min\Big\{ \frac{x^2}{2a}, \frac{x}{2b} \Big\} \qquad \text{for all } x, a, b > 0,$$

we obtain, for some universal numerical constant $c_1 > 0$,

$$\mathbb{P}\big(\|\boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}]\|_{\text{op}} > u\big) \le 2N \exp\Bigg(- c_1\, M_{\text{round}} \cdot \min\Bigg\{ \frac{u^2}{\sup_{\lambda \in [0,1]} \lambda(1-\lambda)}, \frac{u}{1 + \Lambda_t} \Bigg\} \Bigg). \tag{E.1}$$

Since the prefactor $2N$ only enters through a logarithm when the failure probability is inverted (as in the statement of Theorem 1), it can be absorbed into the numerical constants. This yields the concentration inequality claimed in the main text, up to universal constant factors.

(i) Sub-Gaussian coordinates. In the sub-Gaussian channel, we first normalize to unit covariance scale. Let $\tilde{\eta}_{\ell,t,m} := \eta_{\ell,t,m}/\sigma$ and define

$$\tilde{\boldsymbol{J}}_s^{(t)} := \frac{1}{N M_{\text{round}}} \sum_{\ell,m} \tilde{\eta}_{\ell,t,m} \big(\tilde{\eta}_{\ell,t,m}\big)^\top = \sigma^{-2} \boldsymbol{J}_s^{(t)}, \qquad \tilde{\Theta}_t := \mathbb{E}\big[\tilde{\boldsymbol{J}}_s^{(t)}\big] = \sigma^{-2} \Theta_t.$$

By construction, the eigenvalues of $\tilde{\Theta}_t$ are contained in a compact interval $[0, \Lambda_\star]$ with $\Lambda_\star = \mathcal{O}(1)$, and

$$\sup_{\lambda \in \text{spec}(\tilde{\Theta}_t)} \lambda(1-\lambda) \le \tfrac{1}{4}, \qquad 1 + \lambda_{\max}(\tilde{\Theta}_t) \le C$$

for some universal constant C (e.g., $C = 2$ if $\lambda_{\max}(\tilde{\Theta}_t) \le 1$). Applying (E.1) to $\tilde{\boldsymbol{J}}_s^{(t)}$ in place of $\boldsymbol{J}_s^{(t)}$ yields

$$\mathbb{P}\Big(\Big\| \tilde{\boldsymbol{J}}_s^{(t)} - \mathbb{E}\big[\tilde{\boldsymbol{J}}_s^{(t)}\big] \Big\|_{\mathrm{op}} > s \Big) \;\le\; 2N \exp\Big(- c_2\, M_{\mathrm{round}} \cdot \min\{s^2,\, s\} \Big),$$

for some constant $c_2 > 0$. Since $\boldsymbol{J}_s^{(t)} - \mathbb{E}\Big[\boldsymbol{J}_s^{(t)}\Big] = \sigma^2\big(\tilde{\boldsymbol{J}}_s^{(t)} - \mathbb{E}\Big[\tilde{\boldsymbol{J}}_s^{(t)}\Big]\big)$, we have

$$\Big\| \boldsymbol{J}_s^{(t)} - \mathbb{E}\big[\boldsymbol{J}_s^{(t)}\big] \Big\|_{\mathrm{op}} > u \quad\Longleftrightarrow\quad \Big\| \tilde{\boldsymbol{J}}_s^{(t)} - \mathbb{E}\big[\tilde{\boldsymbol{J}}_s^{(t)}\big] \Big\|_{\mathrm{op}} > u/\sigma^2.$$

Substituting $s = u/\sigma^2$ in the previous inequality gives

$$\mathbb{P}\big(\|\boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}]\|_{\mathrm{op}} > u \big) \;\le\; 2N \exp\Big(- c_2\, M_{\mathrm{round}} \cdot \min\Big\{ \tfrac{u^2}{\sigma^4},\, \tfrac{u}{\sigma^2} \Big\} \Big),$$

which is the claimed form in item (i) (up to a readjustment of the numerical constant c_1).

(ii) Bounded / Rademacher channel. In the Rademacher channel, $\chi_j \in \{\pm 1\}$ with $\mathbb{E}[\chi_j] = r$, and hence $\|\eta_{\ell,t,m}\|_2^2 \equiv N$ almost surely. As computed above, this implies $B_{\ell,m}^2 = N B_{\ell,m}$ and

$$\mathbb{E}\big[(B_{\ell,m} - \bar{B}_\ell)^2\big] = N^2\, \Theta_{t,\ell}(I - \Theta_{t,\ell}),$$

so that the variance proxy satisfies

$$v_t := \Big\| \sum_{\ell,m} \mathbb{E}[Z_{\ell,m}^2] \Big\|_{\mathrm{op}} \le \frac{1}{4 M_{\mathrm{round}}},$$

because $\lambda(1-\lambda) \le 1/4$ for all $\lambda \ge 0$. Moreover, in this channel we have the uniform bound $\|Z_{\ell,m}\|_{\mathrm{op}} \le R_t \le c_0/M_{\mathrm{round}}$ for a universal $c_0 > 0$. A Hoeffding-type matrix Bernstein inequality for bounded self-adjoint increments then yields a universal constant $c_* > 0$ such that, for all $u > 0$,

$$\mathbb{P}\big(\|\boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}]\|_{\mathrm{op}} > u \big) \;\le\; 2N \exp\left(- c_*\, M_{\mathrm{round}} \cdot \min\left\{ \frac{u^2}{\sup_{\lambda\in[0,1]} \lambda(1-\lambda)},\; u \right\} \right).$$

Using again $\sup_{\lambda\in[0,1]} \lambda(1-\lambda) = \frac{1}{4}$, we obtain

$$\mathbb{P}\big(\|\boldsymbol{J}_s^{(t)} - \mathbb{E}[\boldsymbol{J}_s^{(t)}]\|_{\mathrm{op}} > u \big) \;\le\; 2N \exp\big(- \tilde{c}\, M_{\mathrm{round}} \cdot \min\{u^2,\, u\} \big),$$

for another universal constant $\tilde{c} > 0$, which is the bound stated in item (ii) up to harmless changes in numerical constants.

(iii) Finite-N refined bound. Set $m := M_{\mathrm{round}}$ and normalize $Y_k := \eta_k/\sqrt{N}$, so that

$$J := \frac{1}{m} \sum_{k=1}^{m} Y_k Y_k^\top, \qquad \Sigma := \mathbb{E}[J].$$

Define $X_k := Y_k Y_k^\top - \Sigma$, which are self-adjoint, mean-zero, and independent, and note that

$$J - \Sigma = \frac{1}{m} \sum_{k=1}^{m} X_k.$$

In the bounded/Rademacher model, $\|Y_k\|_2 = 1$ almost surely, hence

$$\|X_k\|_{\mathrm{op}} \le \|Y_k Y_k^\top\|_{\mathrm{op}} + \|\Sigma\|_{\mathrm{op}} \le 1 + 1 = 2 =: R_\star.$$

Writing $B := Y_k Y_k^\top$, we have $B^2 = B$, so

$$\mathbb{E}[X_k^2] = \mathbb{E}[(B - \Sigma)^2] = \mathbb{E}[B^2] - \Sigma^2 = \mathbb{E}[B] - \Sigma^2 = \Sigma - \Sigma^2.$$

Thus the variance parameter is

$$V_0 := \Big\| \sum_{k=1}^{m} \mathbb{E}[X_k^2] \Big\|_{\mathrm{op}} = m\, v_\star, \qquad v_\star := \|\Sigma - \Sigma^2\|_{\mathrm{op}} = \max_{\lambda\in\mathrm{spec}(\Sigma)} \lambda(1-\lambda) \le \tfrac{1}{4}.$$

A dimension-explicit matrix Bernstein inequality for sums of self-adjoint matrices gives, for any sum deviation $U > 0$,

$$\mathbb{P}\left(\Big\| \sum_{k=1}^{m} X_k \Big\|_{\mathrm{op}} \ge U \right) \;\le\; 2N \exp\left(- \frac{U^2}{2V_0 + \frac{2}{3} R_\star U} \right).$$

Setting $U = mu$ (where u is the average deviation) and using the elementary inequality $\frac{x^2}{a+bx} \ge \frac{1}{2}\min\{\frac{x^2}{a}, \frac{x}{b}\}$, we obtain universal constants $c, C > 0$ such that

$$\mathbb{P}\big(\|J - \Sigma\|_{\mathrm{op}} \ge u \big) \;\le\; 2 \exp\Big(- c\, m\, \min\Big\{ \frac{u^2}{v_\star},\, \frac{u}{R_\star} \Big\} + C \log N \Big),$$

where $C \log N$ accounts for the dimensional prefactor $2N$. In our model, $\Sigma = \frac{\sigma^2}{N} I + r^2 \sum_\mu \pi_t(\mu) u_\mu u_\mu^\top$, so $\lambda_{\max}(\Sigma) \le \frac{\sigma^2}{N} + r^2 \le 1$ and $v_\star \le \min\{\frac{1}{4}, \frac{\sigma^2}{N} + r^2\}$, which yields the finite-N bound stated in item (iii). □

Remark 11. The derived bounds reveal that the concentration speed is governed by the spectral mapping $\lambda \mapsto \lambda(1-\lambda)$. The eigenvalues of the unrescaled population operator $\Sigma = \mathbb{E}[\boldsymbol{J}_s^{(t)}]$ lie in $[0,1]$ and split into an isotropic bulk of order σ^2/N together with at most K signal eigenvalues of order r^2. The variance proxy is therefore controlled uniformly, $v_\star = \max_{\lambda\in\mathrm{spec}(\Sigma)} \lambda(1-\lambda) \le \frac{1}{4}$, while the bulk contribution is negligible, of order σ^2/N, so that the concentration is effectively driven by the K signal directions alone. From a scaling perspective, if the total sample size grows linearly with dimension ($M_{\mathrm{round}} = \Theta(N)$), the deviation probability decays exponentially in N, matching the accuracy predicted by local laws in RMT. Finally, we note that the dimensional term $C \log N$ in the exponent accounts for the ambient dimension; for matrices with rapidly decaying spectra, standard arguments allow replacing N with the intrinsic dimension (effective rank), yielding further refinement.

E.2. BBP threshold lemmata and proofs

Theorem 3 (MP universality + isotropic global law)**.** *Let $Z \in \mathbb{R}^{N\times m}$ be a random matrix with independent entries. We view Z as a collection of m independent sample vectors (columns) $x_1, \dots, x_m \in \mathbb{R}^N$. Assume the entries are standardized and sub-Gaussian:*

$$\mathbb{E}[Z_{ij}] = 0, \qquad \mathbb{E}[Z_{ij}^2] = 1, \qquad \sup_{i,j} \|Z_{ij}\|_{\psi_2} \le B < \infty. \tag{E.2}$$

To facilitate the application of standard universality results, we further assume that the first four moments of Z_{ij} match those of a standard Gaussian (or that the mismatch is negligible for the precision we target). Let

$$S_N := \frac{1}{m} Z Z^\top = \frac{1}{m} \sum_{\mu=1}^{m} x_\mu x_\mu^\top$$

be the sample covariance matrix, and let $G(z) := (S_N - z I_N)^{-1}$ be its resolvent. Set $q_N := N/m$ and assume $q_N \to q \in (0,\infty)$ as $N \to \infty$; q is the aspect ratio denoted α_{MP} in Section 4.1 and equal to N/M_{round} in Theorem 2. Denote by μ_{MP} the Marchenko–Pastur law with aspect ratio q, and by $m_{\mathrm{MP}}(z)$ its Stieltjes transform.

Fix $\delta > 0$ and a large constant $R > (1 + \sqrt{q})^2 + \delta$. Define the bounded spectral domain strictly away from the asymptotic support of μ_{MP}:

$$\mathcal{D}_\delta := \Big\{ z \in \mathbb{C} :\; \mathrm{dist}\big(z, \mathrm{supp}(\mu_{\mathrm{MP}})\big) \ge \delta,\; |z| \le R \Big\}. \tag{E.3}$$

Then, for any finite collection of unit vectors $\mathcal{V} \subset \mathbb{S}^{N-1}$ with $|\mathcal{V}| \le N^{C_0}$, there exists $C = C(\delta, q, B, C_0, R)$ such that, for every $D > 0$ and all N large enough,

$$\mathbb{P}\left(\sup_{z\in\mathcal{D}_\delta} \max_{a,b\in\mathcal{V}} \Big| a^\top G(z)\, b - m_{\mathrm{MP}}(z)\, \langle a, b\rangle \Big| \le C \sqrt{\frac{\log N}{N}} \right) \ge 1 - N^{-D}. \tag{E.4}$$

Proof. The argument combines three standard ingredients of modern random matrix theory: (I) concentration of bilinear forms of the resolvent; (II) identification of the mean in the Gaussian case via the Dyson equation; (III) universality of the mean via the Green Function Comparison Theorem. A final step (IV) upgrades pointwise bounds to uniform ones in z and over $\mathcal{V}$ using rigidity and an ε–net.

Throughout, constants $C, c > 0$ may change from line to line but depend only on (δ, q, B, C_0, R).

Fix deterministic unit vectors $a, b \in \mathbb{R}^N$ and a spectral parameter $z \in \mathcal{D}_\delta$. We first control the random fluctuations of the bilinear form

$$F(z) := a^\top G(z)\, b$$

around its mean $\mathbb{E}F(z)$.

A prerequisite for concentration is the boundedness of the resolvent norm. By the Bai–Yin theorems (Bai & Yin, 1993, Thms. 1 & 2) (or the stronger rigidity estimates in Knowles–Yin (Knowles & Yin, 2017, Lemma 10.1)), with high probability the spectrum of S_N is contained in a small neighborhood of the asymptotic support. Consequently, for $z \in \mathcal{D}_\delta$, the map $Z \mapsto a^\top G(z) b$ is Lipschitz continuous with constant depending on δ^{-1}.

Conditional on this high-probability event, we apply standard concentration inequalities for functions of independent sub-Gaussian variables (see Anderson–Guionnet–Zeitouni (Anderson et al., 2010, Sec. 2.3 & 4.4) or the general Herbst's argument). This yields a sub-Gaussian bound of the form:

$$\mathbb{P}\Big(\Big|a^\top G(z)b - \mathbb{E}[a^\top G(z)b]\Big| > t\Big) \;\le\; 2\exp\big(-c\,N\,\min(t, t^2)\big) \tag{E.5}$$

for all $t > 0$, with $c > 0$ independent of N. Choosing

$$t = C\sqrt{\frac{\log N}{N}}$$

and enlarging C if necessary yields

$$\Big|a^\top G(z)b - \mathbb{E}[a^\top G(z)b]\Big| \;\le\; C\sqrt{\frac{\log N}{N}}$$

with probability at least $1 - N^{-D}$, for any prescribed $D > 0$, provided N is large enough. This step relies on the independence and tail properties of the entries, without requiring gaussianity.

We now specialize to the case where Z has i.i.d. $\mathcal{N}(0,1)$ entries, and denote by $G^{\mathrm{G}}(z)$ the corresponding resolvent. By rotational invariance of the Gaussian measure,

$$\mathbb{E}\,G^{\mathrm{G}}(z) = m_N(z)\,I_N, \qquad m_N(z) := \frac{1}{N}\,\mathbb{E}\,\mathrm{Tr}\,G^{\mathrm{G}}(z).$$

It is a standard consequence of the resolvent method for sample covariance matrices (see, e.g., Pastur–Shcherbina (Pastur & Shcherbina, 2011, Sec. 7.2 and 7.6)) that $m_N(z)$ satisfies the Marchenko–Pastur fixed-point equation up to an error of order $O(1/N)$, uniformly on domains at positive distance from the MP support. More precisely, for $z \in \mathcal{D}_\delta$,

$$1 + z\,m_N(z) - \frac{m_N(z)}{1 + q\,m_N(z)} = O\Big(\frac{1}{N}\Big),$$

where the implicit constant depends on (δ, q) but not on N. Equivalently, $q\,z\,m_N(z)^2 + (z + q - 1)m_N(z) + 1 = O(1/N)$. Since the exact solution $m_{\mathrm{MP}}(z)$ of $1 + z m_{\mathrm{MP}}(z) = \frac{m_{\mathrm{MP}}(z)}{1+q\,m_{\mathrm{MP}}(z)}$ is stable under small perturbations of the equation on $\mathcal{D}_\delta$ (the derivative of the defining RHS does not vanish away from the spectral edges), we obtain

$$|m_N(z) - m_{\mathrm{MP}}(z)| \le \frac{C}{N} \qquad \text{for all } z \in \mathcal{D}_\delta.$$

Hence, for any unit vectors a, b,

$$\Big|\mathbb{E}[a^\top G^{\mathrm{G}}(z)b] - m_{\mathrm{MP}}(z)\,\langle a, b\rangle\Big| = |\langle a, b\rangle| \cdot |m_N(z) - m_{\mathrm{MP}}(z)| \le \frac{C}{N}. \tag{E.6}$$

This deterministic error is asymptotically negligible compared to the fluctuation scale $N^{-1/2}\sqrt{\log N}$ from Step I.

We now remove the Gaussian assumption and return to a general matrix Z satisfying (E.2) and the moment-matching condition up to order 4. Let $G(z)$ and $G^{\mathrm{G}}(z)$ denote the resolvents of S_N in the non-Gaussian and Gaussian cases respectively.

We appeal to the Green Function Comparison Theorem (GFT). While originally formulated for Wigner matrices, the result extends to sample covariance matrices, as detailed in Erdős–Yau (Erdös & Yau, 2017, Thm 16.1 and Remark 16.2). (See also Knowles–Yin Knowles & Yin, 2017 for an alternative approach via continuous interpolation and self-consistent comparison). The GFT asserts that if two ensembles have independent entries with matching first four moments and suitable tail bounds, then expectations of smooth functionals of their resolvents differ asymptotically by a small power of N^{-1}.

In particular, applying this to the bilinear form functional (which is smooth away from the real axis), for each fixed z with $\Im z > 0$ and unit vectors a, b, we have:

$$\Big|\mathbb{E}[a^\top G(z)b] - \mathbb{E}[a^\top G^{\mathrm{G}}(z)b]\Big| \;\le\; C N^{-c} \tag{E.7}$$

for some $c = c(q, B) > 0$. The precise value of c is immaterial for our purposes; the key point is that the error vanishes as $N \to \infty$, uniformly on bounded z–domains away from the spectrum.

Combining the Gaussian mean computation (E.6) with the comparison estimate (E.7), we deduce that, for any fixed $z \in \mathcal{D}_\delta$ and unit a, b,

$$\Big|\mathbb{E}[a^\top G(z)b] - m_{\mathrm{MP}}(z)\,\langle a, b\rangle\Big| \;\le\; C N^{-c'} \tag{E.8}$$

for some $c' > 0$ (e.g. $c' = \min\{1, c\}$).

The bounds in Steps I–III hold for each fixed z and each fixed pair (a, b). We now show that they can be made uniform over all $z \in \mathcal{D}_\delta$ and all $(a, b) \in \mathcal{V} \times \mathcal{V}$.

(a) Rigidity and bounds on the resolvent. By the global Marchenko–Pastur law and sharp bounds on extreme eigenvalues (see, e.g., Bai–Silverstein (Bai & Silverstein, 2010, Thms. 5.9–5.11)), for any fixed $\delta > 0$ we have, with probability at least $1 - N^{-D}$ for each $D > 0$ and N large enough, that all eigenvalues of S_N lie within a $\delta/4$–neighborhood of $\mathrm{supp}(\mu_{\mathrm{MP}})$. On this high-probability event we therefore have

$$\mathrm{dist}(z, \sigma(S_N)) \;\ge\; \frac{\delta}{2} \qquad \text{for all } z \in \mathcal{D}_\delta,$$

and hence

$$\|G(z)\|_{\mathrm{op}} \;\le\; \frac{2}{\delta} \qquad \text{for all } z \in \mathcal{D}_\delta.$$

Differentiating the resolvent with respect to z yields $\partial_z G(z) = (S_N - zI_N)^{-2} = G(z)^2$. Thus, on the same event,

$$\|\partial_z G(z)\|_{\mathrm{op}} \le \|G(z)\|_{\mathrm{op}}^2 \le \frac{4}{\delta^2}, \qquad z \in \mathcal{D}_\delta.$$

Consequently, for any unit vectors a, b, the map

$$z \longmapsto a^\top G(z) b$$

is Lipschitz continuous on $\mathcal{D}_\delta$ with Lipschitz constant bounded by $4/\delta^2$.

(b) ε–net in z and union bound over $\mathcal{V}$. Let $\varepsilon > 0$ be a small mesh size to be chosen later. By construction, the domain is compact and has bounded area. Thus, we can construct a finite ε–net $\mathcal{Z} \subset \mathcal{D}_\delta$ with cardinality

$$|\mathcal{Z}| \;\lesssim\; \varepsilon^{-2}.$$

For each $z \in \mathcal{D}_\delta$ there exists a grid point $z_\star \in \mathcal{Z}$ with $|z - z_\star| \le \varepsilon$. By the Lipschitz bound derived in Step IV(a), we have

$$\Big|a^\top G(z)b - a^\top G(z_\star)b\Big| \;\le\; \frac{4}{\delta^2}\,\varepsilon \qquad \text{for all unit } a, b.$$

The limiting Stieltjes transform $m_{\mathrm{MP}}(z)$ is also Lipschitz on this domain.

Now fix $(a, b) \in \mathcal{V} \times \mathcal{V}$. For each grid point $z_\star \in \mathcal{Z}$, the bounds from Steps I–III imply

$$\Big|a^\top G(z_\star)b - m_{\mathrm{MP}}(z_\star)\,\langle a, b\rangle\Big| \;\le\; C\sqrt{\frac{\log N}{N}}$$

with probability at least $1 - N^{-K}$ (where K can be made arbitrarily large by adjusting the constant in the concentration step). Taking a union bound over all $z_\star \in \mathcal{Z}$ and all $(a, b) \in \mathcal{V} \times \mathcal{V}$ multiplies the failure probability by at most

$$|\mathcal{Z}| \cdot |\mathcal{V}|^2 \;\lesssim\; \varepsilon^{-2}\,N^{2C_0}.$$

Choosing

$$\varepsilon := \sqrt{\frac{\log N}{N}},$$

we have $|\mathcal{Z}| \lesssim N$, so the total number of events is polynomial in N. By choosing the concentration constant sufficiently large, we ensure the overall failure probability is bounded by N^{-D}. On the intersection of

these high-probability events and the rigidity event from part (a), we combine the discrete bound with the Lipschitz approximation to obtain

$$\sup_{z\in D_\delta}\max_{a,b\in\mathcal{V}}\left|a^\top G(z)b - m_{\mathrm{MP}}(z)\,\langle a,b\rangle\right| \;\le\; C\sqrt{\frac{\log N}{N}}$$

for all N large enough, which proves (E.4).

Collecting Steps I–IV completes the proof. □

Remark 12. Theorem 3 establishes the universality of the isotropic Marchenko–Pastur law on a macroscopic scale.[5] While the proof relies on the Green Function Comparison Theorem, which necessitates the matching of the first four moments with a Gaussian ensemble to eliminate lower-order error terms, it is worth noting that the limiting measure μ_{MP} depends only on the first two moments. The higher moments typically influence the subleading fluctuation terms (central limit theorems for linear statistics) rather than the first-order deterministic limit derived here.

Furthermore, the isotropic bound (E.4) carries significant implications for the geometry of the eigenvectors. The fact that the resolvent behaves like a scalar multiple of the identity, $G(z)\approx m_{\mathrm{MP}}(z)I_N$, implies a strong form of *eigenvector delocalization*. Specifically, it suggests that the eigenvectors of S_N are approximately uniformly distributed on the unit sphere $\mathbb{S}^{N-1}$, showing no preference for any specific deterministic direction a or b.[6] Finally, we observe that the error term $O(\sqrt{\log N/N})$ is largely dictated by the union bound over the ε–net; for a single fixed z and fixed vectors, the fluctuations are of order $O(N^{-1/2})$.

Lemma 1 (Uniform quantitative decoupling). *Let $Z\in\mathbb{R}^{N\times M_{\mathrm{round}}}$ satisfy the standing assumptions under which the isotropic local law Theorem 3 holds, with sample covariance S_0 and resolvent $G(z)=(S_0-zI_N)^{-1}$. Fix $\delta\in(0,1)$ and let $\mathcal{D}_\delta$ be the spectral domain from (E.3).*

Let $U=[u_1,\dots,u_K]\in\mathbb{R}^{N\times K}$ have unit–norm columns $u_\mu\in\mathbb{S}^{N-1}$, and write

$$\Gamma := U^\top U\in\mathbb{R}^{K\times K},\qquad \Gamma_{\mu\nu}=\langle u_\mu,u_\nu\rangle.$$

Define

$$\varepsilon_{\mathrm{orth}} := \max_{\mu\neq\nu}|\langle u_\mu,u_\nu\rangle|\in[0,1),\qquad M(z) := U^\top G(z)\,U\in\mathbb{R}^{K\times K}. \tag{E.9}$$

Assume that $K=K(N)$ satisfies $K^2\le N^{C_0}$ for some fixed $C_0>0$ (for example, K fixed or $K\lesssim\log N$).

Then there exists a constant $C_1=C_1(q,\delta,B,C_0,R)<\infty$ such that, for every $D>0$, with probability at least $1-N^{-D}$ and for all N large enough,

$$\sup_{z\in\mathcal{D}_\delta}\left\|M(z)-m_{\mathrm{MP}}(z)\Gamma\right\|_{\mathrm{op}} \;\le\; C_1\,K\sqrt{\frac{\log N}{N}}. \tag{E.10}$$

Consequently,

$$\sup_{z\in\mathcal{D}_\delta}\left\|M(z)-m_{\mathrm{MP}}(z)I_K\right\|_{\mathrm{op}} \;\le\; |m_{\mathrm{MP}}(z)|\,\|\Gamma-I_K\|_{\mathrm{op}}+C_1\,K\sqrt{\frac{\log N}{N}}. \tag{E.11}$$

Moreover, by Gershgorin's theorem,

$$\|\Gamma-I_K\|_{\mathrm{op}}\le(K-1)\,\varepsilon_{\mathrm{orth}}\quad\Longrightarrow\quad \sup_{z\in\mathcal{D}_\delta}\left\|M(z)-m_{\mathrm{MP}}(z)I_K\right\|_{\mathrm{op}}$$

$$\le\; C_0'\,K\,\varepsilon_{\mathrm{orth}}+C_1\,K\sqrt{\frac{\log N}{N}}, \tag{E.12}$$

for some $C_0'=C_0'(q,\delta)$.

In particular:

(i) If K is fixed (independent of N), the right-hand side is $O\big(\varepsilon_{\mathrm{orth}}+\sqrt{\log N/N}\big)$.

(ii) If $K=K(N)$ and $K\sqrt{\log N}/\sqrt{N}\to 0$ (e.g. $K=o(\sqrt{N/\log N})$, in particular $K\lesssim\log N$), then the right-hand side vanishes as $N\to\infty$.

Proof. We first separate the "MP part" from the fluctuations. For $\mu,\nu\in\{1,\dots,K\}$,

$$M_{\mu\nu}(z)=u_\mu^\top G(z)\,u_\nu. \tag{E.13}$$

Add and subtract $m_{\mathrm{MP}}(z)\langle u_\mu,u_\nu\rangle$:

$$M_{\mu\nu}(z)=m_{\mathrm{MP}}(z)\,\langle u_\mu,u_\nu\rangle+R_{\mu\nu}(z),$$
$$R_{\mu\nu}(z) := u_\mu^\top\big(G(z)-m_{\mathrm{MP}}(z)I_N\big)u_\nu. \tag{E.14}$$

In matrix form this is

$$M(z)=m_{\mathrm{MP}}(z)\,\Gamma+R(z),$$
$$R(z) := \big(R_{\mu\nu}(z)\big)_{\mu,\nu=1}^{K}=U^\top\big(G(z)-m_{\mathrm{MP}}(z)I_N\big)U. \tag{E.15}$$

Thus, to prove (E.10), it suffices to bound $\|R(z)\|_{\mathrm{op}}$ uniformly in $z\in D_\delta$.

Consider the finite set of unit vectors

$$\mathcal{V} := \{u_1,\dots,u_K\}\subset\mathbb{S}^{N-1}.$$

By assumption $K^2\le N^{C_0}$ for some fixed $C_0>0$, so in particular $|\mathcal{V}|\le N^{C_0}$ for N large enough. We apply Theorem 3 to this set $\mathcal{V}$. For any $D>0$ there exists $C=C(q,\delta,B,C_0,R)$ such that, with probability at least $1-N^{-D}$ and for all N sufficiently large,

$$\sup_{z\in\mathcal{D}_\delta}\max_{a,b\in\mathcal{V}}\left|a^\top G(z)b-m_{\mathrm{MP}}(z)\langle a,b\rangle\right| \;\le\; C\sqrt{\frac{\log N}{N}}. \tag{E.16}$$

Since each u_μ belongs to $\mathcal{V}$, we may choose $a=u_\mu$ and $b=u_\nu$ and rewrite (E.16) as

$$\sup_{z\in\mathcal{D}_\delta}\max_{1\le\mu,\nu\le K}\left|R_{\mu\nu}(z)\right| \;\le\; C\sqrt{\frac{\log N}{N}}. \tag{E.17}$$

Thus, on the high-probability event where (E.17) holds, we have a uniform entrywise bound on $R(z)$ for all $z\in\mathcal{D}_\delta$.

We now bound the operator norm of $R(z)$ in terms of its entries. Fix $z\in\mathcal{D}_\delta$ and let $(x,y)\in\mathbb{S}^{K-1}\times\mathbb{S}^{K-1}$. Then

$$x^\top R(z)y=\sum_{\mu,\nu=1}^{K}x_\mu\,R_{\mu\nu}(z)\,y_\nu,$$

so that

$$|x^\top R(z)y|\le\Big(\max_{1\le\mu,\nu\le K}|R_{\mu\nu}(z)|\Big)\sum_{\mu=1}^{K}|x_\mu|\sum_{\nu=1}^{K}|y_\nu|.$$

By Cauchy–Schwarz,

$$\sum_{\mu=1}^{K}|x_\mu|\le\sqrt{K}\,\|x\|_2=\sqrt{K},\qquad \sum_{\nu=1}^{K}|y_\nu|\le\sqrt{K}\,\|y\|_2=\sqrt{K},$$

so

$$|x^\top R(z)y|\le K\max_{\mu,\nu}|R_{\mu\nu}(z)|. \tag{E.18}$$

Taking the supremum over all unit vectors x,y yields

$$\|R(z)\|_{\mathrm{op}}=\sup_{\|x\|=\|y\|=1}|x^\top R(z)y|\le K\max_{\mu,\nu}|R_{\mu\nu}(z)|.$$

Combining this with (E.17) we find that, on the high-probability event where (E.17) holds,

$$\sup_{z\in\mathcal{D}_\delta}\|R(z)\|_{\mathrm{op}} \;\le\; C\,K\sqrt{\frac{\log N}{N}}. \tag{E.19}$$

This proves (E.10) with $C_1:=C$.

From the decomposition (E.15), we have

$$M(z)-m_{\mathrm{MP}}(z)I_K=R(z)+m_{\mathrm{MP}}(z)\,\big(\Gamma-I_K\big).$$

[5] The term "macroscopic" refers to the fact that the spectral parameter z remains at a fixed distance δ from the spectrum. In contrast, "local laws" investigate the regime where $\Im z\asymp N^{-1}$, probing the spectrum at the scale of individual eigenvalue spacing.

[6] Rigorous estimates on eigenvector delocalization (e.g., $\sup_i\|u_i\|_\infty\lesssim N^{-1/2}$ up to logarithmic factors) are usually derived by applying isotropic local laws similar to (E.4) but extended down to the microscopic spectral scale $\Im z\sim N^{-1}$.

Taking operator norms and inserting (E.19) gives

$$\sup_{z\in D_\delta}\Big\|M(z)-m_{\mathrm{MP}}(z)I_K\Big\|_{\mathrm{op}} \;\le\; \sup_{z\in D_\delta}\|R(z)\|_{\mathrm{op}} + \sup_{z\in D_\delta}|m_{\mathrm{MP}}(z)|\,\|\Gamma-I_K\|_{\mathrm{op}}.$$

The first term is controlled by (E.19). The Marchenko–Pastur Stieltjes transform $m_{\mathrm{MP}}(z)$ satisfies the usual bound $|m_{\mathrm{MP}}(z)|\le C'(q,\delta)$ on D_δ (this follows directly from the defining equation and the fact that D_δ stays a fixed distance from the MP support). Thus

$$\sup_{z\in D_\delta}\Big\|M(z)-m_{\mathrm{MP}}(z)I_K\Big\|_{\mathrm{op}} \;\le\; C_1\,K\sqrt{\tfrac{\log N}{N}} + C_0'\,\|\Gamma-I_K\|_{\mathrm{op}},$$

which is (E.11).

Finally, we bound $\|\Gamma-I_K\|_{\mathrm{op}}$ in terms of $\varepsilon_{\mathrm{orth}}$. By definition,

$$\Gamma = U^\top U, \qquad \Gamma_{\mu\nu}=\langle u_\mu,u_\nu\rangle,$$

so that $\Gamma_{\mu\mu}=1$ and $|\Gamma_{\mu\nu}|\le\varepsilon_{\mathrm{orth}}$ for $\mu\neq\nu$. Set $B:=\Gamma-I_K$, which has zero diagonal and entries $B_{\mu\nu}=\Gamma_{\mu\nu}$ for $\mu\neq\nu$. Gershgorin's circle theorem states that any eigenvalue λ of B satisfies

$$|\lambda|\le R_\mu := \sum_{\nu\neq\mu}|B_{\mu\nu}| \le \sum_{\nu\neq\mu}\varepsilon_{\mathrm{orth}} = (K-1)\,\varepsilon_{\mathrm{orth}} \quad \text{for some } \mu.$$

Therefore

$$\|\,\Gamma-I_K\,\|_{\mathrm{op}} = \|B\|_{\mathrm{op}} = \max_{\lambda\in\mathrm{spec}(B)}|\lambda| \le (K-1)\,\varepsilon_{\mathrm{orth}},$$

which implies (E.12) upon absorbing the factor $K-1$ and the bound on $|m_{\mathrm{MP}}(z)|$ into C_0'. This concludes the proof. □

Remark 13 (Geometric interpretation and decoupling). Lemma 1 provides a rigorous quantitative manifestation of the *isotropic decoupling* phenomenon. In essence, it asserts that the resolvent $G(z)$ acts on any low-dimensional subspace spanned by U as a scalar multiple of the identity, $m_{\mathrm{MP}}(z)I_N$, up to a geometric correction term encoded by the Gram matrix $\Gamma=U^\top U$. This implies that the correlations between distinct directions u_μ and u_ν induced by the resolvent are asymptotically negligible, provided the directions themselves are not too correlated.[7]

The error bound in (E.12) reveals a competition between two scales: the statistical fluctuation of the eigenvalues (of order $O(\sqrt{\log N/N})$) and the intrinsic geometry of the test vectors (represented by $\varepsilon_{\mathrm{orth}}$). This separation is crucial for applications where the test vectors U may depend on the matrix Z in a weak sense, or when performing a change of basis where approximate orthogonality must be preserved.[8]

We now provide the proof of Theorem 2, whose statement appears in Section 4.1. We only outline the argument, since each part is a direct consequence of standard results in the theory of spiked sample covariance matrices, together with the whitening construction and the decoupling lemma (Lemma 1) proved earlier.

Proof. (D1) Bulk confinement. Write $U:=\mathrm{span}\{u_\mu\}_{\mu=1}^K$ and $W:=U^\perp$, and let P_W be the orthogonal projector onto W. By construction,

$$J=\Sigma^{1/2}S_0\Sigma^{1/2}, \qquad \Sigma=\sigma^2 I_N+\sum_{\mu=1}^K\theta_\mu u_\mu u_\mu^\top.$$

On W we have $\Sigma|_W=\sigma^2 I_W$, so that

$$P_W J P_W = \sigma^2\,P_W S_0 P_W. \tag{E.20}$$

Define the $(N-K)\times(N-K)$ "bulk" sample covariance

$$S_W := P_W S_0 P_W.$$

The rows Z_k are isotropic and satisfy the tail/regularity assumptions under which the anisotropic Marchenko–Pastur local law and eigenvalue rigidity hold (cf. Theorem 3; see also Bai–Silverstein (Bai & Silverstein, 2010, Chs. 3 and 5) and Knowles–Yin (Knowles & Yin, 2017, Thms. 3.6 and 3.12) for the sample–covariance case). Since K is fixed, the effective aspect ratio is

$$q_W := \frac{N-K}{M_{\mathrm{round}}} = q+O\Big(\frac{1}{N}\Big),$$

and the eigenvalues of S_W are, with probability at least $1-\delta$, contained in an interval of the form

$$\Big[(1-\sqrt{q_W})^2-\tilde\varepsilon_N(\delta),\;(1+\sqrt{q_W})^2+\tilde\varepsilon_N(\delta)\Big],$$

where $\tilde\varepsilon_N(\delta)\downarrow 0$ as $N\to\infty$.

Combining this with (E.20) and the relation $q_W=q+O(1/N)$, and absorbing the resulting $O(1/N)$ shift into the error, we obtain

$$\mathrm{spec}(P_W J P_W)\subset\big[\lambda_-(q)-\varepsilon_N(\delta),\;\lambda_+(q)+\varepsilon_N(\delta)\big]$$

for some deterministic $\varepsilon_N(\delta)\downarrow 0$, with probability at least $1-\delta$. A choice of the form (4.11) follows from the quantitative rigidity bounds in Bai and Silverstein (2010), Erdős et al. (2012), Knowles and Yin (2017).

Since $P_W J P_W$ acts on an $(N-K)$–dimensional subspace, Cauchy interlacing (or the min–max principle applied to the decomposition $\mathbb{R}^N=W\oplus U$) implies that at most K eigenvalues of J can lie to the right of this interval (and at most K to the left). This proves (D1).

(D2)–(D4) BBP threshold, outlier locations, and eigenvector alignment. We now identify the outliers and their eigenvectors. For clarity we proceed in two steps: first the "decoupled" case where $\{u_\mu\}$ are orthonormal, then the nearly-orthogonal case $\varepsilon_{\mathrm{orth}}>0$ as a small perturbation.

Orthogonal spikes. Assume temporarily that $\langle u_\mu,u_\nu\rangle=\delta_{\mu\nu}$, so that $U^\top U=I_K$ and $\Sigma=\sigma^2\big(I+U\mathcal{K}U^\top\big)$ with $\mathcal{K}:=\mathrm{diag}(\kappa_1,\dots,\kappa_K)$ and $\kappa_\mu=\theta_\mu/\sigma^2$ as in Theorem 2; the symbol K is reserved for the number of spikes. This is the classical spiked population covariance model (introduced in Johnstone, 2001 and studied in Baik & Silverstein, 2006, Paul, 2007, and Benaych-Georges & Nadakuditi, 2011). In this setting, the eigenvalues of J exhibit the so-called BBP phase transition: each spike κ_μ produces (at most) one outlier eigenvalue of J, which detaches from the upper MP edge $\lambda_+(q)$ if and only if $\kappa_\mu>\sqrt{q}$, and converges to

$$\lambda_{\mathrm{out}}(\kappa_\mu)=\sigma^2(1+\kappa_\mu)\Big(1+\frac{q}{\kappa_\mu}\Big),$$

see e.g. Baik–Silverstein (Baik & Silverstein, 2006, Thm 1.1) or Benaych-Georges–Nadakuditi (Benaych-Georges & Nadakuditi, 2011, Thm 2.7 and Rem. 2.11). This yields (D2) and (D3) in the orthogonal case.

Moreover, for each spike with $\kappa_\mu>\sqrt{q}$, the associated sample eigenvector v_μ aligns nontrivially with the population direction u_μ. In particular, Paul (Paul, 2007, Thm 4) and Benaych-Georges–Nadakuditi (Benaych-Georges & Nadakuditi, 2011, Thm 2.9 and Rem. 2.11) show that

$$|\langle v_\mu,u_\mu\rangle|^2\xrightarrow{P}\gamma(\kappa_\mu,q):=\frac{1-\frac{q}{\kappa_\mu^2}}{1+\frac{q}{\kappa_\mu}},$$

while the projection of v_μ onto u_ν for $\nu\neq\mu$ vanishes in probability. This proves (D4) in the orthogonal case.

Nearly-orthogonal spikes. We now return to the general case where the spike directions satisfy $\varepsilon_{\mathrm{orth}}:=\max_{\mu\neq\nu}|\langle u_\mu,u_\nu\rangle|\to 0$ as $N\to\infty$. Let $U=[u_1,\dots,u_K]$ and recall the block resolvent $M(z):=U^\top G(z)U$ with $G(z)=(S_0-zI_N)^{-1}$. Lemma 1 (uniform quantitative decoupling) shows that, for z in a fixed domain bounded away from the MP bulk,

$$M(z)=m_{\mathrm{MP}}(z)\,\Gamma+R(z), \qquad \|R(z)\|_{\mathrm{op}}\;\le\;C\,K\sqrt{\frac{\log N}{N}},$$

with high probability, where $\Gamma=U^\top U$ is the Gram matrix. When $\varepsilon_{\mathrm{orth}}$ is small, $\Gamma=I_K+E$ with $\|E\|_{\mathrm{op}}=O(K\varepsilon_{\mathrm{orth}})$ by Gershgorin's theorem, and

[7] This behavior is a hallmark of rotationally invariant ensembles. For non-invariant ensembles like the one considered here, the result hinges on the underlying eigenvector delocalization, which ensures that the fixed vectors u_μ do not align with the random eigenbasis of S_N.

[8] In many applications, such as the analysis of outliers or the stability of the resolvent expansion, K represents the number of outliers or a finite rank perturbation. The condition $K\ll\sqrt{N}$ ensures that the cumulative error from the fluctuations does not overwhelm the deterministic signal.

therefore

$$M(z) = m_{\mathrm{MP}}(z) I_K + \widetilde{R}(z), \qquad \|\widetilde{R}(z)\|_{\mathrm{op}} = O\big(\varepsilon_{\mathrm{orth}} + K\sqrt{\tfrac{\log N}{N}}\big).$$

The secular equation controlling the outliers of J can be derived using Woodbury's identity and the matrix determinant lemma (see the master equation for finite-rank perturbations in Benaych-Georges–Nadakuditi (Benaych-Georges & Nadakuditi, 2011, Prop. 5.1)). It takes the form

$$\det\big(I_K + z\, C_\kappa\, M(z)\big) = 0, \qquad C_\kappa := \mathrm{diag}\Big(\frac{\kappa_\mu}{1+\kappa_\mu}\Big), \quad z := \frac{\lambda}{\sigma^2}.$$

The sign follows from the resolvent identity $G(z)S_0 = I_N + zG(z)$, which gives $U^\top G(z) S_0 U = \Gamma + zM(z)$ in the factorization $\det\big(I_K + \mathcal{K} U^\top G(z) S_0 U\big) = 0$ of the perturbation determinant. Replacing $M(z)$ by its asymptotic approximation $m_{\mathrm{MP}}(z) I_K$ (valid for vanishing overlap $\varepsilon_{\mathrm{orth}} \to 0$ by Lemma 1) yields the decoupled scalar equations

$$1 + z c_\mu\, m_{\mathrm{MP}}(z) = 0, \qquad c_\mu := \frac{\kappa_\mu}{1+\kappa_\mu}.$$

The unique solutions $z = z_{\mathrm{out}}(\kappa_\mu)$ to these equations in the domain $z > (1+\sqrt{q})^2$ correspond exactly to the outlier locations $\lambda_{\mathrm{out}}(\kappa_\mu)$ in (4.12), subject to the BBP threshold condition $\kappa_\mu > \sqrt{q}$. For the derivation of these specific locations and thresholds in the spiked covariance model, see, e.g., Baik–Silverstein (Baik & Silverstein, 2006, Thm. 1.1) and the multiplicative perturbation example in Benaych-Georges–Nadakuditi (Benaych-Georges & Nadakuditi, 2011, Thm. 2.7 and Rem. 2.11).

Since $M(z)$ differs from $m_{\mathrm{MP}}(z) I_K$ by an operator of norm $O(\varepsilon_{\mathrm{orth}} + K\sqrt{\log N/N})$ uniformly on the spectral domain, and K is fixed, a standard stability argument for roots of analytic matrix-valued equations implies that the corresponding outlier eigenvalues of J differ from their orthogonal-case limits by at most $O(\varepsilon_{\mathrm{orth}} + N^{-1/2})$ in probability (see, e.g., Benaych-Georges–Nadakuditi (Benaych-Georges & Nadakuditi, 2011, Lemma 6.1)). For the underlying isotropic local laws enabling this control, see Bloemendal–Erdős–Knowles–Yau–Yin (Bloemendal et al., 2014, Thm 2.12 and Intro).

Finally, the eigenvectors associated with these outliers vary smoothly under such small perturbations. Since the outliers are separated from the bulk and from each other (for distinct κ_μ), the Davis–Kahan sinΘ theorem (Davis & Kahan, 1970) yields

$$|\langle v_\mu, u_\mu\rangle|^2 = \gamma(\kappa_\mu, q) + O\big(\varepsilon_{\mathrm{orth}} + N^{-1/2}\big)$$

in probability, while the overlaps with the other u_ν remain negligible. This transfers the orthogonal-case alignment in (D4) to the nearly-orthogonal setting, completing the proof of (D2)–(D4). □

E.3. *The feedback-dependent estimator: Conditional decomposition and residual terms*

This appendix collects the proofs of Proposition 1 and Corollary 1, derives the residual-outlier formula (4.27), and lists explicitly the terms that our treatment of the case $t > 0$ does *not* control rigorously. Throughout, $\mathfrak{F}_{t-1}$ is the history σ-algebra (4.14) and $\mathfrak{G}_t := \mathfrak{F}_{t-1} \vee \sigma(\{w_c(t)\}_{c=1}^L)$.

Proof of Proposition 1. *Decomposition* (4.16). Substituting the client update (3.6) into the server aggregate (3.2) and multiplying by N,

$$\begin{aligned} N\boldsymbol{J}_s^{(t)} &= \frac{1}{L}\sum_{c=1}^{L}\Big[w_c(t)\, N\tilde{\boldsymbol{J}}_c^{(t)} + \big(1-w_c(t)\big) N\boldsymbol{A}^{(t-1)}\Big] \\ &= \frac{1}{L}\sum_{c=1}^{L} w_c(t)\, N\tilde{\boldsymbol{J}}_c^{(t)} + \big(1-\bar{w}_t\big) N\boldsymbol{A}^{(t-1)}, \end{aligned}$$

because $\boldsymbol{A}^{(t-1)}$ does not depend on c. Using the definition (4.30) of $\boldsymbol{A}^{(t-1)}$, i.e. $N\boldsymbol{A}^{(t-1)} = \sum_\nu \hat{\xi}_{s,\nu}^{(t-1)} (\hat{\xi}_{s,\nu}^{(t-1)})^\top$, gives (4.16). The identity is algebraic and involves no approximation.

Property (P1). Each summand of $\boldsymbol{F}_{t-1}$ is a rank-one positive semidefinite matrix and $1-\bar{w}_t \geq 0$, whence $\boldsymbol{F}_{t-1} \succeq 0$ and $\mathrm{rank}\, \boldsymbol{F}_{t-1} \leq \widehat{K}_{\mathrm{rec}}(t-1)$. The vectors $\hat{\xi}_{s,\nu}^{(t-1)}$ are the output of the round-$(t-1)$ server factorization, hence $\mathfrak{F}_{t-1}$-measurable, while $\bar{w}_t$ is $\sigma(\{w_c(t)\})$-measurable; therefore $\boldsymbol{F}_{t-1}$ is $\mathfrak{G}_t$-measurable. For the spectral consequence, write $k := \widehat{K}_{\mathrm{rec}}(t-1)$. Monotonicity of the eigenvalue map under positive semidefinite perturbations gives $\lambda_i(\boldsymbol{P}_t + \boldsymbol{F}_{t-1}) \geq \lambda_i(\boldsymbol{P}_t)$ for every i, while Weyl's inequality $\lambda_{i+j-1}(\boldsymbol{X}+\boldsymbol{Y}) \leq \lambda_i(\boldsymbol{X}) + \lambda_j(\boldsymbol{Y})$ applied with $j = k+1$ and $\lambda_{k+1}(\boldsymbol{F}_{t-1}) = 0$ gives $\lambda_i(\boldsymbol{P}_t + \boldsymbol{F}_{t-1}) \leq \lambda_{i-k}(\boldsymbol{P}_t)$ for $i > k$. This is (4.17). Since, by Theorem 2 (D1) applied conditionally to $\boldsymbol{P}_t$, all but at most K eigenvalues of $\boldsymbol{P}_t$ lie in the band (4.10), all but at most $K+k$ eigenvalues of $\tilde{\boldsymbol{J}}_s^{(t)}$ lie in the same band.

Property (P2). By (2.13) the round-t examples are drawn independently of the past. Taking conditional expectations in $N\tilde{\boldsymbol{J}}_c^{(t)} = (M_c^t)^{-1}\sum_m (\boldsymbol{\eta}_c^{(t)})^m [(\boldsymbol{\eta}_c^{(t)})^m]^\top$ and using the population identity (4.4) at the level of the single client,

$$\mathbb{E}\big[N\tilde{\boldsymbol{J}}_c^{(t)} \,\big|\, \mathfrak{G}_t\big] = \sigma^2 \boldsymbol{I} + Nr^2 \sum_{\mu=1}^{K} \pi_{t,c}(\mu)\, u_\mu u_\mu^\top.$$

Averaging over clients with weights $w_c(t)/L$ and using the definitions (4.15) yields (4.18). The signal-to-noise ratio of spike μ for this population operator is the ratio of the spike strength $Nr^2\pi_t^w(\mu)$ to the isotropic level $\bar{w}_t\sigma^2$, i.e. $Nr^2\pi_t^{\mathrm{eff}}(\mu)/\sigma^2$, which is (4.19). Note that a homogeneous rescaling $\boldsymbol{P}_t \mapsto \bar{w}_t^{-1}\boldsymbol{P}_t$ leaves both q and $\{\kappa_\mu\}$ invariant, which is the precise sense in which the overall plasticity level does not move the BBP threshold.

Property (P3). The first equality in (4.20) is immediate from the definition of $\boldsymbol{F}_{t-1}$. For the second, the sum over ν is dominated by the best-matching index, which realizes $m_{t-1}(\mu)$ by definition, while every other reconstructed archetype has overlap $O(N^{-1/2})$ with u_μ in the nearly-orthogonal regime, contributing $O(K/N)$ in total. For (4.21) we work at the population level: conditionally on $\mathfrak{G}_t$, the finite-rank part of the effective operator is

$$\boldsymbol{\Theta}_t := Nr^2 \sum_\mu \pi_t^w(\mu)\, u_\mu u_\mu^\top + (1-\bar{w}_t)\, N \sum_\nu \hat{u}_\nu \hat{u}_\nu^\top,$$

of rank at most $K + \widehat{K}_{\mathrm{rec}}(t-1)$. The family $\{u_\mu\} \cup \{\hat{u}_\nu\}$ is *not* nearly orthogonal, since $\langle u_\mu, \hat{u}_{\nu(\mu)}\rangle = m_{t-1}(\mu)$ is $O(1)$ for a consolidated archetype; the correct reduction is therefore to diagonalize $\boldsymbol{\Theta}_t$ blockwise, each block being spanned by a matched pair $\{u_\mu, \hat{u}_{\nu(\mu)}\}$, and to apply Theorem 2 to the resulting family of block eigenvectors, which *is* nearly orthogonal up to $O(\varepsilon_{\mathrm{orth}} + N^{-1/2})$. In the orthonormal basis $\{u_\mu, e_\mu\}$ of the block, with $\hat{u}_{\nu(\mu)} = m\, u_\mu + \sqrt{1-m^2}\, e_\mu$ and $m := m_{t-1}(\mu)$, the block reads

$$\boldsymbol{\Theta}_t^{(\mu)} = N\begin{pmatrix} r^2\pi_t^w(\mu) + (1-\bar{w}_t)m^2 & (1-\bar{w}_t)\, m\sqrt{1-m^2} \\ (1-\bar{w}_t)\, m\sqrt{1-m^2} & (1-\bar{w}_t)(1-m^2) \end{pmatrix}, \tag{E.21}$$

whose trace is $N[r^2\pi_t^w(\mu) + (1-\bar{w}_t)]$ and whose determinant is $N^2 r^2 \pi_t^w(\mu)(1-\bar{w}_t)(1-m^2)$. For $m \to 1$ the leading eigenvalue converges to $N[r^2\pi_t^w(\mu) + (1-\bar{w}_t)m^2]$ with eigenvector $u_\mu + O(1-m)$; dividing by the isotropic level $\bar{w}_t\sigma^2$ gives (4.21). Since the spikes are $O(N)$ while the bulk edge is $O(1)$, the multiplicative BBP correction factor $(1+q/\kappa)$ equals $1 + O(N^{-1})$ and the superposition is accurate to leading order; see item (R5) below. □

Proof of Corollary 1. Let $\boldsymbol{S}_t := L^{-1}\sum_c N\tilde{\boldsymbol{J}}_c^{(t)}$ be the *unweighted* rescaled round design, to which Theorem 2 applies conditionally on $\mathfrak{G}_t$ with aspect ratio q, isotropic level σ^2 and spikes $\kappa_\mu^{(t)}$ of (4.2). Since

$$\boldsymbol{P}_t - w_{\min}(t)\,\boldsymbol{S}_t = \frac{1}{L}\sum_{c=1}^{L}\big(w_c(t) - w_{\min}(t)\big)\, N\tilde{\boldsymbol{J}}_c^{(t)} \succeq 0, \qquad w_{\max}(t)\,\boldsymbol{S}_t - \boldsymbol{P}_t \succeq 0,$$

we have the positive semidefinite sandwich $w_{\min}(t)\boldsymbol{S}_t \preceq \boldsymbol{P}_t \preceq w_{\max}(t)\boldsymbol{S}_t$, and Weyl monotonicity gives $w_{\min}(t)\lambda_i(\boldsymbol{S}_t) \leq \lambda_i(\boldsymbol{P}_t) \leq w_{\max}(t)\lambda_i(\boldsymbol{S}_t)$ for every i. On the high-probability event of Theorem 2, the non-outlier eigenvalues of $\boldsymbol{S}_t$ do not exceed $\lambda_+(q) +$

$\varepsilon_N(\delta)$, whereas the outlier generated by a supercritical spike sits at $\lambda_{\text{out}}(\kappa_\mu^{(t)}) = \sigma^2(1+\kappa_\mu^{(t)})(1+q/\kappa_\mu^{(t)})$ up to $O(\varepsilon_{\text{orth}} + N^{-1/2})$. Hence a sufficient condition for the μth eigenvalue of $\boldsymbol{P}_t$ to exceed the bulk of $\boldsymbol{P}_t$ is

$$w_{\min}(t)\,\sigma^2\big(1+\kappa_\mu^{(t)}\big)\Big(1+\frac{q}{\kappa_\mu^{(t)}}\Big) \;>\; w_{\max}(t)\,\sigma^2\big(1+\sqrt{q}\big)^2,$$

that is, $(1+\kappa)(1+q/\kappa) > \rho_t(1+\sqrt{q})^2$ with $\kappa := \kappa_\mu^{(t)}$. Expanding, this is $\kappa + q/\kappa > b_t$ with $b_t = \rho_t(1+\sqrt{q})^2 - (1+q)$, i.e. $\kappa^2 - b_t\kappa + q > 0$. The function $\kappa \mapsto (1+\kappa)(1+q/\kappa)$ attains its minimum $(1+\sqrt{q})^2$ at $\kappa = \sqrt{q}$ and is increasing for $\kappa > \sqrt{q}$; consequently $b_t \geq 2\sqrt{q}$, the discriminant $b_t^2 - 4q$ is nonnegative, and the relevant root is the larger one, $\kappa_c(\rho_t, q) = \frac{1}{2}\big(b_t + \sqrt{b_t^2 - 4q}\big)$, which gives (4.23). For $\rho_t = 1$ one has $b_t = 2\sqrt{q}$ and $\kappa_c = \sqrt{q}$, recovering (D2). Finally, adding $\boldsymbol{F}_{t-1} \succeq 0$ can only increase the eigenvalues by (P1) without displacing the bulk, so the same condition is sufficient for $\tilde{\boldsymbol{J}}_s^{(t)}$. □

Residual outliers generated by imperfect reconstruction.

The second eigenvalue of the block (E.21) equals det / tr up to $O((1-m^2)^2)$, that is,

$$\lambda_\mu^{\text{res}}(t) \simeq N\,\frac{r^2\pi_t^w(\mu)\,(1-\bar{w}_t)}{r^2\pi_t^w(\mu) + (1-\bar{w}_t)}\,\big(1-m_{t-1}(\mu)^2\big),$$

which is (4.27). Its eigenvector lies in $\text{span}\{u_\mu, \hat{u}_{v(\mu)}\}$ and is orthogonal to the leading block direction: it encodes the reconstruction error of round $t-1$, not a new archetype. Because the prefactor is $O(N)$ while the bulk edge is $O(1)$, this residual direction becomes supercritical already for $1 - m_{t-1}(\mu)^2 \gtrsim \sqrt{q}\,\bar{w}_t\sigma^2/(cN)$ with c the harmonic-mean prefactor above, i.e. for reconstruction errors as small as $O((N M_{\text{round}})^{-1/2})$. The unsharpened count $\hat{K}_{\text{MP}}(t)$ is therefore expected to *over*-estimate the number of distinct archetypes in the feedback-driven phase, by an amount controlled by the current reconstruction quality. This is a further reason—beyond the sharpening step itself—why the pipeline drives the reconstruction stage through $\hat{K}_{\text{spec}}(t)$ followed by duplicate merging (2.23) and the quadratic acceptance test (2.27): a residual direction is, by construction, an almost-duplicate of an already accepted archetype and is eliminated at that stage. Consistently, the effective rank reported in Fig. 9 tracks the true number of archetypes rather than an inflated count.

Residual terms not controlled rigorously.

We list, for transparency, the ingredients that separate the analysis above from a fully rigorous treatment of the case $t > 0$, together with the order of magnitude we expect for each of them.

(R1) *Data dependence of the plasticity weights.* The weight $w_c(t)$ is a functional of the round-t batch through the sign-agreement statistic $p_c(t)$ in (4.32) and the quality estimate $\hat{r}_c(t)$ in (4.43); conditioning on $\mathfrak{G}_t$ therefore tilts the conditional law of the batch. Writing $\bar{w}_c(t) := \mathbb{E}[w_c(t) \mid \mathfrak{F}_{t-1}]$ and $\boldsymbol{P}_t^{\text{det}}$ for the operator obtained by replacing $w_c(t)$ with $\bar{w}_c(t)$, one has the deterministic bound

$$\Big\|\boldsymbol{P}_t - \boldsymbol{P}_t^{\text{det}}\Big\|_{\text{op}} \;\leq\; \max_c\Big|w_c(t) - \bar{w}_c(t)\Big| \cdot \max_c\Big\|N\tilde{\boldsymbol{J}}_c^{(t)}\Big\|_{\text{op}},$$

where the second factor is $Nr^2\max_\mu \pi_{t,c}(\mu) + \sigma^2(1+\sqrt{q})^2 + o(1)$ with high probability. The exponential moving average (4.48) implies that only the fraction α of $w_c(t)$ is data-dependent, since $w_c(t-1)$ is $\mathfrak{F}_{t-1}$-measurable; and $w_{c,\text{inst}}(t)$ is a Lipschitz function of the pair $(p_c(t), \hat{r}_c(t))$ away from the clipping boundaries of (4.47). We therefore expect $\max_c |w_c(t) - \bar{w}_c(t)| = O_P\big(\alpha\,(M_c^t)^{-1/2}\big)$, hence a *relative* perturbation of order $\alpha(M_c^t)^{-1/2}$ on both the spikes and the bulk edge, which leaves κ_μ and the threshold unchanged to leading order. We do not prove this concentration: $p_c(t)$ is a functional of $O(N^2)$ strongly dependent sign indicators, and a rigorous bounded-difference argument would require controlling the fraction of near-degenerate entries of the consolidated operator. This is the first point at which our treatment is an approximation rather than a theorem.

(R2) *Quenched versus annealed statements.* All statements are conditional on $\mathfrak{G}_t$. The effective population operator (4.18) is itself random and $\mathfrak{F}_{t-1}$-measurable, so the deterministic equivalent of the round-t spectrum depends on the realized past. A description that is joint across rounds—and in particular an unconditional law of large numbers for the trajectory $t \mapsto \text{spec}(\tilde{\boldsymbol{J}}_s^{(t)})$—would require a local law for a recursively defined, time-dependent ensemble, with free-probability control of the cross terms between $\boldsymbol{P}_t$ and $\boldsymbol{F}_{t-1}$. We do not pursue this here.

(R3) *Reconstruction bias in the feedback directions.* The eigendirections of $\boldsymbol{F}_{t-1}$ are the reconstructed archetypes $\hat{u}_v$, not the true ones. The block reduction of (P3) shows that the leading block eigenvector differs from u_μ by $O(1 - m_{t-1}(\mu))$, so the alignment law (D4) holds for $t > 0$ with an additional bias of this order. Since $m_{t-1}(\mu) \to 1$ as the reconstruction improves, the bias is self-suppressing along a successful trajectory, but it is not uniform in t.

(R4) *Over-counting through residual directions.* As shown above, $\hat{K}_{\text{MP}}(t)$ can exceed the number of distinct archetypes whenever $m_{t-1}(\mu) < 1$. The pipeline is insensitive to this because the downstream pruning removes almost-duplicates, but the identity "supercritical direction = archetype" is not exact for $t > 0$.

(R5) *Mixed additive/multiplicative perturbation.* Denoting by $\boldsymbol{\Sigma}_t$ the conditional population operator (4.18) of the plastic part, the estimator $\tilde{\boldsymbol{J}}_s^{(t)} = \boldsymbol{\Sigma}_t^{1/2}\boldsymbol{S}_0\boldsymbol{\Sigma}_t^{1/2} + \boldsymbol{F}_{t-1}$ combines a multiplicative finite-rank perturbation (the plastic spikes) with an additive one (the feedback). We use the leading-order superposition of the two population contributions, cf. (E.21), instead of the exact joint transform of Benaych-Georges–Nadakuditi. The approximation is accurate because the spikes are $O(N)$ while the bulk edge is $O(1)$, so the BBP correction factor $(1+q/\kappa)$ differs from 1 by $O(N^{-1})$; it would have to be revisited in a regime of weak spikes, $\kappa_\mu = O(1)$.

Items (R1), (R2) and (R5) are the reason why, for $t > 0$, we present Theorem 2 as a controlled approximation rather than as an exact result, in the sense made precise by Remark 6. Items (R3) and (R4) are quantified above and are neutralized operationally by the pruning stage of the reconstruction pipeline.

Appendix F. Technical Details

F.1. Additional experimental details and hyperparameters

F.2. Complexity

We write L (clients/layers), T (rounds), K (archetypes), N (pattern dimension), M_{tot} (total examples), $M_c = \lceil M_{\text{tot}}/(LT)\rceil$ (per-client per-round), S (seeds), I_{prop} (pseudo-inverse propagation steps), n_{rand} (reshuffles for the shuffle null), U (LAM updates), s (blocks of candidate initializations; post-replication $\tilde{K} \approx sL$), and $A \leq$ `max_attempts` (worst-case retries in disentangling) (see Table 2).

Let $M_{\text{round}} = L\,M_c$ and $\tilde{K} \approx sL$. A single round costs

$$\begin{aligned}\text{round_cost} \;=\; O\Big(&LM_cN^2 \;+\; KN^2 \;+\; I_{\text{prop}}N^3 \;+\; C_{\text{spec}}N^3 \\ &+\; A\big[s\hat{K}N \;+\; U(sLN^2 + sL^2N) \;+\; sLN^2 \;+\; (sL)^2N \\ &+\; \tilde{K}KN \;+\; \max\{\tilde{K}, K\}^3\big] \;+\; LM_c \;+\; N^2\Big),\end{aligned} \tag{F.1}$$

where $C_{\text{spec}} = 1$ for MP-edge and $C_{\text{spec}} = 1 + n_{\text{rand}}$ for shuffle-based cuts.

Per seed:

$$O\Big(KN^2 + K^3 + M_{\text{tot}}N + T\cdot\text{round_cost} + M_{\text{tot}}\log M_{\text{tot}}\Big). \tag{F.2}$$

Across S seeds:

$$\begin{aligned}&O\big(S\cdot\texttt{seed_cost}\big) \quad \text{(serial)}\\ &O\big(\lceil S/n_{\text{workers}}\rceil\cdot\texttt{seed_cost}\big) \quad \text{(with concurrency).}\end{aligned} \tag{F.3}$$

Table 2
Main hyperparameters and typical values.

Parameter	Meaning	Typical range	Example default
β	tanh gain (inverse temperature)	$[1, 5]$	2.5
λ	weight of $h^{(2)}$ correction	$[0.05, 0.5]$	0.2
h	coupling of external input $h^{(3)}$	$[0, 0.5]$	0.1
`updates`	LAM steps per round	$[40, 200]$	50
τ_{spec}	post-sharpening eigenvalue threshold on $\widehat{J}^{KS}$	$[0.3, 0.7]$	0.5
ρ	spectral-alignment threshold	$[0.5, 0.8]$	0.6
δ_{dup}	mutual-overlap duplicate threshold	$[0.3, 0.6]$	0.5
w	unsup/Hebb blending	$[0, 1]$	Appendix 4.2
α_{sharp}	logistic sharpening gain	$[0.3, 0.7]$	0.5
`noise_scale`	initial LAM noise amplitude	$[0.05, 0.5]$	0.3
`min_scale`	minimum LAM noise amplitude	$[0.0, 0.1]$	0.02
`prop_iters`	pseudo-inverse propagation steps	$[50, 300]$	200
`prop_eps`	pseudo-inverse propagation ε	$[10^{-3}, 10^{-2}]$	10^{-2}

Table F.2

Block	Asymptotic cost
True archetypes & $J^\star$ build	$O(KN) + O(K^2N + KN^2 + K^3)$
Federated dataset generation	$O(M_{\text{tot}}\, N)$
Round extraction (tensor slicing)	$O(L\, M_c\, N)$
Unsupervised J_{unsup} (client correlators)	$O(L\, M_c\, N^2)$
Blend with memory (Hebbian of $\approx K$ rows)	$O(K\, N^2)$
Pseudo-inverse propagation (polynomial map)	$O(I_{\text{prop}}\, N^3)$
Spectral cut (eigendecomposition)	$O(N^3)$
$\hat{K}$ estimation: MP vs. shuffle	$O(N^3)$ or $O((1{+}n_{\text{rand}})\, N^3)$
Candidate init from eigenvectors	$O(s\, \hat{K}\, N)$
LAM dynamics (multi-layer refinement)	$O(U\, s\, L\, N^2 + U\, s\, L^2\, N)$
Pruning & scoring (overlaps, Rayleigh)	$O(s\, L\, N^2 + (sL)^2\, N + s\, L\, K\, N)$
Extra attempts (worst case)	$\times A$ (multiplicative)
Assignment (Hungarian on $\tilde{K}{\times}K$)	$O(\tilde{K}\, K\, N + \max\{\tilde{K}, K\}^3)$
Coverage / metrics (per round)	$O(L\, M_c) + O(N^2)$
Aggregating across seeds	$O(S\, T)$ (stats) + $O(S\, N^2)$ (saves)

For the default scales ($N \sim 300$, $I_{\text{prop}} \ll N$, moderate s, L), the cubic terms in N dominate: (i) the pseudo-inverse propagation $O(I_{\text{prop}}\, N^3)$ and (ii) the spectral step $O(C_{\text{spec}}\, N^3)$. The TAM stage scales as $O(U\, s\, L\, N^2)$ and becomes competitive only when s approaches its cap (pushing $(sL)^3$ assignment to relevance). Using a top-k eigensolver with warm starts reduces $O(N^3)$ to

$$\text{cost}_{\text{top-}k} \approx O\big(k\, N^2\, n_{\text{iter}}\big), \quad n_{\text{iter}} = O(1\text{–}5),$$

leaving the overall pipeline effectively cubic in N unless $k \ll N$ consistently. The choice of $\hat{K}$ estimator affects constants (shuffle multiplies the spectral cost by $1{+}n_{\text{rand}}$) but not the degree.

Dense operators J, J_{KS}, $J^\star$ take $O(N^2)$ each; candidate pools store $O(s\, L\, N)$ binary states; streaming buffers for data are $O(M_{\text{tot}} N)$ (which can be reduced by per-round streaming).

Communication cost per round. Beyond computation, the per-round communication volume is $O(LN^2)$: each of the L clients uploads one symmetric $N \times N$ Hebbian operator and receives back the single server operator $\widehat{J}_s^{(t)}$ of equal size. This volume is independent of the number of archetypes K and of the number of examples, and can be compressed through low-rank or sparsified operator encodings when N is large. For ease of reference, all the scaling laws discussed in the preceding paragraph are summarized in Table F.2.

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